Bimodality of the elephant random walk: the bias interval, the threshold and rare reversals

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Abstract

We prove the conjecture of Guérin, Laulin, Raschel and Simon that the bimodal initial biases form a (possibly empty) open interval centered at one half for each superdiffusive memory of the elephant random walk. The interval's endpoints vary continuously with memory, and outside it the limit density has one local maximum. There is a unique memory threshold, enclosed near 0.6688128600146, at or below which the interval is empty. The fair density has two inflection points below and at this threshold and four above it. At the threshold the maximum is quartic. The two modes then separate on a square-root scale, while the bias interval opens on a three-halves scale. A fourth-moment argument proves bimodality for memory at least 0.79 without computation. Near memory one, two Landau components approximate the density in total variation and scaled uniform distance, uniformly over the initial bias.

For rare reversals, we give an explicit total variation approximation by a truncated Luria–Delbrück law and a Landau local limit theorem with a rate. The bounds locate every mode. For even length N, the origin is as likely as each end at exactly one reversal probability, with $N\varepsilon = 2\log N - \log\log N - \log 16 + o(1)$. For every even $N \geq 8$, this crossing law has two symmetric modal plateaus, with the center as its only interior minimum.

Keywords: elephant random walk; bimodality; shape transition; random recursive tree; compound Poisson approximation; Landau distribution.
MSC2020 subject classifications: Primary 60G50, Secondary 60E07; 60F05; 60C05; 65G30.

1 Introduction

An elephant and an ant are playing hide and seek. The elephant counts while the ant hides among many identical anthills, then opens its eyes and knows at once which one the ant is in. The ant's shoes are outside the door.

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Fix the first step of a superdiffusive elephant random walk. Guérin, Laulin, Raschel and Simon prove that its limit density is unimodal [10, Theorem 1.1]. Now choose that first step at random. We are mixing two densities (one for each initial direction), and having one peak in each does not mean their mixture has one peak. They ask which initial biases give two, conjecturing an open interval centered at one half [10, Remark 2.1(b)]. We prove their conjecture and locate the unique memory threshold where two peaks first appear under a fair start. A related conjecture, eventual log-concavity, is disproved by Trinas and Valle in a recent preprint [12].

Begin with the finite count law and take differences of neighboring probabilities. Away from the ends, the next row uses positive sums of these differences, so it cannot introduce another change of sign. The ends are the difficulty. Keeping their earlier values in the row lets us include them in the induction. We then pass the resulting shape bounds to the limit. This also explains the two humps in the finite law with rare reversals, where we use the same bounds directly.

1.1 The walk and its parameters

The elephant walk [1] takes steps $X_1, \dots, X_N \in \{+1, -1\}$, with $+1$ meaning right. At each step it chooses uniformly from its earlier steps and repeats the chosen direction, except with probability ε, when it goes the other way. Write U_t for the step chosen at time $t+1$, and call it the parent of that step. We choose all parents and all reversal coins independently, also independently of X_1.

It is convenient to count right steps instead of tracking position. Put $A_N = \#\{i \le N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. There are then $2A_N - N$ net steps to the right, so $A_N = N/2$ puts the walk at the origin. The two ends correspond to $A_N \in \{0, N\}$ (every step in the same direction). We take $\mathbb{P}(X_1 = +1) = \frac{1}{2}$ unless we specify another bias. Reflection then gives $P_N(k) = P_N(N-k)$. For the walk started to the right, we use $\mathbb{P}_+$ and $\mathbb{E}_+$, namely probability and expectation conditional on $X_1 = +1$. For even N put $k_c = N/2$, $C_N = P_N(k_c)$ and $E_N = P_N(N) = P_N(0)$, the center and end masses. The center mass C_N is used only for even N. We call the solution of $C_N = E_N$ the crossing.

All logarithms are natural, $H_k = \sum_{j \le k} 1/j$ is the k-th harmonic number, and γ is Euler's constant. For rare reversals we take $N \to \infty$, restricting to even N in statements about C_N. This is also the limit in $O(\cdot)$ and $o(\cdot)$. Any dependence in the constants implied by $O(\cdot)$ is stated explicitly (otherwise they are absolute).

The reversal probability is $\varepsilon \in (0,1)$, $x = N\varepsilon$, and the crossing is ε_N, with $x_N = N\varepsilon_N$.

For fixed memory put $a = 1 - 2\varepsilon \in (1/2, 1)$ and $S_N = 2A_N - N$. Conditional on $X_1 = 1$, the almost-sure limit of S_N/N^a has density h_a. With $\mathbb{P}(X_1 = 1) = q$, its density is

$$g_{a,q}(u) = q h_a(u) + (1-q) h_a(-u), \qquad g_a = g_{a,1/2}.$$

The copying probability in the original notation is $p = (1+a)/2$. In the finite recurrences we use $r_t(k) = \varepsilon + (1-2\varepsilon)k/t$.

1.2 Fixed memory

Theorem 1.1 (the bias interval and the memory threshold)**.** *For each $a \in (1/2, 1)$ there is $b_a \in [0, 1/2]$ such that $g_{a,q}$ has exactly two local maxima if and only if $b_a < q < 1 - b_a$. Otherwise it is unimodal with a unique local maximum, including at the two endpoints. The interval is empty exactly when $a \le a_*$, where a_* is unique and*

$$0.668812860014597754 < a_* < 0.668812860014601029.$$

The boundary b_a is continuous in a, with $b_a \to 0$ as $a \to 1$. The fair density has exactly two inflection points for $a \le a_*$ and four for $a > a_*$. Bimodality for $a \ge 79/100$ has a proof

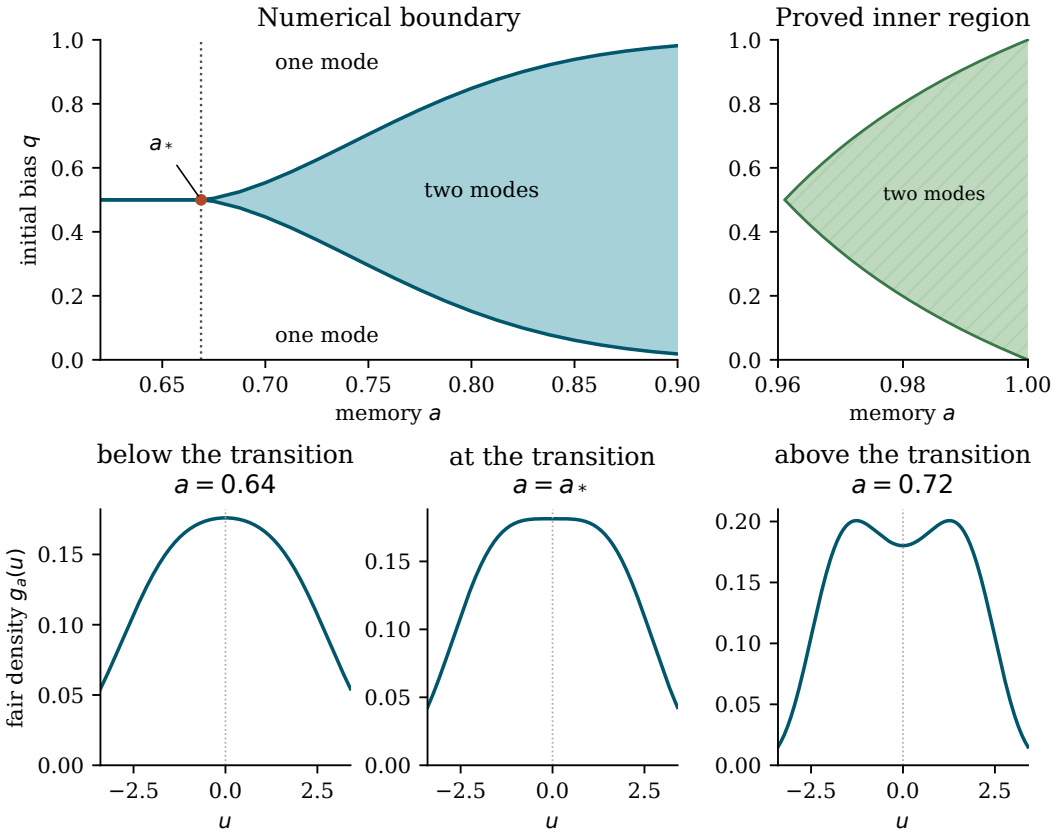


Figure 1: The bias interval and the fair-start transition. The upper left panel shows numerical values of the continuous boundary in Theorem 3.9 (the curves and shading are not certified enclosures). The upper right panel plots the explicit inner region $4\rho_a/(1-4\rho_a) < q < 1-4\rho_a/(1-4\rho_a)$ proved in Corollary 3.10. It does not locate the full boundary. The lower panels show the fair density before, at and after the transition. Its top is quartic at the threshold, whose certified enclosure is given in Theorem 1.1.

without computation. At a_* the top is quartic, and the two fair modes emerge at distance proportional to $\sqrt{a-a_*}$. The bias interval has half-width asymptotic to $Q_* \approx 9.86698769$ times $(a-a_*)^{3/2}$, and strictly widens just above the threshold.

Theorem 3.7 proves the interval assertion. Theorems 3.2–3.5 classify the fair density and enclose its threshold. Proposition 3.11 gives the shape at that threshold. Proposition 3.8 gives the variational equation for b_a, and Theorem 3.9 proves global continuity. Proposition 3.12 gives the three-halves law and local monotonicity of its width.

Theorem 1.2 (the profile near memory one)**.** *Put $\varepsilon = (1-a)/2$ and $A_\varepsilon = 1 + 2\varepsilon\log(1/\varepsilon)$. Let f_L be the Landau density with characteristic function $\exp\{-it\log|t| - \pi|t|/2\}$, and let λ_{L} be its mode. As $\varepsilon \downarrow 0$, the density*

$$\frac{q}{2\varepsilon} f_L\left(\frac{A_\varepsilon - u}{2\varepsilon}\right) + \frac{1-q}{2\varepsilon} f_L\left(\frac{A_\varepsilon + u}{2\varepsilon}\right)$$

approximates $g_{1-2\varepsilon,q}$ in L^1 and in supremum norm multiplied by 2ε, uniformly over $q \in [0,1]$. In particular $b_a \to 0$, with the explicit upper rate $b_a \le 8(1-a) + O((1-a)^2)$ from Corollary 3.10. For a fair start the positive mode is $A_\varepsilon - 2\varepsilon\lambda_{\mathrm{L}} + o(\varepsilon)$ and its height is $(f_L(\lambda_{\mathrm{L}}) + o(1))/(4\varepsilon)$.

Theorem 3.13 and Corollary 3.14 prove these assertions.

Theorem 1.3 (the finite shape bounds)**.** *For every $N \geq 1$, every $0 < \varepsilon < 1$ and every initial bias, the count law has at most two modal plateaus, including its endpoints. If $q \geq 1/2$, it dominates its reflection on the right half and has no strict valley there. For a fair start, $N \geq 3$ and $0 < \varepsilon < 1/2$, the nonzero signs of*

$$D_N(k) = P_N(k+1) - 2P_N(k) + P_N(k-1), \qquad 1 \leq k < N,$$

after equal neighbors are merged, form a subsequence of $(+, -, +, -, +)$.

The proofs are Proposition A.5, Proposition A.6 and Theorem 2.4. Weak convergence is enough to carry the valley bounds to a continuous limit (Lemma A.7). For the curvature bound, we first smooth by a Gaussian and then apply the sign bound.

1.3 Rare reversals

Now let $\varepsilon \to 0$ with $x = N\varepsilon$ of order $\log N$. The center and a specified end cross at about twice as many reversals as in the persistent walk [16]. An early reversal changes an entire family of descendants (all the steps that copy it, directly or indirectly). The calculation is in Section 4.

Let M_N count steps opposite to X_1. Its law $f_N(m) = \mathbb{P}(M_N = m)$ does not depend on X_1, and $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$ (Lemma 4.1). Let CP_N be the law of $\sum_{d=1}^{N-1} d\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. Put $\lambda_m = (m - x\log x)/x$. We use the Landau density f_L of Theorem 1.2.

Theorem 1.4 (compound Poisson and Landau approximations)**.** *For $N \geq 1$ and $0 < \varepsilon < 1$ the total variation distance between f_N and CP_N is at most an explicit $\delta_\Sigma = O(x^2 \log^2 N/N)$ (Theorem 5.8). Suppose that $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Then, uniformly over integers $m \geq 0$,*

$$\left|x f_N(m) - f_L(\lambda_m)\right| = O\Big(\frac{1+\log x}{x} + \frac{x^3 \log^2 N}{N}\Big).$$

The weak convergence $(M_N - x\log x)/x \Rightarrow f_L$ needs only $x \to \infty$ and $x^2 \log^2 N/N \to 0$. Theorem 5.9 gives explicit finite-N bounds.

Theorem 1.5 (mode locations)**.** *Suppose $x \to \infty$ and $x^3 \log^2 N/N \to 0$. For every mode $\kappa_N \geq N/2$ of P_N, and every mode m of f_N,*

$$N - \kappa_N = x\log x + \lambda_{\mathrm{L}} x + o(x), \qquad m = x\log x + \lambda_{\mathrm{L}} x + o(x).$$

Both errors are $O\big(\sqrt{x(1+\log x)} + x^{5/2}\log N/\sqrt{N}\big)$. Here $\lambda_{\mathrm{L}} \approx -0.222783$ is the unique Landau mode (Lemma 5.10, computer-assisted). Corollary 5.12 proves the assertion. For $\log N \leq x \leq 2\log N$, Corollary 5.11 also gives the bound $50\sqrt{x\log x}$ for sufficiently large N (an explicit threshold is in Remark 5.13).

Theorem 1.6 (the center and the crossing)**.** *Let $N \geq 2$ be even.*

(a) The ratio C_N/E_N is strictly increasing in $\varepsilon \in (0,1)$, and the crossing ε_N exists, is unique and lies in $(0, \frac{1}{2})$ (Proposition 4.2).

(b) Uniformly as $\varepsilon \log N \to 0$, $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon \log N))$. Theorem 4.5 gives explicit two-sided bounds.

(c) As $N \to \infty$ over even N, the crossing satisfies $x_N + \log x_N = 2\log N - \log 8 + O(\log^2 N/N)$, and so

$$x_N = 2\log N - \log\log N - \log 16 + O\Big(\frac{\log\log N}{\log N}\Big)$$

(Theorem 4.6).

At the crossing the relative error in (b) is $O(\log^2 N/N)$. For the shape between the humps put $D_N(k) = P_N(k+1) - 2P_N(k) + P_N(k-1)$ and $m_N = N - \kappa_N$ for a mode $\kappa_N \geq N/2$.

Theorem 1.7 (shape at the crossing)**.** *For every even $N \geq 8$, the crossing law has exactly two symmetric intervals of maximizing points. On either half, moving from the end to the center, the probabilities strictly increase up to the maximizing interval and strictly decrease after it. The center is the only interior minimum (Corollary 2.7). The proof uses the finite endpoint-dip certificates in Proposition 5.7. Lean checks these certificates as well as the deduction of the shape statement (Section 6).*

1.4 Proof overview

Signs at the ends. The recurrence alone is not enough at the ends. We append the earlier end values to each row, so that the induction keeps track of where a new sign could enter. In this enlarged row, an entry is either copied or is a positive sum of neighbors from the preceding row. We can therefore bound the signs of the entire row. For first differences this gives at most two modal plateaus. For second differences under a fair start it gives at most five sign blocks.

The half favored by the first step. Take $q \geq 1/2$. The right half has no valley and dominates its reflection. So a second peak exists exactly when the density decreases somewhere on the left. Choose two points there and subtract the left density value from the right one. This difference is affine in the bias. At $q = 1$ it is nonnegative by conditional unimodality. If it is negative for another bias, it remains negative as we move that bias toward one half. Each pair thus contributes an interval of biases, all extending toward one half, and their union is again an interval. Reflecting gives the other half of the bias interval. Analyticity rules out flat pieces. The two-mode bound also restricts the ratio that determines the boundary, which we use to prove boundary continuity.

Passing to the limit. Suppose the limit density had a valley forbidden by the finite bound. Take three short intervals around its two high points and its low point. Their averages still have a strict valley, and weak convergence gives the same inequalities for the finite law once the walk is long enough. Choose the smallest atom in the middle interval and the largest in each outer interval. They give the forbidden finite valley. For two valleys we use five intervals instead (Lemma A.7). For curvature we first smooth the second differences with a Gaussian. Descartes' rule controls their signs. Summation by parts then gives the limit for a fixed Gaussian width, after which we let that width tend to zero.

Locating the threshold. Near the transition, the question reduces to the sign of $g_a''(0)$. We evaluate its Fourier integral and bound the tail using exact moments and the smoothing equation. We use a Taylor model on each interval of a because direct interval evaluation loses cancellations in the integral. In the small interval containing the zero, we also differentiate the smoothing equation, obtaining $0.68 < \partial_a g_a''(0) < 0.70$ and $-0.09 < g_a^{(4)}(0) < -0.07$. The first inequality makes the zero unique. The second identifies the first nonconstant term at the critical maximum, namely the quartic term. At the threshold we have the sharper bounds in Table 1. All estimates used by the certificate are proved in Appendices B and C.

The rare-reversal limit. The finite bounds already give two humps with just one interior minimum at every even crossing with $N \geq 8$. For their locations, view each step

as a vertex joined to its parent. A reversal then affects a whole family. Ignoring repeated flips in a family leads to a compound Poisson approximation, with an explicit bound for the discarded interactions. Its Landau limit locates the peaks on the $x \log x$ scale. We obtain the same profile from the fixed-memory density as memory tends to one. Here the curvature partition matters, since weak convergence alone would not rule out a thin spike at a mode.

1.5 Organization

We prove the finite shape bounds in Section 2 and use them for fixed memory in Section 3. Sections 4 and 5 turn to rare reversals, first the crossing and then the approximation. Section 6 records the proof methods and their inputs, and Section 7 gives the open questions. The appendices contain the longer arguments, including all estimates needed by the certificates.

1.6 Prior work

The counting chain predates the elephant walk. Drezner and Farnum's generalized binomial model [2], in the formulation of James, James and Qi [4, p. 2339], has next-success probability $(1-\theta)p + \theta b/t$. Taking $p = 1/2$ and $\theta = a = 1 - 2\varepsilon$ gives our $r_t(b) = \varepsilon + (1-2\varepsilon)b/t$. For $0 < \varepsilon \leq 1/2$ this is the same transition rule. The fair-start chain also has their initialization. Other initial biases change only that initialization. Heyde [3] established the fixed-parameter limit regimes, including almost-sure convergence for $\theta > 1/2$. James, James and Qi [4, Theorem 2.2] treat time-varying dependence parameters. Drezner and Farnum's abstract identifies bimodal members of this family. We classify the limiting shapes and prove quantitative bounds when reversals are rare.

Fixed-memory results include [7, 8, 9]. Murahara and Tokumitsu [13] give optimal Gaussian local and central limit rates for fixed copying probabilities at most $3/4$. Vishwakarma [14] studies moment recurrences in the three regimes. We use the density and moments of [11, Theorems 1.3 and 1.4] and conditional unimodality from [10, Theorem 1.1]. The local limit in Trinas and Valle's preprint [12, Theorem 3.1] could replace our weak-transfer argument. Their Theorem 4.4 disproves the log-concavity conjecture of [10, Remark 2.3].

Engländer and Volkov's coin-turning walk switches at time n with probability p_n. For $p_n = c/n$ eventually, its occupation fraction tends to $\mathrm{Beta}(c, c)$ [21, Section 4, Theorem 1]. Fan and Qi [17] vary the elephant's memory within a walk. Our reversal probability varies with the final length, giving a triangular array, as in Roy, Takei and Tanemura's varying memory windows [18].

Kürsten's coupling [19] copies a parent with probability $1 - 2\varepsilon$ and otherwise assigns a fair sign. Baur and Bertoin [22] give the urn connection. Our regime is strongly supercritical in Baur's classification [20], following Bertoin [23]. Baur's Corollary 1 treats the largest nonroot clusters. Miyazaki and Takei derive root-cluster limits via the minimal elephant walk [24, Theorems 2.3, 2.4 and Section 2.3]. These results do not estimate our exact central mass.

Bokes, Hlubinová and Singh [27] derived the first-order mutation profile for this counting chain. We bound its error uniformly in the rare-reversal regime and use it to locate the crossing.

The untruncated law is the Lea–Coulson Luria–Delbrück law. Pakes proved its weak Landau limit [28, p. 993]. Möhle gave a uniform lattice local limit for integer parameters, without a rate [29, Corollaries 4.1–4.2]. Kessler and Levine give the small-parameter profile, Landau limit and dip threshold, with mode near $\mu N(\log \mu N - 0.22)$ [25, Section 2 and Eqs. (14), (18), (19)]. Bertoin proves the same weak limit for vertices cut off from a

recursive-tree root at cut probability of order $k^{-1/4}$ on a tree of size k [31, Proposition 1]. The Iksanov–Möhle coupling [32, Proposition 1] represents the Meir–Moon subtree cuts [33] by independent sizes of law $1/(j(j+1))$ up to first passage. The same weights appear in our expected family counts. Schweinsberg [30] and Bertoin [31] obtain the stable root-cluster fluctuations at retention $1 - c/\log n$. We add total variation and local error bounds for the truncated law, transfer them to the elephant walk and locate its modes.

Broadcasting papers study root-color reconstruction [5, 6], color-count limits and martingale error bounds [34, 35]. We estimate the tie probability as the flip probability tends to zero. Gut and Stadtmüller's count [36, (1.5)] has next-step probability pk/n, whereas ours has $\varepsilon + (1-2\varepsilon)k/n$, so their chain cannot leave zero. The irreversible mutation expansion of Kessler and Levine [26, Eqs. (10), (13), (14)] has the same first derivative at $\varepsilon = 0$ as our law, but not our uniform relative bound at $N\varepsilon \asymp \log N$.

Dekking and Kong [46, Theorem 4.1] classify the Markov binomial law as unimodal, bimodal or trimodal. It depends on the previous state, so it differs from the walk here. Trinas [47] studies last returns to zero at fixed superdiffusive memory. His leading constant involves $g_{a,q}(0)$, which is independent of q by reflection. The bias changes the shape away from zero even though it leaves that value unchanged.

2 Finite shape bounds

The second difference of P_N changes sign at most four times. At an even crossing with $N \geq 8$, this will give two intervals of modes and a unique interior minimum, without taking a limit. Appendix A treats first differences for arbitrary initial bias and proves the corresponding bounds for the limit density.

For the fair-start second differences, $0 < \varepsilon < \frac{1}{2}$ and $a = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + ab/t$. For $t \geq 1$ and every integer k we write $P_t(k) = \mathbb{P}(A_t = k)$, which is 0 unless $0 \leq k \leq t$, and

$$D_t(k) = P_t(k+1) - 2P_t(k) + P_t(k-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \geq 2$.

Lemma 2.1 (recurrence for the second difference)**.** *For every $t \geq 1$ and every integer k,*

$$P_{t+1}(k) = \big(1 - r_t(k)\big)P_t(k) + r_t(k-1)\,P_t(k-1), \tag{2.1}$$

$$D_{t+1}(k) = \big(1 - r_t(k+1)\big)D_t(k) + r_t(k-2)\,D_t(k-1). \tag{2.2}$$

Also $P_t(k) = P_t(t-k)$ and $D_t(k) = D_t(t-k)$. The same two recurrences hold with $f_t(m)$ in place of $P_t(k)$.

Proof. Given $A_t = k$, the copied step is $+1$ with probability k/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{k}{t} + \varepsilon(1-\frac{k}{t}) = r_t(k)$. This gives (2.1) for $0 \leq k \leq t+1$, where at $k = 0$ and at $k = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1-r_t(t+1))P_t(t+1)$, and this term is 0 because $P_t(-1) = P_t(t+1) = 0$. For other k every term vanishes. The symmetry of P_t is Lemma 4.1, and it gives the symmetry of D_t. The chain M of Lemma 4.1 has the same transition probabilities, so (2.1) holds for f_t as well.

For (2.2) put $\varsigma = a/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (2.1) at $k+1$, k and $k-1$ into the definition of $D_{t+1}(k)$ and collect the coefficients of $P_t(k+1)$, $P_t(k)$, $P_t(k-1)$ and $P_t(k-2)$. They are

$$1 - r_t(k+1), \qquad 3r_t(k) - 2, \qquad 1 - 3r_t(k-1), \qquad r_t(k-2).$$

The right side of (2.2) has the coefficients $1 - r_t(k+1)$, $-2 + 2r_t(k+1) + r_t(k-2)$, $1 - r_t(k+1) - 2r_t(k-2)$ and $r_t(k-2)$. The second ones agree because $2r_t(k+1) + r_t(k-2) = 3r_t(k)$, and the third ones agree because $r_t(k+1) + 2r_t(k-2) = 3r_t(k-1)$. □

Trinas and Valle [12, (2.2)] use (2.1), the forward equation of Lemma 4.1, for a superdiffusive local limit theorem. Their higher-difference recurrence [12, (4.10)] supports a certified fixed-memory study of log-concavity. We use (2.2) to count sign changes. Its coefficients are positive for $2 \le k \le t-1$, so new sign changes can enter only at the ends. The next lemma is the elementary fact behind this argument.

For a finite real list z, define its sign pattern $\operatorname{sp}(z)$ by deleting the zeros of z, replacing each remaining entry by its sign and merging equal neighbors. Thus $\operatorname{sp}(z)$ lists the alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 2.2 (positive sums of neighbors)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \le i \le n-1$ with $A_i \ge 0$ and $B_i \ge 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\operatorname{sp}(w) = \operatorname{sp}(\zeta)$ and $\operatorname{sp}(\eta) \preceq \operatorname{sp}(\zeta)$.*

We prove this in Appendix H.1.

Lemma 2.3 (sign changes next to an end)**.** *For every $t_1 \ge 2$ we have $\operatorname{sp}(s_2, s_3, \dots, s_{t_1}) \preceq \Pi_3$. Also, for every $0 \le \varepsilon \le 1$, the numbers s_3 and s_4 do not both vanish.*

We prove this in Appendix H.2.

Theorem 2.4 (at most five blocks)**.** *Let $N \ge 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\operatorname{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \dots, N-1\}$. The same holds for the second difference of f_N, and for f_N it holds for $N = 2$ as well.

We prove this in Appendix H.3.

A negative-to-positive change at $k_0 < k_1$ and its reflection give $-,+,+,-$ at $k_0 < k_1 < N-k_1 < N-k_0$. A negative value between the two positive ones would give the five-sign pattern excluded by Theorem 2.4. To exclude a zero, trace (2.2) backward. Both coefficients are strictly positive, so both parents must vanish or have opposite signs. Opposite signs give the forbidden pattern, so both parents must vanish. Repeating this back to $t = 5$ forces $s_3 = s_4 = 0$, contrary to Lemma 2.3. So the convex stretch has no gaps.

Theorem 2.5 (no holes)**.** *Let $N \ge 5$ and $0 < \varepsilon < \frac{1}{2}$. Let $1 \le k_0 < k_1 \le N/2$ be integers with $D_N(k_0) < 0 < D_N(k_1)$. Then $D_N(k) > 0$ for every $k_1 \le k \le N - k_1$.*

We prove this in Appendix H.4.

Lemma 2.6 (a criterion for a central minimum)**.** *Let $N \ge 6$ be even and $0 < \varepsilon < \frac{1}{2}$. If $P_N(1) > \max\{E_N, C_N\}$, then $P_N(k) > C_N$ for $1 \le k \le N-1$, $k \ne k_c$, and $D_N(k_c) > 0$.*

Proof. Suppose that $P_N(k) \le C_N$ for some $1 \le k < k_c$, and let j be the first minimizer of P_N on $\{1, \dots, k_c\}$. Then $2 \le j < k_c$ and $P_N(j-1) > P_N(j) \le P_N(j+1)$, so $D_N(j) > 0$. The first maximizer k_0 on $\{0, \dots, j\}$ has value at least $P_N(1)$, which exceeds both endpoint values of this interval. Thus $1 \le k_0 < j$ and $D_N(k_0) < 0$. Theorem 2.5 gives $D_N(k) > 0$ for $j \le k \le N - j$.

Put $\delta(k) = P_N(k+1) - P_N(k)$. These first differences are strictly increasing on $\{j-1, \dots, N-j\}$, and symmetry gives $\delta(k_c) = -\delta(k_c - 1)$. Hence $\delta(k_c - 1) < 0$, and every difference from j to $k_c - 1$ is negative. This makes $P_N(j) > P_N(k_c)$, a contradiction. Reflection proves the other half, and $D_N(k_c) = 2(P_N(k_c - 1) - P_N(k_c)) > 0$. □

Corollary 2.7 (the shape at every even crossing)**.** *For every even $N \ge 8$, at the crossing,*

$$P_N(k) > C_N = E_N \quad (1 \le k \le N-1,\ k \ne k_c), \qquad D_N(k_c) > 0.$$

There are integers $1 \le l \le r < k_c$ such that P_N strictly increases from 0 to l, is constant from l to r, and strictly decreases from r to k_c. Its two intervals of global maximizers are $\{l, \dots, r\}$ and $\{N-r, \dots, N-l\}$.

Proof. At the crossing $P_N(1) > E_N = C_N$ by Proposition 5.7, so Lemma 2.6 gives the minimum and central curvature. For the humps, let $p = \mathrm{sp}(D_N(1), \dots, D_N(k_c))$. It ends in $+$, and symmetry makes the full pattern p followed by its reverse, with the central positive blocks merged. If p has j blocks, the full pattern has $2j-1 \le 5$ by Theorem 2.4. Thus p is a subsequence of $(+,-,+)$.

The first differences $\delta(k) = P_N(k+1) - P_N(k)$ on $0 \le k < k_c$ therefore have at most three successive monotonic stretches, namely nondecreasing, nonincreasing, nondecreasing. Their first value is positive, and their last is $-D_N(k_c)/2 < 0$. The first stretch stays positive and the last stays negative. The middle stretch passes through zero at most once, possibly with consecutive zeros. This gives the claimed rise, plateau and fall. Symmetry gives the other hump. □

Corollary 2.8 (a convexity window from the peak height)**.** *Let $N \ge 8$ be even and $x = x_N$. Let $[l, r]$ be the left interval of maximizers in Corollary 2.7, and put*

$$M = P_N(r), \qquad w_N = r - 1 + M^{-1}, \qquad k_v = \min\{k_c, \lceil w_N \rceil - 1\}.$$

Then $D_N(k) > 0$ for every integer $k_v \le k \le N - k_v$. This window is nonempty.

Proof. Let j_0 be the first integer after l with $D_N(j_0) > 0$. Corollary 2.7 gives $D_N(l) < 0 < D_N(k_c)$, so $j_0 \le k_c$. The constant plateau gives $j_0 > r$. By its definition P_N is concave on $[r, j_0]$. Hence

$$\frac{M(j_0 - r + 1)}{2} \le \sum_{k=r}^{j_0 - 1} P_N(k) \le \frac{1 - P_N(k_c)}{2} < \frac{1}{2},$$

where the first inequality sums the chord from $P_N(r) = M$ to the nonnegative value $P_N(j_0)$. Thus $j_0 < w_N$, and consequently $j_0 \le k_v \le k_c$. Theorem 2.5, applied at $l < j_0$, gives positivity on $[j_0, N - j_0]$, which contains $[k_v, N - k_v]$. □

The endpoint-dip certificates in Proposition 5.7 supply the finite input to both corollaries. Knowing the peak height already gives a convexity window, even when we have no useful bound on the mode's location (the window may consist only of the center). Heuristically, Theorem 5.9(b) and Corollary 5.11 give $M = P_N(r) = \frac{f_L(\lambda_L)}{2x}(1 + o(1))$, hence, with $m_N = r$, $w_N = m_N + \big(2/f_L(\lambda_L) + o(1)\big)x$. Since $f_L(\lambda_L) \approx 0.1807$, this is about $m_N + 11x$, far below k_c for large N. At $N = 4000$, the computed value of $w_N - m_N$ is about $9.9x$. These asymptotics explain the expected width. The preceding chord argument proves the window for every even length.

3 The limit density at fixed memory

Fix $\varepsilon \in (0, \frac{1}{4})$, so $a = 1 - 2\varepsilon \in (\frac{1}{2}, 1)$. Conditional on $X_1 = 1$, S_k/k^a converges almost surely to $\mathcal{L}_a$ [8, Theorem 3.7]. The fair-start limit $\mathcal{L}_a^\circ$ has the law of $\xi\mathcal{L}_a$, where ξ is a fair sign independent of $\mathcal{L}_a$ [11, (1.3) and (1.6)]. Guérin, Laulin and Raschel prove that both limits have positive, bounded and smooth densities [11, Theorem 1.3], denoted by h_a and g_a as before. Guérin, Laulin, Raschel and Simon prove that $\mathcal{L}_a$ is unimodal [10, Theorem 1.1].

3.1 Memory near one

Corollary 3.1 (the fixed-memory limit near one)**.** *For the elephant walk with $X_1 = 1$ and fixed memory $p = (1+a)/2$, $a \in (\frac{1}{2}, 1)$, let $\mathcal{L}_a = \lim_{k\to\infty} S_k/k^a$ as in [8, Theorem 3.7]. If Z has characteristic function $\mathbb{E}e^{itZ} = \exp\{-it\log|t| - \pi|t|/2\}$, continuously extended at $t = 0$, then*

$$\frac{1 - \mathcal{L}_{1-2\varepsilon}}{2\varepsilon} + \log(1/\varepsilon) \implies Z \qquad (\varepsilon \downarrow 0).$$

Proof. Fix $0 < \varepsilon < 1/8$, put $a = 1 - 2\varepsilon$, and use one infinite walk with $X_1 = 1$. By [8, (2.4), Appendix A], $\mathcal{M}_k = \Gamma(k)S_k/\Gamma(k+a)$ is a martingale with limit $\mathcal{L}_a$. It is Bercu's martingale divided by $\Gamma(1+a)$. Its increments are

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \frac{\Gamma(k+1)}{\Gamma(k+1+a)}\left(X_{k+1} - \frac{aS_k}{k}\right).$$

Log-convexity of gamma gives $1 \le k^a\Gamma(k)/\Gamma(k+a) \le (1+a/k)^{1-a} \le 1 + 1/k$. Since $S_k = k - 2M_k$, Lemma E.2 gives

$$\mathbb{E}\big[1 - a^2(S_k/k)^2\big] \le 4\varepsilon + \frac{4\mathbb{E}M_k}{k} \le 4\varepsilon(1 + \log k).$$

Orthogonality of the increments therefore gives square-mean convergence and

$$\mathbb{E}|\mathcal{L}_a - \mathcal{M}_n|^2 \le C\varepsilon \sum_{k\ge n} k^{-2a}(1+\log k) \le C\varepsilon n^{1-2a}(1+\log n),$$

with C independent of $a \in [3/4, 1)$, since $2a - 1 \ge 1/2$.

Take $n = \lfloor \varepsilon^{-4/3} \rfloor$ and $x = n\varepsilon$. Then $x \to \infty$ and $x^3\log^2 n/n = n^2\varepsilon^3\log^2 n \to 0$, so Theorem 5.9(c) gives $Z_\varepsilon = (M_n - x\log x)/x \Rightarrow Z$. The tail bound gives $\mathbb{E}|(\mathcal{L}_a - \mathcal{M}_n)/\varepsilon|^2 = O(\varepsilon^{1/3}(1+\log(1/\varepsilon))) = o(1)$, using $n^{4\varepsilon} \to 1$. Also $|\mathcal{M}_n - n^{-a}S_n| \le n^{-a} = o(\varepsilon)$. Finally,

$$n^{-a}S_n = e^{2\varepsilon\log n}[1 - 2\varepsilon(\log x + Z_\varepsilon)],$$
$$\frac{1 - n^{-a}S_n}{2\varepsilon} + \log(1/\varepsilon) = Z_\varepsilon + o_{\mathbb{P}}(1).$$

The second equality uses $\log x = \log n + \log\varepsilon$, tightness of Z_ε, and $\varepsilon\log^2 n \to 0$. Replacing $n^{-a}S_n$ by $\mathcal{L}_a$ proves the claim. □

Theorem 3.13 proves the stronger uniform density approximation and total variation convergence. Trinas and Valle find $\operatorname{Var}\mathcal{L}_a = 2(1-a) + o(1-a)$ [12, Eq. (4.5)]. In both results, $\mathcal{L}_a$ concentrates at 1 as $a \to 1$. For Theorem 3.3, however, the exact second and fourth moments are enough.

3.2 The number of modes

A probability density h on $\mathbb{R}$ is *unimodal* if for some m it is nondecreasing on $(-\infty, m]$ and nonincreasing on $[m, \infty)$. Guérin, Laulin, Raschel and Simon ask for a unique local maximum [10], which also rules out a flat top. The density g_a is a mixture of the unimodal density of $\mathcal{L}_a$ and its reflection, but mixing unimodal densities can give more than one peak [41]. We first give the shape for a fair start, then let the starting bias vary. There are no flat intervals by Lemma A.4, so we can speak of individual modes rather than plateaus.

Theorem 3.2 (at most two modes)**.** *Let $a \in (\frac{1}{2}, 1)$. Exactly one of the following holds.*

(a) The density g_a is nondecreasing on $(-\infty, 0]$ and nonincreasing on $[0, \infty)$.

(b) There is $\kappa > 0$ such that g_a is nondecreasing on $[0,\kappa]$ and nonincreasing on $[\kappa,\infty)$, and $g_a(0) < g_a(\kappa)$.

In case (b), g_a is nondecreasing on $(-\infty,-\kappa]$ and nonincreasing on $[-\kappa,0]$, and it is not unimodal. All these monotonicities are strict. The unique local maximum is 0 in case (a), and the only local maxima are $\pm\kappa$ in case (b). In case (a), there is $d > 0$ such that g_a is strictly concave on $(-d,d)$ and strictly convex on each tail. In case (b), there are $0 < d_1 < \kappa < d_2$ such that g_a is strictly convex on $(-d_1,d_1)$ and on each tail beyond $\pm d_2$, and strictly concave on $(-d_2,-d_1)$ and (d_1,d_2). Thus it has exactly two inflection points in case (a) and four in case (b).

We cannot pass directly from the second differences of P_N to the curvature of g_a. First smooth with a Gaussian and apply Descartes' rule to the resulting exponential sum. Now let $N \to \infty$, keeping the width fixed, and only then send the width to 0. Appendix A proves the lemmas used below. A real function v has *values at most* Π if $\operatorname{sp}(v(y_1),\dots,v(y_j)) \preceq \Pi$ for all $y_1 < \cdots < y_j$, and *increments at most* Π if $\operatorname{sp}(v(y_2)-v(y_1),\dots,v(y_j)-v(y_{j-1})) \preceq \Pi$ for all such points. We put $\Pi_4 = (+,-,+,-)$.

Proof. Fix a, so that $\varepsilon = (1-a)/2 \in (0,\frac14)$.

Step 1, smoothing. Fix $\sigma > 0$ and let $\varphi(y) = e^{-y^2/(2\sigma^2)}/(\sigma\sqrt{2\pi})$. For $N \ge 3$ put $\eta = 2N^{-a}$ and $z_k = (2k-N)N^{-a}$, so that $S_N/N^a = z_{A_N}$, and let

$$V_N(y) = \eta^{-2} \sum_{k=1}^{N-1} D_N(k)\, \varphi(y - z_k).$$

Since $\varphi(y-z_k) = \varphi(y) e^{-z_k^2/(2\sigma^2)} e^{y z_k/\sigma^2}$ and the z_k increase with k, Lemma A.1 and Theorem 2.4 show that V_N has values at most Π_5. (If every $D_N(k)$ vanishes, then $V_N = 0$ and every pattern is empty.)

Step 2, the limit $N \to \infty$. Let $u(y) = \mathbb{E}\varphi(y - \mathcal{L}_a^\circ)$. The derivatives of φ are bounded, so u is smooth and $u^{(j)}(y) = \mathbb{E}\varphi^{(j)}(y - \mathcal{L}_a^\circ)$. We show that $V_N(y) \to u''(y)$ for every y. Put $P_N(k) = 0$ for $k \notin \{0,\dots,N\}$. Then $D_N(k) = 0$ unless $-1 \le k \le N+1$, and summation by parts gives

$$\sum_{k=-1}^{N+1} D_N(k)\, \varphi(y - z_k) = \sum_{k=0}^{N} P_N(k)\, \Delta\varphi(y - z_k), \qquad \Delta\varphi(t) = \varphi(t+\eta) - 2\varphi(t) + \varphi(t-\eta).$$

Taylor's formula gives $|\eta^{-2}\Delta\varphi(t) - \varphi''(t)| \le \eta^2 \|\varphi''''\|_\infty/12$. By symmetry the four terms with $k \in \{-1,0,N,N+1\}$ on the left are at most $\varphi(0)\,(6P_N(0) + 2P_N(1))$ in absolute value. Here $P_N(0) \le (1-\varepsilon)^{N-1}$, and the proof of Theorem 5.5 gives $f_N(1) \le \varepsilon N(1-\varepsilon)^{N-2}$ and $f_N(N-1) \le \varepsilon(1-\varepsilon)^{N-2}$, so that $P_N(1) \le \varepsilon N(1-\varepsilon)^{N-2}$. As $\eta^{-2} = N^{2a}/4$, these four terms times η^{-2} tend to 0. Hence

$$V_N(y) = \mathbb{E}\varphi''\big(y - S_N/N^a\big) + o(1) \longrightarrow \mathbb{E}\varphi''(y - \mathcal{L}_a^\circ) = u''(y),$$

because $S_N/N^a \to \mathcal{L}_a^\circ$ in law and φ'' is bounded and continuous. By Lemma A.2, u'' has values at most Π_5.

Step 3, integrating the sign bound. By dominated convergence, $u'(y) \to 0$ as $y \to \pm\infty$. So u has increments at most Π_4, by Lemma A.3.

Step 4, the limit $\sigma \to 0$. Write $u = u_\sigma$. As g_a is bounded and continuous, $u_\sigma(y) = \int g_a(y-t)\varphi(t)\,dt$ tends to $g_a(y)$ for every y. So the increments of u_σ tend to those of g_a, and Lemma A.2 shows that g_a has increments at most Π_4. Lemma A.4 also makes g_a'' bounded and continuous, so $u_\sigma'' = g_a'' * \varphi \to g_a''$ pointwise. Hence g_a'' has values at most Π_5.

Step 5, symmetry. Let $0 \le y_1 < \cdots < y_j$, let $d_i = g_a(y_{i+1}) - g_a(y_i)$, and let p be the pattern of the d_i. As g_a is even, the points $-y_j < \cdots < -y_1 < y_1 < \cdots < y_j$, with $-y_1$ left out if $y_1 = 0$, have the increments $-d_{j-1}, \ldots, -d_1$, then 0 if $y_1 > 0$, then $d_1, \ldots, d_{j-1}$. Their pattern is the reverse of $-p$ followed by p. It has $2|p|$ entries and is $\preceq \Pi_4$. So p is empty, $(+)$, $(-)$ or $(+,-)$. In particular there are no $0 \le y < y' < y''$ with $g_a(y) > g_a(y') < g_a(y'')$.

If no increment on $[0,\infty)$ is positive, (a) holds by evenness. Otherwise $g_a(y_1) < g_a(y_2)$ for some $0 \le y_1 < y_2$. Some increment on $[y_2,\infty)$ is negative, since otherwise $g_a \ge g_a(y_2) > 0$ there, against $\int g_a = 1$. If $g_a(y_4) < g_a(y_3)$ with $y_2 \le y_3 < y_4$, then g_a cannot rise after y_4, so it attains its maximum over $[0,\infty)$ at some $\kappa \in [0, y_4]$. If $0 \le y < y' \le \kappa$ and $g_a(y) > g_a(y')$, then $g_a(y') < g_a(\kappa)$, and (y, y', κ) is a fall followed by a rise. If $\kappa \le y < y'$ and $g_a(y) < g_a(y')$, then (κ, y, y') is one. So g_a is nondecreasing on $[0,\kappa]$ and nonincreasing on $[\kappa,\infty)$. If $g_a(0) = g_a(\kappa)$, then g_a would be nonincreasing on $[0,\infty)$, against $g_a(y_1) < g_a(y_2)$. So (b) holds.

As $g_a(0) < g_a(\kappa)$ in (b), the two cases exclude each other. In (b), symmetry gives the monotonicity on $(-\infty, 0]$. If g_a were unimodal with mode m, then $m \ge 0$ would give $g_a(-\kappa) \le g_a(0)$ and $m < 0$ would give $g_a(0) \ge g_a(\kappa)$, while $g_a(\pm\kappa) > g_a(0)$. Finally a local maximum y other than the points named would make g_a constant on a one-sided neighborhood of that point. Lemma A.4 excludes this. Each monotonicity is strict as well. Otherwise two distinct points would have the same value, forcing a constant interval between them, which the same lemma rules out.

Step 6, curvature. The Fourier bound in Lemma A.4 gives $g_a(y), g_a'(y) \to 0$ as $y \to \infty$. Evenness and the Π_5 bound give at most three sign blocks for g_a'' on $(0,\infty)$. Its last sign must be positive. Otherwise g_a' would eventually be nonincreasing with limit zero, making g_a eventually nondecreasing. This contradicts $g_a > 0$ and $g_a \to 0$. Analyticity and strict monotonicity imply that the first nonconstant term in the even Taylor series of g_a at zero is negative in case (a) and positive in case (b). The same signs therefore hold for g_a'' near zero. In case (b), g_a'' must also be negative somewhere, since g_a' is positive near zero and negative somewhere beyond κ. The positive-half-line sign patterns are consequently $(-,+)$ in case (a) and $(+,-,+)$ in case (b). Analyticity excludes an interval of zero curvature and makes the resulting convexities strict. In case (b), $g_a'(d_1) > 0 > g_a'(d_2)$, so $d_1 < \kappa < d_2$. □

Theorem 3.3 (two modes for memory near one)**.** *If $\frac{4}{5} \le a < 1$, that is $0 < \varepsilon \le \frac{1}{10}$, then case (b) of Theorem 3.2 holds. So g_a has exactly two modes.*

Proof. In case (a), Khinchine's representation [41] gives $\mathcal{L}_a^\circ \overset{d}{=} UZ$, with U uniform on $(0,1)$ and independent of Z. Hence Cauchy–Schwarz gives $\mathbb{E}(\mathcal{L}_a^\circ)^4 \ge \frac{9}{5}[\mathbb{E}(\mathcal{L}_a^\circ)^2]^2$. The moment recursion of [11, Theorem 1.4] instead gives

$$\frac{\mathbb{E}\mathcal{L}_a^4}{(\mathbb{E}\mathcal{L}_a^2)^2} = \frac{6(2a^2+2a-1)}{4a-1}\,\mathrm{Beta}(2a,2a),$$

where Beta is the beta function. For $a \ge \frac{3}{4}$, its integral and $t(1-t) \le \frac{1}{4}$ give $\mathrm{Beta}(2a,2a) \le 4^{-2a+3/2}\,\mathrm{Beta}(\frac{3}{2},\frac{3}{2}) = \pi 2^{-4a}$. Put $R(a) = (2a^2+2a-1)/(4a-1)$. On $[\frac{3}{4},1]$ we have $(\log R)' \le 2 < 4\log 2$. The first inequality reduces to $4a(4a^2+a-2) \ge 0$. Thus $6\pi R(a) 2^{-4a}$ decreases there, and for $a \ge \frac{4}{5}$ the displayed moment ratio is at most

$$\frac{282\pi}{55\,2^{16/5}} < \frac{141}{80} < \frac{9}{5},$$

using $\pi < 22/7$ and $2^{1/5} > 8/7$. Even moments are unchanged by symmetrization, so this contradicts case (a). Theorem 3.2 then gives case (b). □

Corollary 3.4 (a larger computation-free range)**.** *The conclusion of Theorem 3.3 holds for* $79/100 \le a < 1$.

Proof. The preceding upper bound $6\pi R(a)2^{-4a}$ decreases for $a \ge 3/4$. At $a = 79/100$, use $2^{4/25} > 111/100$, which follows from the exact integer inequality $111^{25} < 16 \cdot 100^{25}$. Substitute this bound and $\pi < 22/7$ to get

$$6\pi R(79/100)2^{-79/25} < \frac{33517}{18648} < \frac{9}{5}.$$

So the kurtosis is again too small for a unimodal density, proving the claim. □

To find where the fair density changes shape, we examine its curvature at the origin. The shape theorem and certified signs first confine the transition to a short interval. Within it, the certificate shows that the central curvature strictly increases with memory, so it can vanish only once.

Theorem 3.5 (the threshold, computer-assisted)**.** *Put*

$$z_- = 1838418790980431/2748779069440000 = 0.6688128600145977 54\ldots,$$
$$z_+ = 45960469774511/68719476736000 = 0.6688128600146010 28\ldots,$$

so that $z_+ - z_- < 3.3 \cdot 10^{-15}$.

(a) There is exactly one $a_* \in (\frac{1}{2}, 0.6712]$ *with* $g''_{a_*}(0) = 0$*, and* $z_- < a_* < z_+$.

(b) For $\frac{1}{2} < a \le a_*$*, case (a) of Theorem 3.2 holds. For* $a_* < a < 1$*, case (b) holds. Thus* a_* *is the unique transition from one mode to two, and the critical density is unimodal.*

(c) We have $g''_a(0) < 0$ *for* $\frac{1}{2} < a < a_*$ *and* $g''_a(0) > 0$ *for* $a_* < a \le 0.6712$.

Thus the transition is unique even inside the bracket. For $[\frac{4}{5}, 1)$ we also have the moment proof of Theorem 3.3, which needs no computation. For a biased start we lose symmetry, and instead use first differences.

Remark 3.6 (the certificate and finite-length thresholds)**.** Appendix B bounds the Fourier integral for $g''_a(0)$ by Taylor models in a and a tail estimate from the fixed-point equation. It uses scaled moments near $a = \frac{1}{2}$ and a Hankel test for $a \ge 0.6705$. The certificate takes about 2.8 hours of CPU time. On the exact finite law, $D_N(N/2)$ changes sign at $a \approx 0.6563$ for $N = 4000$ and at $a \approx 0.6592$ for $N = 8000$. These two numbers are computed and are not used in any proof.

3.3 Arbitrary initial bias

Write h_a for the density of $\mathcal{L}_a$ and $g_{a,q}(x) = qh_a(x) + (1-q)h_a(-x)$ for initial bias $\mathbb{P}(X_1 = 1) = q$. Thus $g_{a,1/2} = g_a$.

Theorem 3.7 (the interval of bimodal biases)**.** *For every* $a \in (1/2, 1)$ *and* $q \in [0, 1]$*, the density* $g_{a,q}$ *has at most two local maxima. There is* $b_a \in [0, 1/2]$ *such that it has exactly two local maxima if and only if*

$$b_a < q < 1 - b_a.$$

Outside this open interval, including its endpoints, the density is unimodal with a unique local maximum. The interval is empty exactly when $a \le a_*$*, in which case* $b_a = 1/2$*. At finite time, for every* $0 < \varepsilon < 1$*,* $q \in [0, 1]$ *and* $N \ge 1$*, the count law has at most two modal plateaus, including its endpoints.*

Proof. The mode bound. Let P_N here denote the count law with first-step bias q. The finite assertion follows from Proposition A.5. In particular, the count law cannot have two successive strict valleys. Neither can $g_{a,q}$. If it did, weak convergence of S_N/N^a and averages over five short intervals would give five finite-law atoms with those same forbidden inequalities (Lemma A.7). There are also no constant intervals, for any bias, by Lemma A.4. Hence three local maxima $x < y < z$ would give $x < u < y < v < z$ with

$$g_{a,q}(u) < \min\{g_{a,q}(x), g_{a,q}(y)\}, \qquad g_{a,q}(v) < \min\{g_{a,q}(y), g_{a,q}(z)\}.$$

For each pair, choose a point in a one-sided maximizing neighborhood of the smaller maximum. There is a point with strictly smaller value, since the density is not constant on an interval. The resulting two valleys contradict the limit bound.

The bias interval. Fix a, write $f_q = g_{a,q}$ and $h = h_a$, and call $x < y < z$ a strict valley if $f_q(y) < \min\{f_q(x), f_q(z)\}$. Proposition A.6 and the same weak-transfer lemma show that, for $q \geq 1/2$, f_q has no strict valley on $[0, \infty)$ and $f_q(-x) \leq f_q(x)$ for $x \geq 0$. Consequently it has a strict valley if and only if $f_q(x) > f_q(y)$ for some $x < y \leq 0$. Such a descent gives the valley $(x, y, -x)$ by reflection dominance. Conversely, a valley must start at $x < 0$. If its middle point y is positive, the no-valley property at $(0, y, z)$ gives $f_q(0) \leq f_q(y) < f_q(x)$.

At $q = 1$, conditional unimodality [10] excludes a valley, so h is nondecreasing on $(-\infty, 0]$. For fixed $x < y \leq 0$, the difference $f_q(x) - f_q(y)$ is affine in q and nonpositive at $q = 1$. If positive at $q \geq 1/2$, it remains positive for every $r \in [1/2, q]$. The valley biases in $[1/2, 1]$ are therefore downward closed. They are relatively open, since the strict inequalities at three fixed points persist under a small change of bias, and they exclude 1. Reflection gives an open interval $(b_a, 1 - b_a)$, possibly empty.

A strict valley gives two local maxima. Integrability and positivity supply an outer point on each side below the corresponding witness height, and we maximize on the two resulting compact intervals. Conversely, two local maxima give a strict valley because there is no flat interval. The mode bound excludes a third maximum. A continuous integrable density without a strict valley is unimodal, and absence of flat intervals makes its maximum unique. Finally the bias interval is nonempty exactly when it contains $1/2$, so Theorem 3.5 identifies the empty case. □

This proves the initial-bias conjecture of [10, Remark 2.1(b)], with its half-width $q(a) = 1/2 - b_a$. Trinas and Valle's stronger local limit [12, Theorem 3.1], in an unrefereed arXiv preprint, would also transfer the finite inequalities. Weak convergence suffices for the proof here.

3.4 The bias boundary

To find the endpoint of the bias interval, we compare the slopes of the conditional density at reflected points. This reduces the problem to a one-dimensional infimum.

Proposition 3.8 (a variational equation for the bias boundary)**.** *For $a \in (1/2, 1)$ put $U_a = \{t > 0 : h_a'(t) > 0\}$. With the infimum over an empty set interpreted as $+\infty$,*

$$b_a = \min\left\{\frac{1}{2}, \quad \inf_{t \in U_a} \frac{h_a'(-t)}{h_a'(t) + h_a'(-t)}\right\}. \tag{3.1}$$

In particular $b_a \to 0$ as $a \uparrow 1$. If the infimum is attained at $t > 0$ with $h_a'(t) > 0$ and $b_a < 1/2$, then, with $q = 1 - b_a$,

$$g_{a,q}'(-t) = g_{a,q}''(-t) = 0.$$

At this point the smaller peak disappears.

Proof. The proof of Theorem 3.7 shows that $h'_a(-t) \geq 0$ for $t \geq 0$. For $q \geq 1/2$ bimodality is equivalent to a decrease somewhere on the negative half-line, hence to $g'_{a,q}(-t) < 0$ for some $t > 0$. Here

$$g'_{a,q}(-t) = qh'_a(-t) - (1-q)h'_a(t).$$

This expression can be negative only if $t \in U_a$, in which case it is negative exactly when $1 - q > h'_a(-t)/(h'_a(t) + h'_a(-t))$. Taking the union over t gives (3.1), with equality excluded as in the interval theorem. At an interior minimizer the ratio has derivative zero. Differentiating its numerator and denominator gives the two displayed equations. Finally Corollary 3.14 puts every fixed $q \in (0,1)$ inside the interval for all sufficiently large a. Thus $0 \leq b_a < q$ eventually for every $q > 0$, which proves the limit. □

Theorem 3.9 (continuity of the bias boundary)**.** *The function $a \mapsto b_a$ is continuous on $(1/2, 1)$ and extends continuously to $b_1 = 0$. If $a > a_*$ and $b_a > 0$, the infimum in (3.1) has a unique minimizer $t_a \in (0, \kappa_a)$. This minimizer depends continuously on a wherever $a > a_*$ and $b_a > 0$.*

Proof. For $a > a_*$, the curvature partition in Theorem 3.2 gives $g'_a(t) > 0$ on $(0, \kappa_a)$ and $g'_a(t) < 0$ on (κ_a, ∞). Put

$$D_a(t) = h'_a(t) + h'_a(-t), \qquad R_a(t) = \frac{g'_a(t)}{D_a(t)} \quad (0 < t < \kappa_a).$$

We have $D_a > 0$, $0 < R_a \leq 1/2$, and $1/2 - b_a = \sup_{0<t<\kappa_a} R_a(t)$. Also $R_a(0+) = 0$. To see this, h'_a is nonnegative to the left of zero and positive just to its right, so its first nonzero Taylor term has even degree. This term cancels in $2g'_a(t) = h'_a(t) - h'_a(-t)$ and adds in $D_a(t)$.

The ratio cannot have a strict valley. More precisely,

$$R_a(v) \geq \min\{R_a(u), R_a(w)\} \qquad (0 < u < v < w < \kappa_a). \tag{3.2}$$

Otherwise, choose $s \in (0, 1/2)$ strictly between the middle value and both outer values. With $q = 1/2 + s$, $g'_{a,q}(-t) = D_a(t)(s - R_a(t))$ then has signs $-, +, -$ at $-w, -v, -u$, and is positive sufficiently close to zero on the left. Positivity of the density and its zero limits at infinity supply a positive derivative farther left and a negative derivative farther right. There would be three local maxima, contrary to Theorem 3.7.

At κ_a the limit of R_a is either zero or $1/2$. The case that needs checking is $D_a(\kappa_a) = 0$. The analytic function $h'_a(-t) \geq 0$ has a zero of even order there, whereas $2g'_a(t)$ has a zero of odd order. In $D_a(t) = 2h'_a(-t) + 2g'_a(t)$ the smaller order dominates, giving respectively the limits zero and $1/2$ for R_a. If $b_a > 0$, the latter is impossible. Thus R_a vanishes at both ends and attains its positive maximum in the interior. Equation (3.2) and analyticity make this maximum unique, since two maximizers would force a constant interval. They also make R_a strictly increasing before its maximizer and strictly decreasing after it.

Lemma A.4 gives joint continuity of all the derivatives used here. The signs of g'_a on either side of its unique positive zero give continuity of κ_a. Fix $a_0 > a_*$ with $b_{a_0} > 0$, and choose $0 < l < t_{a_0} < r < \kappa_{a_0}$. The strict inequalities $R_a(l), R_a(r) < R_a(t_{a_0})$ persist near a_0. Equation (3.2) bounds every value outside $[l, r]$ below $R_a(t_{a_0})$, so a maximum on $[l, r]$ is global. Uniform convergence on this fixed compact interval proves continuity of the maximum and, by uniqueness, of its location. If instead $b_{a_0} = 0$, any fixed point where $R_{a_0} > 1/2 - \delta$ persists nearby, giving $0 \leq b_a < \delta$. This proves continuity above a_*. Below it $b_a = 1/2$, Proposition 3.12 gives continuity at a_*, and Proposition 3.8 gives the limit at one. □

We can use this argument to enclose the boundary numerically, since three comparisons restrict the search to a bounded interval. Verified inequalities $R_a(l), R_a(r) < R_a(t)$ with $0 < l < t < r < \kappa_a$ trap the maximum in $[l, r]$. If $L \le R_a(t)$ and $R_a \le U$ throughout that interval, then $1/2 - U \le b_a \le 1/2 - L$. If these inequalities hold uniformly, the enclosure covers a whole parameter interval. This does not yet prove that the maximum increases with a.

Corollary 3.10 (a rate for the bias boundary near one)**.** *Put*

$$\rho_a = \frac{\Gamma(1+a)^2}{(2a-1)\Gamma(2a)} - 1.$$

If $\rho_a < 1/12$*, then*

$$0 \le b_a \le \frac{4\rho_a}{1-4\rho_a}.$$

In particular, as $a \uparrow 1$*,* $b_a \le 8(1-a) + O((1-a)^2)$*.*

Proof. The conditional limit has mean $\mu_a = 1/\Gamma(1+a) > 0$ and variance $\rho_a \mu_a^2$, by the moment formulas of [11, Theorem 1.4]. Consider the three adjacent intervals of length μ_a centered at $-\mu_a$, 0 and μ_a. Chebyshev's inequality puts at least $1 - 4\rho_a$ of the conditional law in the right interval and at most $4\rho_a$ in the middle. For a start to the left, reflect these intervals. After mixing, the left interval has mass at least $(1-q)(1-4\rho_a)$ and the right at least $q(1-4\rho_a)$. The middle still has mass at most $4\rho_a$.

For $4\rho_a/(1-4\rho_a) < q < 1 - 4\rho_a/(1-4\rho_a)$, the middle has less mass than either outer interval. All three intervals have the same length. So, by continuity, we can choose a point in each with the middle value strictly smaller than the other two. This rules out unimodality. The two-mode bound in Theorem 3.7 gives exactly two modes, and therefore the stated bound on b_a. Finally, the logarithmic derivative at $a = 1$ of $1 + \rho_a$ is $2\psi(2) - 2 - 2\psi(2) = -2$. Thus $\rho_a = 2(1-a) + O((1-a)^2)$, which gives the last assertion. □

3.5 The density at the threshold

We next ask how far apart the modes are just after the transition. The needed constants are derivatives at the threshold. Write

$$C_* = \partial_a g_a''(0)|_{a=a_*}, \qquad B_* = -g_{a_*}^{(4)}(0), \qquad H_* = h_{a_*}'(0).$$

Table 1 gives strict certified enclosures. The coefficient Q_* is defined in Proposition 3.12.

Constant	Lower bound	Upper bound
C_*	0.6889503945	0.6889503947
B_*	0.079207784468	0.079207784470
H_*	0.09707494510086	0.09707494510088
Q_*	9.86698768	9.86698770
$\sqrt{6C_*/B_*}$	7.224131934	7.224131936

Table 1: The constants in the critical expansions. The enclosure is uniform on a parameter interval centered at the lower endpoint of the certified bracket for a_* (Appendix C.4).

Proposition 3.11 (the critical shape)**.** *The density at the threshold satisfies*

$$g_{a_*}(u) = g_{a_*}(0) - \frac{B_*}{24}u^4 + O(u^6) \qquad (u \to 0).$$

For $a > a_*$ let $\kappa_a > 0$ be the positive mode of g_a. As $a \downarrow a_*$,

$$\kappa_a = \sqrt{\frac{6C_*}{B_*}}\sqrt{a-a_*} + o\big(\sqrt{a-a_*}\big),$$
$$g_a(\kappa_a) - g_a(0) = \frac{3C_*^2}{2B_*}(a-a_*)^2 + o\big((a-a_*)^2\big).$$

Proof. On a neighborhood of a_*, the certificate bounds $|\phi_a(t)|$ by Ct^{-8} and $|\partial_a\phi_a(t)|$ by $4t^{-7}$ for large t (Appendix C.4). On a bounded frequency interval we instead use the moment series, which converges locally uniformly, as does its parameter derivative. Together these bounds justify dominated Fourier inversion. Thus the spatial derivatives through order six are continuous in a, and we can differentiate the central curvature once in the parameter. Restricting to a smaller parameter interval makes the Taylor remainders through order four uniform. At criticality, evenness, $g''_{a_*}(0) = 0$ and $g^{(4)}_{a_*}(0) = -B_* < 0$ give the first assertion.

Put $\delta = a - a_* > 0$. Uniform Taylor expansion gives

$$\frac{g'_a(u)}{u} = C_*\delta + o(\delta) - \frac{B_* + o(1)}{6}u^2 + O(u^4).$$

For $u = k\sqrt{\delta}$ this has the sign of $C_* - B_*k^2/6$ when k is fixed away from $\sqrt{6C_*/B_*}$. The unique positive mode lies between any two such choices with opposite signs, by Theorem 3.2. Squeezing gives the mode expansion. Substitute it into

$$g_a(u) - g_a(0) = \tfrac{1}{2}\big(C_*\delta + o(\delta)\big)u^2 - \tfrac{1}{24}\big(B_* + o(1)\big)u^4 + O(u^6)$$

to obtain the height expansion. □

Proposition 3.12 (how the bias interval opens)**.** *With the constants of Table 1, put*

$$Q_* = \frac{\sqrt{2}\,C_*^{3/2}}{3H_*\sqrt{B_*}}, \qquad Q_* \approx 9.86698769.$$

As $a \downarrow a_*$,

$$\frac{1}{2} - b_a = Q_*(a-a_*)^{3/2} + o\big((a-a_*)^{3/2}\big). \tag{3.3}$$

On some interval immediately above a_*, b_a is strictly decreasing. For $a > a_*$ sufficiently close to a_*, the infimum in (3.1) is attained. Every minimizer t_a satisfies

$$t_a = \sqrt{\frac{2C_*}{B_*}}\sqrt{a-a_*} + o\big(\sqrt{a-a_*}\big) = \frac{\kappa_a}{\sqrt{3}} + o\big(\sqrt{a-a_*}\big).$$

At $q = 1 - b_a$, the stationary point is $-t_a$.

Proof. Proposition 3.11 supplies the required regularity. The refined certificate in Appendix C.4 encloses $\pi^{-1}\int_0^\infty t\,\mathrm{Im}\,\phi_a(t)\,dt = h'_a(0)$ uniformly on an interval containing a_*, giving the positive H_* in Table 1.

Write $\delta = a - a_*$. By (3.1),

$$\frac{1}{2} - b_a = \max\left\{0, \sup_{t\in U_a}\frac{g'_a(t)}{h'_a(t) + h'_a(-t)}\right\}.$$

The fair-shape theorem makes the numerator nonpositive for $t \geq \kappa_a$. Proposition 3.11 gives $\kappa_a = O(\sqrt{\delta})$, and $H_* > 0$ makes both conditional derivatives positive on this shrinking interval. Uniform Taylor expansion gives, for bounded $v \geq 0$,

$$\frac{g'_a(v\sqrt{\delta})}{h'_a(v\sqrt{\delta}) + h'_a(-v\sqrt{\delta})} = \frac{\delta^{3/2}}{2H_*}\left(C_*v - \frac{B_*v^3}{6}\right) + o(\delta^{3/2}).$$

The cubic is largest at the unique positive point $v = \sqrt{2C_*/B_*}$. Maximizing on $[0, \kappa_a/\sqrt{\delta}]$, whose right endpoint tends to $\sqrt{6C_*/B_*}$, proves (3.3). The maximum is positive and attained inside $(0, \kappa_a)$, since the ratio vanishes at both endpoints. Uniform convergence and the unique maximum of the limiting cubic also force every scaled maximizer to tend to $\sqrt{2C_*/B_*}$. This proves the assertion about t_a. The bounds on Q_* follow by squaring its formula and inserting the three rational enclosures for C_*, B_* and H_*.

For strict monotonicity, choose a small fixed $r > 0$. For a sufficiently close to a_*, the denominator of the ratio is positive on $[0, r]$ and $\kappa_a < r$. The maximum can therefore be taken over this fixed compact interval. The differentiated Fourier bounds justify the expansions $\partial_a g'_a(t) = C_* t + o(t)$ and $\partial_a(h'_a(t) + h'_a(-t)) = O(1)$, uniformly as $(a, t) \to (a_*, 0)$, while $g'_a(t) = O((|a - a_*| + t^2)t)$. The parameter derivative of the ratio is consequently positive for $0 < t \le r$ after shrinking the parameter interval and r. At $a > a_*$ its maximum is positive and occurs at $t > 0$. Evaluating at that maximizer at any larger nearby memory makes the maximum strictly larger. Hence b_a strictly decreases there. □

This also explains why two different powers appear. Near zero, the derivative is approximately $C_*(a - a_*)t - B_* t^3/6$. Its positive part has width of order $\sqrt{a - a_*}$ but height of order $(a - a_*)^{3/2}$. The modes separate on the width scale. The bias must overcome the height to remove the smaller peak.

3.6 Density approximation near memory one

Theorem 3.13 (the two peaks near memory one)**.** *Let h_a denote the density of $\mathcal{L}_a$, and put*

$$A_\varepsilon = 1 + 2\varepsilon \log(1/\varepsilon), \qquad K_\varepsilon(z) = 2\varepsilon h_{1-2\varepsilon}(A_\varepsilon - 2\varepsilon z),$$
$$H_\varepsilon(z) = 4\varepsilon g_{1-2\varepsilon}(A_\varepsilon - 2\varepsilon z)\mathbf{1}\{z < A_\varepsilon/(2\varepsilon)\}.$$

As $\varepsilon \downarrow 0$, both profiles converge uniformly to f_L, and $\|K_\varepsilon - f_L\|_1 \to 0$. If $\kappa_a > 0$ is the positive mode of g_a, then

$$\kappa_{1-2\varepsilon} = 1 + 2\varepsilon \log(1/\varepsilon) - 2\varepsilon\lambda_{\mathrm{L}} + o(\varepsilon), \qquad 4\varepsilon g_{1-2\varepsilon}(\kappa_{1-2\varepsilon}) \longrightarrow f_L(\lambda_{\mathrm{L}}).$$

For $q \in [0, 1]$, define the two-component density

$$\widetilde{g}_{\varepsilon,q}(x) = \frac{q}{2\varepsilon} f_L\left(\frac{A_\varepsilon - x}{2\varepsilon}\right) + \frac{1-q}{2\varepsilon} f_L\left(\frac{A_\varepsilon + x}{2\varepsilon}\right).$$

Both bounds hold uniformly over the initial bias.

$$\sup_{q\in[0,1]} 2\varepsilon \|g_{1-2\varepsilon,q} - \widetilde{g}_{\varepsilon,q}\|_\infty \longrightarrow 0, \qquad \sup_{q\in[0,1]} \|g_{1-2\varepsilon,q} - \widetilde{g}_{\varepsilon,q}\|_1 \longrightarrow 0.$$

Proof. Write $f = f_L$, $m = \lambda_{\mathrm{L}}$ and $B_\varepsilon = A_\varepsilon/(2\varepsilon) \to \infty$. Lemma 5.10 gives a unique mode m and strict concavity near it. Fourier inversion gives continuity and zero tail limits for the density f.

The bridge gives weak convergence of the unimodal densities K_ε to f. Their modes tend to m. Indeed, for $b < m$, a short interval about m has larger limiting mass than an equal-length interval between b and m. A mode left of b would force the reverse ordering. Reflection excludes modes bounded to the right of m. Away from m, eventual monotonicity squeezes the density between adjacent interval averages, giving $K_\varepsilon(z) \to f(z)$ for every $z \neq m$. The identity

$$\int |K_\varepsilon - f| = 2 - 2\int \min(K_\varepsilon, f)$$

and dominated convergence now give L^1 convergence.

If k is a nonnegative unimodal density with mode b, then

$$k(y) \le \frac{1}{y-b} \qquad (y > b),$$

because its integral over $[b, y]$ is at least $(y-b)k(y)$. Since the modes of K_ε remain bounded, $\sup_{y \ge B_\varepsilon} K_\varepsilon(y) \to 0$. The fair-density identity gives, for $z < B_\varepsilon$,

$$H_\varepsilon(z) = K_\varepsilon(z) + K_\varepsilon(2B_\varepsilon - z).$$

It follows that $H_\varepsilon(z) \to f(z)$ for every $z \neq m$.

For small enough ε, the strict-shape theorem gives one positive hump of $g_{1-2\varepsilon}$ with a concavity interval between two convex intervals. After the affine change of variables, H_ε is unimodal and has the same convex–concave–convex partition up to B_ε. We make no convexity assertion across the truncation point. The preceding mode argument, or its pointwise version, shows that its mode m_ε tends to m.

Choose two fixed triples, one on each side of m, in the neighborhood where f is strictly concave. Each strict midpoint-concavity inequality persists for H_ε. Neither convex part of H_ε can contain its corresponding triple. Hence its concavity interval contains a fixed neighborhood $[L, R]$ of m for all sufficiently small ε. For fixed $L < \ell < x < m$ in this neighborhood, concavity gives

$$H_\varepsilon(m_\varepsilon) \le H_\varepsilon(x) + \frac{m_\varepsilon - x}{x - \ell}\{H_\varepsilon(x) - H_\varepsilon(\ell)\}.$$

First let $\varepsilon \downarrow 0$ and then $x \uparrow m$. This bounds the limiting upper peak height by $f(m)$. The inequality $H_\varepsilon(m_\varepsilon) \ge H_\varepsilon(x)$ gives the reverse bound. Thus $H_\varepsilon(m_\varepsilon) \to f(m)$. This is where we need the secant bound. A thin spike could have vanishing mass and still change the peak height, so weak convergence and unimodality would not have been enough.

To make convergence uniform away from m, take a finite grid in a compact interval. Monotonicity bounds each value between the values at neighboring grid points. Close to m, use two endpoints and the peak height instead. Outside a large fixed interval, H_ε is bounded by its endpoint values and f tends to zero. These three bounds give $H_\varepsilon \to f$ uniformly on $\mathbb{R}$. In the folding identity the reflected tail tends uniformly to zero, so $K_\varepsilon \to f$ uniformly below B_ε too. Above B_ε, the tail bounds make both functions uniformly small.

Finally, $m_\varepsilon = (A_\varepsilon - \kappa_{1-2\varepsilon})/(2\varepsilon)$ and $H_\varepsilon(m_\varepsilon) = 4\varepsilon g_{1-2\varepsilon}(\kappa_{1-2\varepsilon})$. These identities give the stated mode and height limits. Reflection and mixing do not increase the supremum error or the density L^1 error. For $0 \le q \le 1$, we have

$$\big\|q(K_\varepsilon - f) + (1-q)(K_\varepsilon - f)(2B_\varepsilon - \cdot)\big\|_r \le \|K_\varepsilon - f\|_r, \qquad r = 1, \infty.$$

An affine change of variables gives the two assertions uniform in q. □

Corollary 3.14 (both peaks under a biased start)**.** *Fix $0 < \eta \le 1/2$. For all sufficiently small ε, uniformly over $q \in [\eta, 1-\eta]$, the density $g_{1-2\varepsilon,q}$ has exactly two local maxima $x^-_{\varepsilon,q} < x^+_{\varepsilon,q}$, with*

$$\begin{aligned} x^+_{\varepsilon,q} &= A_\varepsilon - 2\varepsilon\lambda_{\mathrm{L}} + o(\varepsilon), \qquad 2\varepsilon g_{1-2\varepsilon,q}(x^+_{\varepsilon,q}) = q f_L(\lambda_{\mathrm{L}}) + o(1), \\ x^-_{\varepsilon,q} &= -A_\varepsilon + 2\varepsilon\lambda_{\mathrm{L}} + o(\varepsilon), \quad 2\varepsilon g_{1-2\varepsilon,q}(x^-_{\varepsilon,q}) = (1-q) f_L(\lambda_{\mathrm{L}}) + o(1). \end{aligned}$$

The error terms are uniform for the stated biases.

Proof. Take a small closed interval about λ_{L} in the scaled coordinates of the positive peak. The approximation in Theorem 3.13 is qf_L there, up to an error tending uniformly

to zero. The reflected component is part of this error, since its argument $2B_\varepsilon - z$ goes to infinity. Now $q \geq \eta$, so the strict gap between the value at λ_{L} and the two endpoint values persists for all sufficiently small ε, uniformly in the bias. The maximum on this interval is therefore attained inside it.

Apply the reflected argument with $1 - q \geq \eta$. This gives a maximum in the negative interval, which is eventually disjoint from the positive one. By Theorem 3.7 these are the only two maxima. The intervals can be chosen arbitrarily small, giving the position estimates, and the uniform density approximation then gives the heights. □

If a bias tends to zero or one, the density approximation still holds, but one component may become smaller than its absolute error. This is why we keep both weights bounded below when locating the modes. Corollary 3.10 gives a separate bound on b_a.

4 The center-end crossing

We use the fair-start walk of Section 1.1 and call a reversal a flip. Let M_t count the $i \leq t$ with $X_i \neq X_1$, and put

$$r_t(b) = \varepsilon + (1 - 2\varepsilon)\frac{b}{t}, \qquad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 4.1. *The process $(M_t)_{t\geq 1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b + 1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1. For $0 \leq k \leq N$, $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$. In particular $C_N = f_N(k_{\mathrm{c}})$, $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$, and $P_N(k) = P_N(N-k)$.*

Proof. Given $X_1, \ldots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t. So $X_{t+1} \neq X_1$ with conditional probability $(1-\varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1, which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N - k$. Finally $M_N \leq N - 1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1}(1 - r_t(0)) = (1-\varepsilon)^{N-1}$. □

The end mass is therefore the same as for the persistent walk with switch probability ε. We next show that increasing the flip probability raises the center relative to the end.

Proposition 4.2. *Let $N \geq 2$ and $1 \leq k \leq N - 1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0,1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $P_N(k) \geq 2E_N$ for $\varepsilon \geq \frac{1}{2}$, and in particular $C_N \geq 2E_N$ there.*

Proof. Begin by comparing a path of M with the path that never flips. A path of M has probability equal to a product of $N - 1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s. The path that never flips has probability $f_N(0) = (1-\varepsilon)^{N-1}$, which is one factor $1 - \varepsilon$ for each step. So we divide each factor by $1 - \varepsilon$. With $y = b_s/s \in [0,1)$ the two kinds of factors become $\frac{\varepsilon + (1-2\varepsilon)y}{1-\varepsilon}$ and $1 - y\frac{1-2\varepsilon}{1-\varepsilon}$, whose derivatives in ε are $\frac{1-y}{(1-\varepsilon)^2} > 0$ and $\frac{y}{(1-\varepsilon)^2} \geq 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1-y) + (1-\varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε, and it is strictly increasing if the path has an up step. Summing over the paths with $k \geq 1$ up steps proves the claim for $f_N(k)/f_N(0)$. By Lemma 4.1, $P_N(k)/E_N = [f_N(k) + f_N(N-k)]/f_N(0)$, and both k and $N - k$ are at least 1. So the claim for $P_N(k)/E_N$ follows.

It remains to locate the crossing. The ratio C_N/E_N is continuous and strictly increasing in ε. As $\varepsilon \to 0$ it tends to 0, since $C_N \to 0$ (at $\varepsilon = 0$ the chain never leaves

0) and $E_N \to \frac{1}{2}$. For $\varepsilon \geq \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1-\varepsilon, \varepsilon]$, so each path of M has probability at least $(1-\varepsilon)^{N-1} = 2E_N$. Every m with $0 \leq m \leq N-1$ is the end of some path, so $f_N(m) \geq 2E_N$. Both k and $N-k$ lie in $\{1, \dots, N-1\}$, so $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)] \geq 2E_N$, and in particular $C_N \geq 2E_N$ there. So the ratio equals 1 at exactly one ε, and this ε lies in $(0, \frac{1}{2})$. □

4.1 The first flip and the center

At $\varepsilon = 0$, the chain M stays at 0. To find the first-order term in ε, allow one first flip and then turn off further flips. The remaining evolution is Pólya's urn.

Lemma 4.3 (Pólya's urn)**.** *Let $2 \leq v \leq N$. At time v an urn holds one ball of one color and $v-1$ of the other. At each time $k = v, \dots, N-1$ we draw a ball uniformly and return it with one more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first color. Then $\beta_v(d) = \binom{N-d-1}{v-2} / \binom{N-1}{v-1}$ for $1 \leq d \leq N-v+1$, and $\beta_v(d) = 0$ for other d. Also β_v is nonincreasing on $\{1, \dots, N-v+1\}$ and $\sum_d d\,\beta_v(d) = N/v$. For $1 \leq d \leq N-1$,*

$$\sum_{v=2}^{N} \beta_v(d) = \frac{N}{d(d+1)}, \qquad \sum_{v=2}^{N} (v-2)\,\beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. Fix an order of the draws. We compute its probability and then count how many orders give the same final count. At time k the urn holds k balls. So if s of them are of the first color, the next draw is of that color with probability s/k. Take one order of the $N-v$ draws in which i draws are of the first color. The draws of the first color contribute the numerators $1, 2, \dots, i$, the other draws contribute $v-1, v, \dots, N-i-2$, and the denominators are $v, v+1, \dots, N-1$. So this order has probability $\frac{i!\,(v-1)v\cdots(N-i-2)}{v(v+1)\cdots(N-1)} = \frac{(v-1)\,i!\,(N-i-2)!}{(N-1)!}$, whatever the order is. There are $\binom{N-v}{i}$ orders. With $d = i+1$ this gives $\beta_v(d) = \omega_d (v-1)\binom{N-v}{d-1}$ with $\omega_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, and writing out the factorials shows that this is $\binom{N-d-1}{v-2} / \binom{N-1}{v-1}$. From the last form, $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d} \leq 1$ for $2 \leq d \leq N-v+1$, so β_v is nonincreasing on its support. The fraction of balls of the first color is a martingale that starts at $1/v$. So the mean number of such balls at time N is N/v.

For the two sums we use the first form of $\beta_v(d)$ and put $i = N-v$, so that $v - 1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2\binom{N-1-i}{2}$. The Chu–Vandermonde identity says that $\sum_i \binom{i}{j}\binom{N-1-i}{b} = \binom{N}{j+b+1}$ (sort the subsets of $\{1, \dots, N\}$ of size $j+b+1$ by their element number $j+1$, which we call $i+1$). With $j = d-1$ and $b = 1, 2$ it shows that the two sums are $\omega_d \binom{N}{d+1}$ and $2\omega_d \binom{N}{d+2}$, and these are the values in the statement. □

Define the hitting probability $V_s(b)$ from state b at time s by $V_N(b) = \mathbf{1}\{b = k_c\}$ and $V_s(b) = r_s(b) V_{s+1}(b+1) + (1 - r_s(b)) V_{s+1}(b)$ for $s < N$. So $V_s(b) = \mathbb{P}_+(M_N = k_c \mid M_s = b)$ whenever $\mathbb{P}_+(M_s = b) > 0$, and $V_s(b)$ is a polynomial in ε. This defines it at $\varepsilon = 0$ as well (so we do not need to condition on an event of probability zero). For $1 \leq t \leq k_c$ let

$$\pi_t = \beta_{t+1}(k_c) = \frac{t}{N-t} \prod_{i=1}^{t-1} \frac{k_c - i}{N-i}, \qquad c_1 = \sum_{t=1}^{k_c} \pi_t, \qquad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \leq t \leq k_c}$ is a probability vector.

The coefficient in part (b) below was found by Bokes, Hlubinová and Singh. With equal switching probabilities ε and one initial sensitive cell, the resistant-cell count of Bokes, Hlubinová and Singh [27, Section 2 and Eq. (1)] is M. Their first-switch and Pólya-urn argument gives $\varepsilon N/(m(m+1))$ as the first-order part of $f_N(m)$ [27, Eqs. (6)–(10)], hence $\frac{4\varepsilon}{N+2}$ at $m = k_c$. We include the derivation because parts (a) and (c) give the weights needed for the uniform bounds in Theorem 4.5 and Appendix D.

Proposition 4.4 (first order, as in [27, Eqs. (6)–(10)])**.** *Let $N \geq 2$ be even.*

(a) The center is $C_N = \sum_{t=1}^{k_c} \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1)$.

(b) The center C_N is a polynomial in ε with $C_N(0) = 0$ (extended as a polynomial to $\varepsilon = 0$). At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C_N'(0) = c_1 = \frac{4}{N+2}$ exactly.

(c) The weights satisfy $\sum_t w_t t = 3 - \frac{6}{k_c+2}$, and $w_t \leq 2t2^{-t}$ for $1 \leq t \leq k_c$.

Proof. (a) We condition on the time of the first flip. Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \leq t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at k_c with probability $V_{t+1}(1)$. A path with no flip ends at $0 \neq k_c$. So C_N is the sum of $\varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$ over $1 \leq t \leq N-1$. Only $t \leq k_c$ contribute. Indeed at time $t+1$ there are t steps equal to X_1, this number never decreases, and at the center it is k_c. So $V_{t+1}(1) = 0$ for $t > k_c$.

(b) Every transition probability is affine in ε, so C_N and $V_{t+1}(1)$ are polynomials in ε. By (a), $C_N(0) = 0$ and $C_N'(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t+1$ on, M counts the balls of the first color in the urn of Lemma 4.3 with $v = t+1$, and $V_{t+1}(1) = \beta_{t+1}(k_c) = \pi_t$. Since $\beta_v(k_c) = 0$ for $v > k_c + 1$, Lemma 4.3 with $d = k_c$ gives $c_1 = \sum_{v=2}^{N} \beta_v(k_c) = \frac{N}{k_c(k_c+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 4.3, with $d = k_c$ and $v = t+1$, gives $\sum_t (t-1)\pi_t = \frac{2N(k_c-1)}{k_c(k_c+1)(k_c+2)}$. Dividing by $c_1 = \frac{N}{k_c(k_c+1)}$ gives $\sum_t w_t(t-1) = \frac{2(k_c-1)}{k_c+2} = 2 - \frac{6}{k_c+2}$, and adding $\sum_t w_t = 1$ gives the first claim. For the second claim, each factor $\frac{k_c-i}{N-i}$ in π_t is at most $\frac{1}{2}$, so $\pi_t \leq \frac{t}{N-t}2^{1-t}$. Hence $w_t = \frac{N+2}{4}\pi_t \leq \frac{N+2}{2(N-t)}\, t2^{-t}$. For $t \leq k_c$ we have $N - t \geq k_c$ and $\frac{N+2}{2k_c} = 1 + \frac{2}{N} \leq 2$, so $w_t \leq 2t2^{-t}$. □

To first order, one flip at step $t+1$ starts Pólya's urn with t balls of one color and one of the other. As $N \to \infty$, the latter fraction tends to Beta$(1,t)$, with density $t2^{1-t}$ at $\frac{1}{2}$. Thus $\sum_t t2^{1-t} = 4$ explains $4/(N+2)$, and $w_t \leq 2t2^{-t}$ restricts this term to early flips.

4.2 Bounds on the central mass

Theorem 4.5. *Let $N \geq 2$ be even.*

(a) For $0 < \varepsilon < \frac{1}{2}$, $C_N \geq \frac{4\varepsilon}{N+2}(1-\varepsilon)^{2-12/(N+4)} \geq \frac{4\varepsilon}{N+2}(1-2\varepsilon)$.

(b) The bound $C_N \leq \frac{4\varepsilon}{N+2}\, N^{2\varepsilon} e^{5200\varepsilon}$ holds whenever $0 < \varepsilon(1+\log N) \leq \frac{1}{100}$.

In particular $\log(1-2\varepsilon) \leq \log C_N - \log \frac{4\varepsilon}{N+2} \leq \varepsilon(2\log N + 5200)$ in the range of (b).

The last, weaker bound in (a) also holds for $\frac{1}{2} \leq \varepsilon < 1$, because its right side is not positive there.

Idea of the proof. Appendix D represents M at time s, after its first flip, as Pólya's urn with αs extra balls of each color, where $\alpha = \varepsilon/(1-2\varepsilon)$ (Lemma D.1). These extra balls pull the minority fraction toward $\frac{1}{2}$. Dropping them gives (a). For (b), we compare with an urn having $\zeta(s+10)$ extra balls, where $\zeta = 2\varepsilon/(1-4\varepsilon)$. Each step multiplies its hitting probability by at most about $e^{\zeta/s}$, whose product is about $e^{\zeta H_N} \approx N^{2\varepsilon}$.

4.3 Locating the crossing

By Theorem 4.5, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1-x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2\log N - \log 8$. So let

$$\Lambda_N = 2\log N - \log 8, \qquad G(x) = x + \log x,$$

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique, since G increases from $-\infty$ to ∞ on $(0,\infty)$.

Table 2: Elephant crossings x_N and the root g_N of $G(g) = \Lambda_N$. These are numerical illustrations. We computed x_N in double precision from the recursion in Lemma 4.1, using Brent's method with absolute tolerance 10^{-12} in x. For g_N the tolerance was about 10^{-15}.

N	x_N	g_N	$x_N/\log N$
50	3.940487	4.288636	1.007
400	7.629217	7.843768	1.273
1600	10.240565	10.340051	1.388
3200	11.546834	11.610464	1.431
12800	14.158845	14.182921	1.497
51200	16.778468	16.786946	1.547

Theorem 4.6. *As $N \to \infty$ over even N, the following hold.*

(a) The crossing satisfies $x_N + \log x_N = \Lambda_N + O(\log^2 N/N)$, so $x_N = g_N + O(\log^2 N/N)$.

(b) The crossing satisfies $x_N = 2\log N - \log\log N - \log 16 + O(\log\log N/\log N)$.

Proof. (a) *Step 1. Both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε. By Proposition 4.2, F is strictly increasing and x_N is its only zero. We show that

$$|F(x) - G(x) + \Lambda_N| \le \rho_N \quad (1 \le x \le 2\Lambda_N), \qquad \text{with } \rho_N = O(\log^2 N/N). \tag{$*$}$$

Let $1 \le x \le 2\Lambda_N$ and $\varepsilon = x/N$. Then $\varepsilon \le 4\log N/N$, so for large N both parts of Theorem 4.5 apply, and together they say that $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon \log N)$. Here $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2\log N - \log(1 + 2/N)$, and $\log(1 + 2/N) \le 2/N \le 2x\log N/N$ because $x \ge 1$. For the end we use $\log(1 - y) = -y + O(y^2)$ for $0 \le y \le \frac{1}{2}$ in $\log E_N = -\log 2 + (N-1)\log(1 - \frac{x}{N})$. So

$$\log C_N = \log x + \log 4 - 2\log N + O\Big(\frac{x\log N}{N}\Big), \qquad \log E_N = -\log 2 - x + O\Big(\frac{x^2}{N}\Big).$$

We subtract. Since $\log 4 + \log 2 = \log 8$, the main terms give $x + \log x - 2\log N + \log 8 = G(x) - \Lambda_N$, and both errors are $O(\log^2 N/N)$ because $x \le 2\Lambda_N \le 4\log N$. This is ($*$).

Keeping the constants gives an explicit ρ_N. Suppose that $400\log N(1 + \log N) \le N$. Then $\varepsilon(1 + \log N) \le 4\log N(1 + \log N)/N \le \frac{1}{100}$, so the inequality after part (b) of Theorem 4.5 applies. With $\log(1 - y) \ge -y - y^2$ for $0 \le y \le \frac{1}{2}$ and $x \le 4\log N$, the error term for $\log C_N$ above lies between $-(3 + 8\log N)/N$ and $(8\log^2 N + 20800\log N)/N$, and the one for $\log E_N$ lies between $-16\log^2 N/N$ and $4\log N/N$. So ($*$) holds with $\rho_N = (24\log^2 N + 20800\log N)/N$.

Step 2. Trap the zero. We have $G' = 1 + 1/x \ge 1$ and $G(g_N) = \Lambda_N$. So $G(x) - \Lambda_N < -\rho_N$ for $x < g_N - \rho_N$, and $G(x) - \Lambda_N > \rho_N$ for $x > g_N + \rho_N$. By ($*$), $F(x) < 0$ for $1 \le x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \le 2\Lambda_N$. Both ranges are nonempty for large N, because $\Lambda_N - \log\Lambda_N \le g_N \le \Lambda_N$ (note that $G(\Lambda_N - \log\Lambda_N) \le \Lambda_N \le G(\Lambda_N)$). Since F is increasing, its zero x_N lies between the two ranges, so $|x_N - g_N| \le \rho_N$. Then x_N lies in $[1, 2\Lambda_N]$, and $F(x_N) = 0$ in ($*$) gives $|G(x_N) - \Lambda_N| \le \rho_N$. With the explicit ρ_N above, these two bounds hold for every even N with $\log N \ge 30$. Indeed $e^{\log N} \ge \log^{12} N/12!$ gives $24\log^4 N + 20800\log^3 N \le N$, so $400\log N(1 + \log N) \le N$ and $\rho_N \le 1/\log^2 N$, and both ranges are nonempty because $g_N \ge \Lambda_N - \log\Lambda_N > 50$.

(b) We solve $g_N = \Lambda_N - \log g_N$ by one substitution. Since $\Lambda_N - \log\Lambda_N \le g_N \le \Lambda_N$, the ratio g_N/Λ_N is $1 + O(\log\Lambda_N/\Lambda_N)$, so $\log g_N = \log\Lambda_N + O(\log\Lambda_N/\Lambda_N)$. Also $\Lambda_N =$

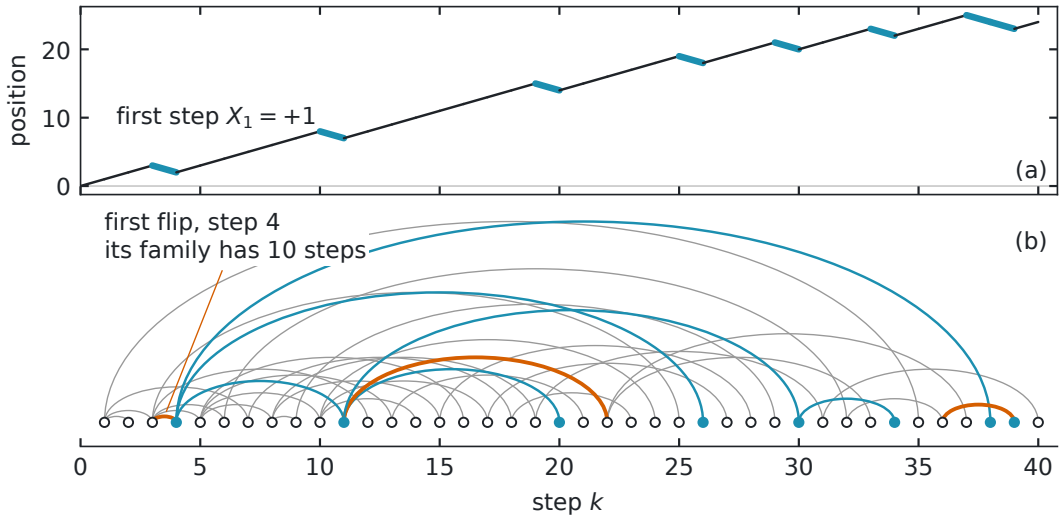


Figure 2: One elephant walk with $N = 40$ steps and $\varepsilon = 0.07$. (a) The position against the step. The steps that disagree with the first step are drawn heavier. (b) The memory tree of the same walk. Step $k+1$ copies the step at the left end of its arc, and the arcs drawn heavier are flips. A step disagrees with the first step exactly when the path from it to step 1 crosses an odd number of flips (Lemma 5.1). The first flip, at step 4, turns its family of ten steps. The flip at step 22, inside that family, turns its own family of three steps back, and the flip at step 39 turns only that step.

$2\log N - \log 8$ gives $\log \Lambda_N = \log 2 + \log\log N + O(1/\log N)$. So $g_N = 2\log N - \log 8 - \log 2 - \log\log N + O(\log\log N/\log N)$, which is (b) for g_N, and (a) gives it for x_N. □

Table 2 gives crossings for six values of N of the form $50 \cdot 2^k$. The limit 2 of $x_N/\log N$ is approached slowly, because of the term $-\log\log N$.

Why twice as many reversals? An endpoint has mass about $\frac{1}{2}e^{-x}$. In the persistent walk, let θ be the switch probability and $T = N\theta$. Its runs have typical length N/T, and the right-step count spreads over about $N/\sqrt{T}$ values. Its central mass is of order $\sqrt{T}/N$, which balances e^{-T} at $T \approx \log N$. For the elephant walk, follow the descendants of an early reversal (the steps that copy it, directly or indirectly). Their number grows as in Pólya's urn. The middle is reached when this family takes up about half the walk. The early reversal costs order ε, and selecting the exact middle adds a factor $1/N$. Thus the center has mass about $4\varepsilon/N = 4x/N^2$, balancing the endpoint at $x \approx 2\log N$. Most of the roughly $2\log N$ reversals there occur too late to produce a large family.

5 The modes of the finite walk

In this section and Appendices E and F, "Assume (H)" means

$$N \geq 64, \qquad 0 < \varepsilon \leq \tfrac{1}{4}, \qquad x = N\varepsilon \geq 1. \tag{H}$$

We invoke this condition only where stated. Recall that $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$. For a mode $\kappa_N \geq N/2$ of P_N we put $m_N = N - \kappa_N$, and $m_{\max}$ is the largest mode of f_N.

5.1 Families and the first flip

For $k_0 \geq 1$, the chain $(M_k^{[k_0]})_{k \geq k_0}$ starts at $M_{k_0}^{[k_0]} = 0$ with transitions $r_k(b)$. It counts disagreeing steps when the first k_0 steps are forced to agree. Thus $M^{[1]}$ has the law of M. Put $Y_d = M_d^{[1]}$ and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \geq 2$, $t \geq j-1$. In a random recursive tree on $[N] = \{1, \dots, N\}$, vertex $k+1$ joins a uniform vertex of $[k]$, independently for $k = 1, \dots, N-1$. The family $\mathcal{T}_v$ consists of v and all its descendants.

Lemma 5.1 (the memory tree)**.** *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k+1$ is U_k. For $2 \leq v \leq N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \geq k_0$, is a copy of the chain $M^{[k_0]}$.*

Proof. The parent U_k of $k+1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k+1$ to the root is $k+1$ followed by the path from U_k, so $\sigma(k+1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k, $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \cdots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \geq k_0$, then $k+1$ is odd with conditional probability $(1-\varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. □

A single flip at v turns its family (Figure 2). For $k \geq v$, a family $\mathcal{T}_v$ containing s vertices of $[k]$ gains vertex $k+1$ with probability s/k. Lemma 4.3 gives $|\mathcal{T}_v|$ law β_v, mean N/v, and an average $N/(d(d+1))$ families of size d. If v alone is marked, $\mathcal{T}_v$ is the odd set. At fixed N, multiple marks have probability $O(\varepsilon^2)$ and no mark gives $M_N = 0$. Thus the first-order part of $f_N(m)$ in ε is $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$ for $1 \leq m \leq N-1$. This is [27, Eq. (10)] with one initial cell, and it is also proved in Lean for M (Section 6).

To go beyond first order we cut the tree at the first flip.

Lemma 5.2 (splitting off a family)**.** *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \leq v \leq N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S, the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v, which is uniform on $[v-1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \geq 2$ has its parent outside S. So, given this event, each element of S other than v picks its parent freely among the smaller elements of S, v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. □

Lemma 5.3 (the first flip)**.** *Let $V = \min\{v \geq 2 : \mu_v = 1\}$, with $V = \infty$ and $|\mathcal{T}_V| = 0$ if there is no mark. If $V < \infty$, let $\bar{Y}$ be the number of vertices u of $\mathcal{T}_V$ with $\sigma(u)$ even, and let $\bar{Z}$ be the number of odd vertices outside $\mathcal{T}_V$. Then for $2 \leq j \leq N$, $1 \leq d \leq N-j+1$ and integers y, z,*

$$\mathbb{P}\big(V = j,\, |\mathcal{T}_V| = d,\, \bar{Y} = y,\, \bar{Z} = z\big) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)\,\mathbb{P}(Y_d = y)\,\mathbb{P}\big(Z_{N-d}^{(j)} = z\big).$$

In particular $\mathbb{P}(V = j,\, |\mathcal{T}_V| = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and for $m \geq 1$

$$f_N(m) = \sum_{j=2}^{N} \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d)\,\mathbb{P}\big(d - Y_d + Z_{N-d}^{(j)} = m\big),$$

where Y_d and $Z_{N-d}^{(j)}$ are independent.

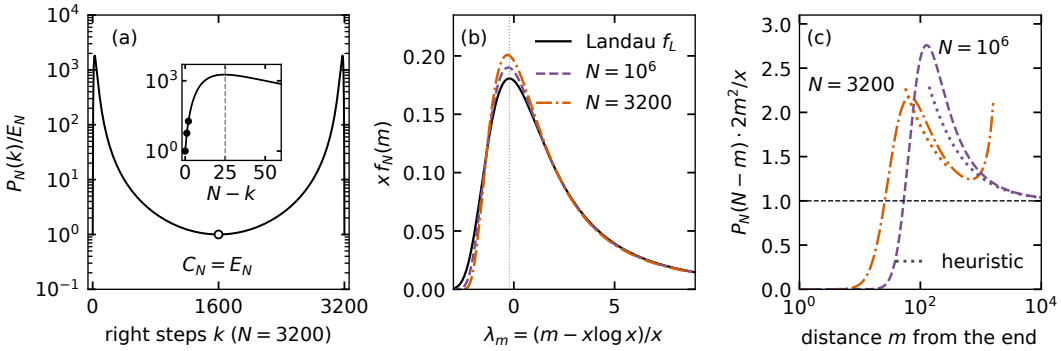


Figure 3: The elephant walk at its crossing. (a) The law P_N for $N = 3200$ at its crossing $x = 11.5468$, divided by the end E_N. The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 values, with dots at N, $N-1$ and $N-2$. The end is below its neighbor, and the modes are at distance 25 from each end (dashed). (b) The law of M_N, scaled as $xf_N(m)$, against $\lambda_m = (m - x\log x)/x$, with the Landau density f_L. The dotted line is its mode λ_{L}. (c) The tail $P_N(N-m)$ divided by $x/(2m^2)$. The dashed line is the reference value 1, and the dotted curves are the heuristic correction $1 + x(2\log m + 2\gamma - 3)/m$ for $2x\log x \le m \le N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N-m)$. At $N = 10^6$ we use $x = g_N = 22.4407$.

Proof. Marks and tree are independent, and $|\mathcal{T}_j|$ has the law β_j, which gives the law of $(V, |\mathcal{T}_V|)$. Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \dots, N$ are still i.i.d., and by Lemma 5.2 the trees on S and on $[N]\setminus S$ are independent random recursive trees. The vertices $1, \dots, j-1$ carry no mark, so j is odd, and $u \in S$ is even exactly when the path from u up to j, with j excluded, has an odd number of marks. So by Lemma 5.1 for the tree on S rooted at j, the count $\bar{Y}$ has the law of Y_d. Paths from outside S stay outside S, and the tree on $[N] \setminus S$ has the unmarked vertices $1, \dots, j-1$ first. So by Lemma 5.1 the count $\bar{Z}$ has the law of $Z^{(j)}_{N-d}$. The two counts are functions of the two trees and of the marks on the two disjoint sets S and $[N] \setminus S$, so they are independent, and their joint law depends on S only through d. This gives the joint law. On $\{V = j\}$ we have $M_N = d - \bar{Y} + \bar{Z}$, and $M_N = 0$ if there is no mark. Summing over j, d and the pairs (y, z) with $d - y + z = m$ gives the formula for $f_N(m)$. □

Only the law of each pair $(Y_d, Z^{(j)}_{N-d})$ enters the lemma. Henceforth $(Y_d)_{d\ge1}$ is a copy of $M^{[1]}$, and $(Z^{(j)}_t)_{t\ge j-1}$ is a copy of $M^{[j-1]}$, independent of Y and $(V, |\mathcal{T}_V|)$. The latter walk starts with $j-1$ steps agreeing with X_1.

Lemma 5.4 (unimodality given the first step [10, Section 2, proof of Theorem 1.1])**.** *The function f_N is unimodal.*

The proof of [10, Theorem 1.1] treats every N and $p = 1-\varepsilon \in [0,1]$, for $N - 2M_N$ given $X_1 = +1$. Section 6 records the Lean proof. Thus f_N is nondecreasing on $\{0, \dots, m_{\max}\}$ and nonincreasing on $\{m_{\max}, \dots, N\}$. Its fair mixture P_N can have two humps when the mode of f_N is far from the center.

5.2 The end and its neighbor

The heuristic threshold is $x > 2$ (Figure 3(a), inset). Given $X_1 = +1$, the end needs no flip, whereas $N-1$ needs one flip that is never copied. A flip at step j is never copied with probability about j/N. Summing $\varepsilon j/N$ over j gives the factor $x/2$.

Theorem 5.5 (endpoint dip)**.** *Let $N \geq 2$ and $0 < \varepsilon < 1$, and put $\rho_1 = P_N(N-1)/E_N$. Then*

$$\rho_1 > \frac{x}{2(1-\varepsilon)}.$$

If moreover $N \geq 3$ and $\varepsilon \leq \frac{1}{2}$, then

$$\rho_1 \leq \frac{x(1+1/N)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

Hence the end is strictly below its neighbor if $x \geq 2(1-\varepsilon)$. If $N \geq 3$, $\varepsilon \leq \frac{1}{2}$ and the right side of the second bound is below 1, the end is strictly above its neighbor.

Proof. By Lemma 4.1, $\rho_1 = \big(f_N(1) + f_N(N-1)\big)/f_N(0)$ with $f_N(0) = (1-\varepsilon)^{N-1}$. We bound the two terms in turn.

The term $f_N(1)$. On the event $M_N = 1$ the chain moves up exactly once, say from time $j-1$ to time j with $2 \leq j \leq N$. Before that it stays at 0 for $j-2$ steps, each with probability $1 - r_t(0) = 1 - \varepsilon$. The move up has probability ε. After it the chain stays at 1 at the times $k = j, \dots, N-1$, each with probability $1 - r_k(1) = (1-\varepsilon)(1 - c'/k)$, where $c' = (1-2\varepsilon)/(1-\varepsilon) \leq 1$. Multiplying and summing over j gives

$$f_N(1) = \varepsilon(1-\varepsilon)^{N-2} \sum_{j=2}^{N} \vartheta_j, \qquad \vartheta_j = \prod_{k=j}^{N-1} \Big(1 - \frac{c'}{k}\Big).$$

For the lower bound we use $c' \leq 1$. Then $\vartheta_j \geq \prod_{k=j}^{N-1}(1 - 1/k) = (j-1)/(N-1)$, and these lower bounds sum to $N/2$ over $2 \leq j \leq N$. So $f_N(1)/f_N(0) \geq x/(2(1-\varepsilon))$. For the upper bound let $N \geq 3$ and $\varepsilon \leq \frac{1}{2}$, so that $c' \geq 0$. Then $1 - c'/k \leq e^{-c'/k}$ and $\sum_{k=j}^{N-1} 1/k \geq \log(N/j)$ give $\vartheta_j \leq (j/N)^{c'}$. As $(t/N)^{c'}$ is nondecreasing in t, we can compare the sum with an integral, namely

$$\sum_{j=2}^{N} \vartheta_j \leq \int_2^{N+1} \Big(\frac{t}{N}\Big)^{c'} dt \leq \frac{N(1+1/N)^{c'+1}}{c'+1} \leq \frac{N(1+1/N)^2(1-\varepsilon)}{2-3\varepsilon},$$

using $c' \leq 1$ and $1/(c'+1) = (1-\varepsilon)/(2-3\varepsilon)$. So $f_N(1)/f_N(0) \leq x(1+1/N)^2/(2-3\varepsilon)$.

The term $f_N(N-1)$. The event $M_N = N-1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \geq 2$ has probability $r_k(k-1) = 1 - \varepsilon - (1-2\varepsilon)/k$. This lies strictly between 0 and 1, as every $r_k(b)$ with $0 < b < k$ does, and it lies in $[\frac{1}{2}, 1-\varepsilon]$ when $\varepsilon \leq \frac{1}{2}$. So $f_N(N-1) > 0$, and $f_N(N-1) \leq \varepsilon(1-\varepsilon)^{N-2}$ when $\varepsilon \leq \frac{1}{2}$. This makes the lower bound strict and gives the term $\varepsilon/(1-\varepsilon)$ in the upper bound. (For $N = 2$ the values 1 and $N-1$ coincide, and the argument is the same.) □

Corollary 5.6 (the dip at the crossing)**.** *Let $N \geq 2$ be even and $0 < \varepsilon < 1$, and suppose that $x \geq 2(1-\varepsilon)$ and $C_N \leq E_N$. Then $C_N \leq E_N < P_N(N-1)$, and P_N is not unimodal. In particular, if $x_N \geq 2(1-\varepsilon_N)$, then at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal.*

Proof. Theorem 5.5 gives $P_N(N-1) > E_N \geq C_N$. Suppose that P_N is nondecreasing on $\{0, \dots, k_0\}$ and nonincreasing on $\{k_0, \dots, N\}$. It is symmetric about k_c, so we may replace k_0 by $N - k_0$ and assume $k_0 \leq k_c$. Then $P_N(k) \leq C_N$ for every $k \geq k_c$, including $k = N-1$, which is a contradiction. □

By Proposition 4.2, $C_N \leq E_N$ holds exactly when $\varepsilon \leq \varepsilon_N$, and $x \geq 2(1-\varepsilon)$ says that $\varepsilon \geq 2/(N+2)$. So the corollary covers the ε with $2/(N+2) \leq \varepsilon \leq \varepsilon_N$. In particular the condition $x_N \geq 2(1-\varepsilon_N)$ says that $\varepsilon_N \geq 2/(N+2)$. By Proposition 4.2 it holds exactly when $C_N \leq E_N$ at $\varepsilon = 2/(N+2)$. We now show that this is so for every even $N \geq 8$.

Proposition 5.7 (the endpoint dip for $N \geq 8$, computer-assisted)**.** *For every even $N \geq 8$ we have $C_N < E_N$ at $\varepsilon = 2/(N+2)$, and hence $x_N > 2(1-\varepsilon_N)$. So for every even $N \geq 8$, at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal. For $N = 2$, 4 and 6 the condition $x_N \geq 2(1-\varepsilon_N)$ fails, and no ε meets the hypotheses of Corollary 5.6.*

Proof. The range $N \geq 1688$. Take $\varepsilon = 2/N$. Then $\varepsilon(1+\log N) \leq \frac{1}{100}$, so Theorem 4.5(b) gives $C_N \leq \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \geq -y - y^2$ for $0 \leq y \leq \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1-\frac{2}{N})^{N-1} \geq \frac{1}{2} e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 4.2 gives $\varepsilon_N > 2/N$. Hence $C_N < E_N$ at $\varepsilon = 2/(N+2)$ as well.

The range $8 \leq N < 1688$. For $N = 8$ the ratio C_N/E_N at $\varepsilon = 2/(N+2)$ is exactly $\frac{67903}{71680} < 1$. For even $10 \leq N \leq 2000$ the script `certify_center.py` encloses ε_N in an interval, by computing $C_N - E_N$ in ball arithmetic at its two ends and using Proposition 4.2, and it checks by an exact rational comparison that the lower end of the enclosure lies above $2/(N+2)$. So $\varepsilon_N > 2/(N+2)$, and Proposition 4.2 gives $C_N < E_N$ at $\varepsilon = 2/(N+2)$. The ratio of the lower end to $2/(N+2)$ is at least 1.107, with its minimum at $N = 10$.

The three smallest N. At $\varepsilon = 2/(N+2)$ the ratio C_N/E_N is exactly 2, $\frac{5}{3}$ and $\frac{508}{405}$ for $N = 2$, 4 and 6, so $\varepsilon_N < 2/(N+2)$ there. These ratios and the one for $N = 8$ are computed in exact rational arithmetic by `certify_elephant.py`. The last claim of the proposition about $N \geq 8$ is Corollary 5.6. □

5.3 Compound Poisson and Landau approximations

Every step that disagrees with X_1 belongs to a flipped family. Recall the compound Poisson law CP_N from Section 1.3. The expected number of flipped vertices with a family of size d is $x/(d(d+1))$. If no flipped step has a flipped descendant (as is likely when $x^2 \log N/N$ is small), then M_N is exactly the sum of the flipped family sizes. We approximate this sum by independent Poisson family counts.

The compound Poisson law and its Landau limit are known (Section 1.6). Here we bound the total variation and local errors explicitly, keeping the jump cutoff at $N-1$ in the local bound. Kölbig and Schorr [37] compute f_L and its mode. Lemma 5.10 gives a certified enclosure of the mode.

Theorem 5.8 (compound Poisson approximation)**.** *For $N \geq 1$ and $0 < \varepsilon < 1$,*

$$\mathrm{TV}(f_N, \mathrm{CP}_N) \leq \delta_\Sigma := \varepsilon^2(NH_N - 2N + 1) + 2\varepsilon^2(N-1) + 4\varepsilon^2 \sum_{k=0}^{N-2} (H_k^2 + H_k).$$

Also $\delta_\Sigma \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$.

Here $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_k |P(k) - Q(k)|$, so the bound holds for every mass and event, and $\delta_\Sigma = O(x^2 \log^2 N/N)$. Appendix F.1 couples marks with Poisson variables and reveals parents successively. Changing one parent affects family sizes along only two root paths, whose depth second moments have order $\log^2 N$.

Families of size up to x contribute nearly $x \log x$. Those of order x give fluctuations of order x. Centering M_N at $x \log x$ and dividing by x gives λ_m and the index-one stable density f_L. Appendix F shows that e^{Ψ_L} is the characteristic function of a law of class L, so f_L is unimodal by [38].

Theorem 5.9 (Landau limit)**.** *(a) Let $x \geq 1$ and $N \geq 4x$. For every integer m, with $\mathrm{CP}_N(m) = 0$ for $m < 0$,*

$$\left| x\,\mathrm{CP}_N(m) - f_L(\lambda_m) \right| \leq E_0 := \frac{47 + 10.4 \log x}{x} + \frac{2.7\,x}{N} \leq 47\Big(\frac{1+\log x}{x} + \frac{x}{N}\Big).$$

(b) Let $0 < \varepsilon \le \frac{1}{4}$ and $x \ge 1$. For every integer $m \ge 0$,

$$\bigl|x f_N(m) - f_L(\lambda_m)\bigr| \le E_0 + x\delta_\Sigma \le 47\Bigl(\frac{1+\log x}{x} + \frac{x}{N}\Bigr) + x\delta_B.$$

(c) Let $N_n \ge 1$ and $0 < \varepsilon_n < 1$ for $n \ge 1$, and put $x_n = N_n\varepsilon_n$. If $x_n \to \infty$ and $x_n^2 \log^2 N_n / N_n \to 0$, then $(M_{N_n} - x_n \log x_n)/x_n$ converges in law to the law with density f_L.

At the crossing, the disagreement count is about $x \log x$, with Landau fluctuations of order x (Figure 3(b)). Theorem 4.6 supplies both conditions of (c). For the untruncated law, (c) is Pakes' theorem [28, p. 993]. Bertoin proves an analogue for vertices cut from a random recursive tree's root in another regime [31, Proposition 1]. The contribution in (c) is the transfer to the walk. We know no earlier form of (a) or (b).

Appendix F proves the local bound by expanding the compound Poisson exponent at zero and applying Fourier inversion. The expansion gives the centering $x \log x$, and its remainder gives the rate.

To locate the modes we need to know that f_L is curved at its maximum.

Lemma 5.10 (the Landau mode)**.** *(a) The unique mode λ_{L} of f_L satisfies*

$$-0.22278299 < \lambda_{\mathrm{L}} < -0.22278297.$$

The density is nondecreasing on $(-\infty, \lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}}, \infty)$.

(b) The derivative vanishes at the mode, $f_L'(\lambda_{\mathrm{L}}) = 0$, and $-0.0792 \le f_L''(\lambda) \le -0.0786$ for $-0.2229 \le \lambda \le -0.2227$.

(c) For every real λ, $f_L(\lambda_{\mathrm{L}}) - f_L(\lambda) \ge \Delta_L \min\{1, (\lambda - \lambda_{\mathrm{L}})^2\}$ with $\Delta_L = 1/50$.

Appendix F.3 proves the lemma. Parts (a) and (b) also have a Lean proof from class-L unimodality. The constant in (c) uses a finite ball-arithmetic cover. Lean checks the deduction from the cover's gap bounds and unimodality (Appendix I).

Corollary 5.11 (mode locations)**.** *(a) Assume (H) and use Δ_L from Lemma 5.10(c). Let $E_1 = E_0 + x\delta_B$ be the bound of Theorem 5.9(b), and put*

$$E' = E_1 + \frac{16x^2 H_N}{N^2} + \frac{\|f_L''\|_\infty}{8x^2}.$$

If $E' < \Delta_L/2$ and $m_{\max} \le N/4$, then every mode $\kappa_N \ge N/2$ of P_N satisfies $|m_N - x\log x - \lambda_{\mathrm{L}} x| \le x\sqrt{2E'/\Delta_L}$, and so does every mode of f_N.

(b) For all sufficiently large N, even or odd, and every ε with $\log N \le x \le 2\log N$, every mode $\kappa_N \ge N/2$ of P_N has

$$\bigl|m_N - x\log x - \lambda_{\mathrm{L}} x\bigr| \le 50\sqrt{x\log x},$$

and so has every mode of f_N in place of m_N. In particular this holds at the crossing, with $x = x_N$, for every large even N (Theorem 4.6(b)).

In (a) every constant is explicit. (Here $\|f_L''\|_\infty \le 16/\pi^4$, since the characteristic function of f_L has modulus $e^{-\pi|t|/2}$.) Remark 5.13 gives an explicit threshold for (b).

Corollary 5.12 (a wider rare-reversal range)**.** *Suppose $x \to \infty$ and $x^3 \log^2 N/N \to 0$. For every mode $\kappa_N \ge N/2$ of P_N, and every mode m of f_N,*

$$N - \kappa_N = x\log x + \lambda_{\mathrm{L}} x + o(x), \qquad m = x\log x + \lambda_{\mathrm{L}} x + o(x).$$

Each error is $O\bigl(\sqrt{x(1+\log x)} + x^{5/2}\log N/\sqrt{N}\bigr)$.

Proof. The hypotheses give $E' \to 0$ in Corollary 5.11(a). They also give $\varepsilon \log N \to 0$ and $x \log^2(x+1) = o(N)$, so Lemma E.5 puts $m_{\max} \le N/4$ eventually. That corollary applies, with

$$E' = O\left(\frac{1+\log x}{x} + \frac{x^3 \log^2 N}{N}\right).$$

Taking the square root gives both error bounds. □

So each hump of the law at the crossing sits about $x \log x - 0.2228\, x$ steps from its end. The proof is in Appendix F.4. It compares f_N at a mode with f_N at the integer nearest to $x \log x + \lambda_{\mathrm{L}} x$, and uses Theorem 5.9(b) and Lemma 5.10(c).

Remark 5.13 (size of the approximation error)**.** The bounds of this subsection are far from sharp at the sizes we can compute. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there the computed law has $|f_N(m)/\mathrm{CP}_N(m) - 1| \le 3.6 \cdot 10^{-4}$ for $1 \le m \le 10^4$. At every tested crossing m_N equals the mode of CP_N. The certified global gap is $\Delta_L = 0.02$. The constants in the local bound still require a large x, and the sufficient threshold $N_2 = \lceil e^{30000} \rceil$ in Corollary 5.11(b) is not practical. The displayed approximation errors are numerical observations in double precision and are not used in any proof.

6 Verification

Table 3 records how each claim is proved and what the proof uses. Some Lean results are implications, so their hypotheses still need proofs for this walk. In particular, Lean does not check the transcendental enclosures at the transition. We compute those separately with interval arithmetic.

Claim	How it is established	Trusted base
Finite shape bounds and the shape at every even crossing	Written proof and Lean, including finite integer certificates	Lean kernel and exact rational arithmetic
Fair-limit shape, inflection counts, bias interval and boundary continuity	Written proofs, with Lean deductions for shape and the interval	Cited limit law, conditional unimodality and Fourier representation
Unique memory threshold	Interval certificate and written Fourier estimates, followed by a Lean implication	Arb transcendental arithmetic and the stated uniform envelopes
Quartic critical top, emerging modes and bias width	Written Taylor and maximization arguments using certified derivatives	The same Fourier envelopes and interval enclosures
Compound Poisson and Landau approximations	Written proofs, certified constants and Lean mode deductions	Fourier inversion, compound Poisson identity, Efron–Stein and Arb
Near-one profile	Written weak bridge and Lean density deductions	Conditional unimodality, fair curvature, Landau facts and Scheffé's theorem [44]

Table 3: The proofs and what they require. The numerical illustrations are not used in proofs. Appendix I lists the records needed to reproduce the checks and the exact scope of each formal result.

We use Lean 4.33.1 with Mathlib pinned to `0df444a360ea`. The declaration audit allows only `propext`, `Classical.choice` and `Quot.sound`. Lean checks the integer certificates by exact outward rounding, without `native_decide`. The total variation

comparison, martingale weak bridge, boundary continuity and critical expansions have written proofs. Arb supplies the sharper local-limit constants and the gap $\Delta_L = 1/50$, whose inequalities are hypotheses of the formal mode deduction.

For the transition, running `python -O` does not remove the acceptance tests. We also check the two threshold endpoint signs by direct evaluation at the real parameter values. This avoids the parameter Taylor models, but still uses Arb and the same analytic tail estimate.

7 Open problems

We have bounded the shapes, but several questions about the transitions remain.

The full bias boundary. Does the bimodal interval keep widening, namely is b_a strictly decreasing on $(a_*, 1)$? It does so near a_* (Proposition 3.12), and Figure 1 suggests the same elsewhere. We have continuity and a unique minimizer where $a > a_*$ and $b_a > 0$ (Theorem 3.9). We also have $b_a \le 8(1-a) + O((1-a)^2)$ as $a \to 1$ (Corollary 3.10). For a certified boundary, we would need the three-point bounds following Theorem 3.9 to hold on whole parameter intervals, with estimates near both endpoints. Computing more points alone does not bound what happens between them.

A finite-length threshold. For even N, does the central curvature have exactly one zero a_N^{crit} in $(0, 1)$, and does $a_N^{\mathrm{crit}} \to a_*$? The computed zero is near 0.6563 for $N = 4000$ and 0.6592 for $N = 8000$ (Remark 3.6), still below a_* in both cases. This suggests slow convergence, but gives neither uniqueness nor a rate. In reversal coordinates the proposed limit is from $(1 - a_N^{\mathrm{crit}})/2$ to $(1 - a_*)/2$. Our weak convergence and mode bounds do not prove it.

Between rare reversals and fixed memory. The wider range in Corollary 5.12 assumes $x \to \infty$ and $x^3 \log^2 N/N \to 0$. The fixed-memory density has the same Landau profile as $a \to 1$ (Theorem 3.13), though we have no rate for that limit. Can a single approximation cover the intermediate scales? The errors computed in Remark 5.13 are much smaller than our bounds. Derivative estimates could also describe how the smaller peak disappears and give an asymptotic equivalent for b_a.

The Drezner–Farnum model and higher dimensions. Drezner and Farnum observed bimodal laws when the innovation probability p varies. Varying this probability changes the transition rule, whereas varying the first-step bias q only mixes the two reflected laws. So the mixture argument does not give the shapes for $p \neq 1/2$. For higher dimensions, densities are available from the fixed-point theory of [11]. The difficulty is that points are no longer ordered along a line, as our sign argument requires. We would also need to specify what a central depression means (the usual axis-step walk is not rotationally symmetric). Our computations cover neither extension.

Acknowledgments. I thank Peter Kagey, whose Problem 131 of the Open Problem Collection [15] led to the crossing question, and Aidan Deshong for helpful conversations. The opening riddle points to the same idea as the crossing argument, namely to look for the first reversal.

A note on the use of AI

I wrote the original draft of this paper and its mathematics. I used Claude Opus 5.5 (Anthropic, through Claude Code) for literature searches, citation checks and referee-

style reviews that informed the framing and abstract. I read every source cited in the paper, except that for Drezner and Farnum [2] and Heyde [3] I relied on their abstracts and the account of James, James and Qi [4]. I proposed and drafted the following additions: Lemma A.4; the strictness and inflection-point clauses of Theorem 3.2; the fourth-moment extension of Theorem 3.3 to $a \geq 4/5$; the Jensen lower bound in Theorem 4.5(a); and the weaker weak-convergence hypothesis in Theorem 5.9(c). I used GPT-6 Astra (OpenAI, through Codex) to revise the manuscript and to revise the proofs of these additions. Proofs aside from the aforementioned were not altered by either model. The associated Lean files in kagey-problems/problem131/lean/PaperD were originally written by me and edited using GPT-6 Astra through Codex. Section 6 distinguishes the formal proofs, their named classical hypotheses and the separate numerical checks. I take responsibility for the claims, proofs and computations in the paper.

Code and data

The tagged repository contains the source, certificate records and Lean files. The submission package also contains the numerical records and checksums. For the assumptions and formal coverage, see Section 6.

A Sign patterns and regularity

We use the notation of Sections 2 and 3.2 to prove the lemmas for Theorem 3.2. Deleting entries can only remove signs from a list's pattern. An alternating list p satisfies $p \preceq q$ for every alternating list q with the same first sign and at least as many entries.

Lemma A.1 (Descartes' rule for exponential sums)**.** *Let $\lambda_1 < \cdots < \lambda_n$ and $b_1, \ldots, b_n$ be real, with $b = (b_1, \ldots, b_n) \neq 0$, and put $F(y) = \sum_{k=1}^{n} b_k e^{\lambda_k y}$. Then F has finitely many real zeros, and $\operatorname{sp}(F(y_1), \ldots, F(y_j)) \preceq \operatorname{sp}(b)$ for all $y_1 < \cdots < y_j$. The same holds for ψF for every positive function ψ.*

Proof. Let $Z(F)$ be the number of distinct real zeros of F, and $S(b)$ the number of sign changes of b. We show $Z(F) \leq S(b)$ by induction on $S(b)$. If $S(b) = 0$, then F has the sign of the nonzero b_k everywhere. If $S(b) \geq 1$, choose i such that the last nonzero b_k with $k \leq i$ and the first nonzero b_k with $k > i$ have opposite signs, and choose $\lambda_i < \kappa < \lambda_{i+1}$. Then

$$G(y) = \frac{d}{dy}\left(e^{-\kappa y} F(y)\right) = \sum_{k=1}^{n} b_k (\lambda_k - \kappa) e^{(\lambda_k - \kappa) y}.$$

The coefficient $b_k(\lambda_k - \kappa)$ has the sign of $-b_k$ for $k \leq i$ and of b_k for $k > i$, so the coefficients of G have $S(b) - 1$ sign changes. By Rolle's theorem G vanishes between any two zeros of F, so $Z(F) \leq Z(G) + 1 \leq S(b)$.

If b_{k_0} is the first nonzero b_k, then $e^{-\lambda_{k_0} y} F(y) \to b_{k_0}$ as $y \to -\infty$. The zeros of F cut the line into at most $S(b) + 1$ open intervals, on each of which F has one sign. With equal neighbors merged, these signs form an alternating list that starts with the sign of b_{k_0} and has at most $S(b) + 1$ entries, so it is $\preceq \operatorname{sp}(b)$. The nonzero values $F(y_1), \ldots, F(y_j)$ lie in these intervals in order. A positive factor ψ changes no sign. □

Lemma A.2 (pointwise limits)**.** *Let v_n and v be real functions on $\mathbb{R}$ with $v_n(y) \to v(y)$ for every y, and let Π be an alternating list. If every v_n has values at most Π, then so has v. If every v_n has increments at most Π, then so has v.*

Proof. Let $y_1 < \cdots < y_j$, and let I be the set of indices i with $v(y_i) \neq 0$. For all large n, $v_n(y_i)$ has the sign of $v(y_i)$ for every $i \in I$. Then $\operatorname{sp}(v(y_1), \ldots, v(y_j))$ is the pattern of the

list $(v_n(y_i))_{i\in I}$, which is a subsequence of $\mathrm{sp}(v_n(y_1),\dots,v_n(y_j)) \preceq \Pi$. The increments of v_n at $y_1,\dots,y_j$ tend to those of v, and the same argument applies to them. □

Lemma A.3 (integrating the sign bound)**.** *Let u be twice continuously differentiable on $\mathbb{R}$, with $u'(y)\to 0$ as $y\to\pm\infty$. If u'' has values at most Π_5, then u' has values at most Π_4, and u has increments at most Π_4.*

Proof. It is enough to treat points $y_1<\cdots<y_j$ at which the values of u' are nonzero and alternate in sign. Let s be the sign of $u'(y_1)$. Choose $y_0<y_1$ with $|u'(y_0)|<|u'(y_1)|$, and $y_{j+1}>y_j$ with $|u'(y_{j+1})|<|u'(y_j)|$. For $0\le i\le j$ the mean value theorem gives $w_i\in(y_i,y_{i+1})$ with $u''(w_i)(y_{i+1}-y_i)=u'(y_{i+1})-u'(y_i)$. The right side has the sign s for $i=0$, the sign of $u'(y_{i+1})$ for $1\le i\le j-1$, and the sign of $-u'(y_j)$ for $i=j$. In each case this is $(-1)^i s$. So u'' takes $j+1$ alternating signs, starting with s, at $w_0<\cdots<w_j$. This list is $\preceq\Pi_5$, so $j+1\le 5$, and $j+1\le 4$ if s is $-$. In both cases the list of the j signs of $u'(y_1),\dots,u'(y_j)$ is $\preceq\Pi_4$.

For the increments at $y_1<\cdots<y_j$, each nonzero $u(y_{i+1})-u(y_i)$ has the sign of $u'(\xi_i)$ for some $\xi_i\in(y_i,y_{i+1})$, by the mean value theorem. So the pattern of the increments is a pattern of values of u'. □

Lemma A.4 (entire densities and parameter continuity)**.** *For every $a\in(\frac12,1)$ and $1<r<1/a$, the characteristic function ϕ_a of $\mathcal{L}_a$ satisfies*

$$|\phi_a(t)|\le 2e^{-c|t|^r}\qquad(t\in\mathbb{R})$$

for some $c=c(a,r)>0$. The same c works on a sufficiently small neighborhood of any fixed a. Every spatial derivative of h_a is jointly continuous in (a,u). For every $q\in[0,1]$, the density of the limit with first-step bias q extends to an entire function and is not constant on an interval of positive length.

Proof. First, we check weak continuity in a. On a compact $J\subset(1/2,1)$, the martingale $\mathcal{M}_n(a)=\Gamma(n)S_n/\Gamma(n+a)$ from the proof of Corollary 3.1 satisfies

$$\sup_{a\in J}\mathbb{E}|\mathcal{L}_a-\mathcal{M}_n(a)|^2\le C_J\sum_{k\ge n}k^{-2\min J}\longrightarrow 0.$$

The walk's centered next step has conditional variance at most one, and the gamma ratios are uniformly bounded on J. The finite laws are continuous in a, so $\phi_a(t)$ is jointly continuous, uniformly on compact frequency intervals.

The fixed-point equation of [11, Theorem 1.1 and (2.10)] gives

$$|\phi_a(t)|\le\int_0^1|\phi_a(tv^a)|\,|\phi_a(t(1-v)^a)|\,dv.$$

Here ϕ_a is the characteristic function conditional on a positive first step. The random sign disappears after taking absolute values, since $|\phi_a(-s)|=|\phi_a(s)|$. Fix a_0 and $1<r<1/a_0$, and choose J about a_0 with $r\max J<1$. Put $D(v)=v^{r\max J}+(1-v)^{r\max J}-1$. We have $D(v)>0$ for $0<v<1$. Dominated convergence therefore permits $K>1$ such that

$$R:=2\int_0^1 e^{-K^r(\log 2)D(v)}\,dv<1.$$

The law of $\mathcal{L}_{a_0}$ has a density [11, Theorem 1.3], so the Riemann–Lebesgue lemma gives $T>0$ with $|\phi_{a_0}(t)|<2^{-K^r}$ for $|t|\ge T$. Joint continuity, after shrinking J, gives $|\phi_a(t)|\le 2^{1-K^r}$ for $a\in J$ and $T\le|t|\le KT$. Set $c=(\log 2)/T^r$ and $b(t)=2e^{-ct^r}$ for $t\ge 0$. On $[0,T]$ we have $b\ge 1$, and on $[T,KT]$ we have $b\ge 2^{1-K^r}$. Thus $|\phi_a|\le b$ on $[0,KT]$.

For $t \geq KT$,

$$\frac{\int_0^1 b(tv^a)b(t(1-v)^a)\,dv}{b(t)} \leq 2\int_0^1 e^{-ct^r D(v)}\,dv \leq R < 1.$$

Suppose the interval on which $|\phi_a| \leq b$ holds has finite supremum $t_* \geq KT$. By continuity it holds on $[0, t_*]$ as well. At the endpoint the fixed-point inequality gives the strict bound $|\phi_a(t_*)| \leq Rb(t_*) < b(t_*)$. Continuity then extends the interval past t_*, a contradiction. The bound thus holds on $[0, \infty)$, and evenness extends it to $\mathbb{R}$.

Since $r > 1$, the Fourier integral converges absolutely and locally uniformly for complex arguments. Indeed, for every $B < \infty$, $\int_{\mathbb{R}} e^{B|t|}|\phi_a(t)|\,dt < \infty$. The Fourier integral therefore defines an entire function. After any fixed number of spatial derivatives, the bound on J still supplies an integrable majorant, giving joint continuity by dominated convergence. For initial bias q, apply the triangle inequality to $q\phi_a(t) + (1-q)\phi_a(-t)$ to get the same bound. A density of integral one cannot be constant on $\mathbb{R}$. If it were constant on an interval, the identity theorem would make it constant everywhere. □

A.1 First differences for a biased start

Proposition A.5 (two modes for every initial bias)**.** *Let $0 < \varepsilon < 1$, let $q \in [0,1]$ be the probability that the first step is positive, and let $P_n(k)$ be the law of the number of positive steps after $n \geq 1$ steps. Extend P_n by zero outside $\{0, \dots, n\}$. The first differences $d_n(k) = P_n(k+1) - P_n(k)$ satisfy*

$$\operatorname{sp}\big(d_n(-1), d_n(0), \dots, d_n(n)\big) \preceq (+,-,+,-).$$

In particular, P_n has at most two modal plateaus, including endpoint modes. Moreover, there are no integers $i_0 < \cdots < i_4$ for which

$$P_n(i_0) > P_n(i_1) < P_n(i_2) > P_n(i_3) < P_n(i_4).$$

Proof. Put $a = 1 - 2\varepsilon$ and $r_t(k) = \varepsilon + ak/t$. The forward equation is

$$P_{t+1}(k) = (1 - r_t(k))P_t(k) + r_t(k-1)P_t(k-1), \qquad P_1 = (1-q, q).$$

Subtracting adjacent equations gives

$$d_{t+1}(k) = r_t(k-1)d_t(k-1) + (1 - r_t(k+1))d_t(k), \qquad 1 \leq k \leq t-1. \tag{A.1}$$

Both coefficients are at least $\min(\varepsilon, 1-\varepsilon) > 0$, since $r_t(j)$ is a convex combination of ε and $1-\varepsilon$ for $0 \leq j \leq t$. It remains to check the two ends.

Suppose first that $q < 1$. Since $P_t(0) = (1-q)(1-\varepsilon)^{t-1} > 0$, define

$$\xi_t = \frac{P_t(1)}{P_t(0)}, \qquad b = \frac{1-2\varepsilon}{1-\varepsilon} < 1, \qquad A = \frac{\varepsilon}{1-\varepsilon} > 0.$$

The boundary equation gives

$$\xi_{t+1} = (1 - b/t)\xi_t + A, \qquad \xi_1 = \frac{q}{1-q}.$$

Thus $\delta_t = \xi_{t+1} - \xi_t$ obeys

$$\delta_{t+1} = \frac{t-b}{t+1}\delta_t + \frac{A}{t+1}. \tag{A.2}$$

Once an increment is nonnegative, every later increment is positive. Thus ξ_t first decreases and then increases (either part may be empty). For $\ell_t = d_t(0) = P_t(0)(\xi_t - 1)$ this proves

$$\operatorname{sp}(\ell_1, \dots, \ell_N) \preceq (+,-,+).$$

If $q \le 1/2$, then $\xi_1 \le 1$ and the sharper bound is $\mathrm{sp}(\ell_1, \dots, \ell_N) \preceq (-,+)$. In particular, a positive ℓ_i cannot be followed by a negative ℓ_j. Zeros do not affect either conclusion.

Write $u_t = d_t(t-1)$. Reflection of the walk gives $u_t(q) = -\ell_t(1-q)$ and $d_t^{(1-q)}(t-1-k) = -d_t^{(q)}(k)$. For $0 < q \le 1/2$, the right boundary history therefore has pattern contained in $(-,+,-)$. For $q = 0$ it is nonpositive throughout. The conditional count is at most $t-1$, so $u_t = -P_t(t-1) \le 0$.

Fix N and, for $1 \le t \le N$, form the augmented row

$$Z_t = (\ell_N, \dots, \ell_{t+1}, d_t(0), \dots, d_t(t-1), u_{t+1}, \dots, u_N).$$

For $q \le 1/2$, its first row is contained in $(+,-,+,-)$. To include possible zeros at $\ell_1 = u_1 = 2q-1$, argue as follows. If Z_1 contained four ordered signs $-,+,-,+$ and the second lay in the reversed left history, the first two would give a positive left edge followed in time by a negative one, which is impossible. Otherwise the last three lie in the right history and have signs $+,-,+$. For $q > 0$, their reflected negatives give the forbidden pattern $(-,+,-)$ in the left history at bias $1-q$. For $q = 0$, they contradict the nonpositive right history. This also covers $N = 1$.

Every entry of Z_{t+1} is a nonnegative combination of adjacent entries of Z_t. The interior entries use (A.1). The two new boundary entries and the remaining future edge values are copied, with coefficients $(1,0)$ or $(0,1)$. Each nonzero entry has a parent of the same sign. Parents of successive opposite signs are strictly ordered, so the new row cannot have an extra alternating sign. Thus $\mathrm{sp}(Z_N) \preceq (+,-,+,-)$. Reflection and sign reversal preserve this pattern, proving the claim also for $q \ge 1/2$, including $q = 1$.

Finally, extension by zero adds $d_N(-1) = P_N(0) \ge 0$ on the left and $d_N(N) = -P_N(N) \le 0$ on the right. Neither enlarges the pattern. Every modal plateau contributes a positive-to-negative change, so there are at most two. If the last displayed inequalities in the statement held, each of their four intervals would contain an adjacent difference of the same strict sign, by telescoping. These four differences would have signs $-,+,-,+$, a contradiction. □

Proposition A.6 (the dominant half)**.** *Under the hypotheses of Proposition A.5, suppose $q \ge 1/2$. Then $P_n(n-k) \le P_n(k)$ for $2k \ge n$, and the right half has no strict valley. That is,*

$$P_n(j) \ge \min\{P_n(i), P_n(k)\} \qquad \text{if } i < j < k \text{ and } 2i+1 \ge n.$$

Equivalently, its first differences contain no negative value followed by a positive one. Reflection gives the corresponding left-half result when $q \le 1/2$.

Proof. Write the bias explicitly as $P_{n,q}$ and put

$$A_n(k) = P_{n,q}(k) - P_{n,1-q}(k) = P_{n,q}(k) - P_{n,q}(n-k).$$

The forward equation gives $A_{n+1}(k) = (1 - r_n(k))A_n(k) + r_n(k-1)A_n(k-1)$. Induction gives $A_n(k) \ge 0$ for $2k \ge n$. Both parents lie in the preceding upper half unless k is the new center, where $A_{n+1}(k) = 0$ by reflection. Terms outside the support vanish. In particular, the central first difference of an odd row is nonnegative.

Restrict the augmented row Z_t in the preceding proof to spatial indices $2k+1 \ge t$. At $t = 1$ its pattern is contained in $(+,-)$. The central entry is $2q-1 \ge 0$, followed by the future right boundary values $u_s(q) = -\ell_s(1-q)$. Since $1-q \le 1/2$, their pattern is $(+,-)$ by (A.2). The positive two-parent recurrence preserves absence of a negative entry followed by a positive one. The sole possible parent outside the previous half occurs at the new central difference of an odd row. That entry is nonnegative, so it cannot begin the forbidden pair. More explicitly, if a negative entry at index i of row $t+1$ satisfies

$2i+1 \geq t+1$, then $2i > t$, so both its parents satisfy $2k+1 \geq t$. Parents of a later positive entry follow them strictly, as in the preceding proof. Induction therefore excludes the pair in every restricted augmented row. A strict valley would contain such two adjacent differences by telescoping, which proves the claim. □

Lemma A.7 (weak transfer of shape)**.** *Let probability laws on lattices $s_n + \eta_n\mathbb{Z}$, with $\eta_n > 0$ and $\eta_n \to 0$, converge weakly to a law with continuous density g. If their lattice mass sequences, including zero masses, exclude two successive strict valleys, then so does g. If they exclude a strict valley among positive lattice points, then g has no strict valley on $[0,\infty)$. For reflection-symmetric lattices, if the mass at each positive lattice point is at least that at its reflection, then $g(x) \geq g(-x)$ for $x \geq 0$.*

Proof. For a bounded interval J of positive length, let $K_n(J)$ be its lattice indices and let $\mu_n(J)$ be its probability. Lattice counting gives $|\eta_n \# K_n(J) - |J|| \leq 2\eta_n$, so these sets are eventually nonempty. Since the limiting law has no atoms, weak convergence gives

$$B_n(J) := \frac{\mu_n(J)}{\eta_n \# K_n(J)} \longrightarrow \frac{1}{|J|}\int_J g(x)\,dx.$$

Here $B_n(J)$ averages the scaled atom masses on J. If g had two successive strict valleys, choose five disjoint equal-length intervals about the witnesses. By continuity, their averages retain all four strict inequalities. The corresponding B_n satisfy these inequalities for large n. Choose a point of maximum mass in the first, third and fifth intervals, and one of minimum mass in each of the other two. The five points are ordered, and their scaled masses retain all four comparisons. This is a contradiction.

Three positive intervals give the one-valley assertion by the same maximum–minimum–maximum selection. If the first witness is 0, continuity moves it slightly to the right while preserving the strict inequalities. Finally, for any interval $J \subset (0,\infty)$ the reflection hypothesis gives $\mu_n(-J) \leq \mu_n(J)$. Weak convergence passes this inequality to the limiting interval masses. Shrink J to $x > 0$ and use continuity to get $g(-x) \leq g(x)$. At $x = 0$ there is equality. □

B Certifying the memory threshold

This appendix describes the certificate for Theorem 3.5. Appendix C proves the estimates it uses. Fix $a \in (\frac{1}{2}, 1)$ and write $\phi(t) = \mathbb{E}e^{it\mathcal{L}_a}$. Put $\theta_j = 1$ for even j and $\theta_j = a$ for odd j, let $m_0 = m_1 = 1$ and $m_n = (na - \theta_n)^{-1}\sum_{j=1}^{n-1}\theta_j m_j m_{n-j}$ for $n \geq 2$. Then $\mathbb{E}\mathcal{L}_a^n = n!\,c_n$ with $c_n = m_n/\Gamma(1+an)$ [11, Theorem 1.4]. As $\mathbb{E}e^{t|\mathcal{L}_a|} < \infty$ for real t [11, Corollary 2.11], $\phi(t) = \sum_n c_n (it)^n$ for all real t. By [11, Theorem 1.1 and (2.10)], $\mathcal{L}_a$ has the law of $V^a\mathcal{L}_a' + \xi(1-V)^a\mathcal{L}_a''$, with V uniform on $[0,1]$, $\xi = \pm 1$, $\mathcal{L}_a'$ and $\mathcal{L}_a''$ distributed as $\mathcal{L}_a$, and all four independent. Since $|\phi(-s)| = |\phi(s)|$, this gives

$$|\phi(t)| \leq \int_0^1 |\phi(tv^a)|\,|\phi(t(1-v)^a)|\,dv \qquad (t \in \mathbb{R}). \tag{B.1}$$

The density g_a of $\mathcal{L}_a^\circ$ has the characteristic function $\operatorname{Re}\phi$.

Step 1, one value of a. If $m_j \leq AB^j$ for $1 \leq j \leq N$, with $A = ((N+1)a - 1)/N$, then $m_n \leq AB^n$ for all $n \geq 1$, by induction, since $(na-1)/(n-1)$ increases in n. This bounds the tail of the moment series and of its derivatives by geometric series. Taylor's formula then bounds $|\phi|$ on cells covering $[0, T_3]$ by s_j, from finitely many coefficients. The next step turns these bounds into a bound for all t. Let $T_3 \geq 2^a T_2$, with T_2 an end point of a cell, and $\beta > 3$ with $a\beta > 1$. Let C be the largest $s_j r_j^\beta$ over the cells $[l_j, r_j]$ that meet

$[T_2, T_3]$, and

$$R = T_2^{-1/a} \sum_{r_j \le T_2} s_j \Big(1 - \tfrac{1}{2}(r_j/T_2)^{1/a}\Big)^{-a\beta} \big(r_j^{1/a} - l_j^{1/a}\big) + \frac{2^{a\beta} C T_2^{-\beta}}{a\beta - 1}.$$

If $R < 1$, then $|\phi(t)| \le Ct^{-\beta}$ for all $t \ge T_3$ (Lemma C.4). Substituting the proposed bound into (B.1) gives a strictly smaller one. Continuity then prevents a first failure. It is an explicit form of Step 4 in the proof of [11, Theorem 2.5]. By Fourier inversion g_a then has two continuous derivatives, and $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\phi(t)\,dt$. We integrate the moment series term by term on $[0, T_3]$, and the two error terms are explicit.

Step 2, intervals of a. Direct interval evaluation in a is too coarse here, since the series cancels terms of size about 10^{60}. On $I = [a_0 - w, a_0 + w]$ we put $\rho = w/\vartheta$ with $0 < \vartheta < 1$ and expand in $a - a_0$. The coefficients are computed at the exact point a_0 by power series arithmetic. If h is analytic with $|h| \le M$ on the disc $|z - a_0| \le \rho$, then Cauchy's estimate bounds the remainder after degree K by $M\vartheta^{K+1}/(1-\vartheta)$ on I. To bound M, we enclose $c_n(z)$ on a square containing the disc, using absolute bounds. Beyond degree N we use the complex form of Step 1 with the product formula for $|\Gamma(x)/\Gamma(x+iy)|^2$ [45, 5.8.3]. This gives a polynomial G and a number η with $|g_a''(0) - G(a - a_0)| \le \eta$ for every $a \in I$. It also shows that $a \mapsto g_a''(0)$ is continuous on I (Proposition C.7).

Step 3, a near $\frac{1}{2}$. Put $y = 2a - 1$ and $\mu_n = m_n y^{\lfloor n/2 \rfloor}$. Then $\mu_2 = a$, and for $n \ge 3$ the recursion for μ_n has the factor y in its terms with n even and j odd, and no other change. So every μ_n is analytic for $\operatorname{Re} a > \frac{1}{4}$ (Lemma C.8). The variable $X = \mathcal{L}_a/\sigma$ with $\sigma^2 = \mathbb{E}\mathcal{L}_a^2 = 1/(y\Gamma(2a))$ has $\mathbb{E}X^n/n! = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor}/\Gamma(1 + an)$ for even n, and the same times $\sqrt{y\Gamma(2a)}$ for odd n. It satisfies the same fixed-point equation. Steps 1 and 2 in this variable reach $a = \frac{1}{2}$ by continuation of the scaled moment bounds. They enclose $\sigma^3 g_a''(0)$, the second derivative at zero of the density of the fair-start scaled variable, whose sign agrees with $g_a''(0)$ for $a > \frac{1}{2}$. The unscaled density g_a is not defined at $a = \frac{1}{2}$.

Step 4, the Hankel test. This step uses no Fourier analysis.

Lemma B.1 (a Hankel test)**.** *Let $s_j = (2j+1)\mathbb{E}\mathcal{L}_a^{2j}$. Suppose that for some $k \ge 1$ and $x \in \mathbb{R}^{k+1}$ we have $\sum_{i,j \le k} x_i x_j s_{i+j} < 0$ or $\sum_{i,j \le k} x_i x_j s_{i+j+1} < 0$. Then case (a) of Theorem 3.2 fails, so case (b) holds.*

Proof. In case (a), $\mathcal{L}_a^\circ$ is unimodal about 0. By Khinchine's theorem [41] it has the law of UZ, with U uniform on $(0,1)$ and independent of Z. So $s_j = (2j+1)\mathbb{E}(\mathcal{L}_a^\circ)^{2j} = \mathbb{E}W^j$ with $W = Z^2 \ge 0$, and the two sums are $\mathbb{E}(\sum_i x_i W^i)^2 \ge 0$ and $\mathbb{E}W(\sum_i x_i W^i)^2 \ge 0$. □

The script `certify_threshold.py` checks every hypothesis in ball arithmetic [40]. It evaluates R with a as a ball over I and evaluates G on subintervals of I, bisecting until $G \pm \eta$ has one sign on each. For the Hankel test it takes k and a rational x from a floating-point eigenvector at the center of a box of a, and checks the sum on the whole box with a Taylor model as in Step 2. Floating-point choices of N, the precision and x only affect whether the subsequent certificate succeeds. The cells have half-width $\frac{1}{32}$, the Taylor order in t is 12, $\beta = 4$, and the degree K is chosen so that $M\vartheta^{K+1} \le e^{-60}$. Table 4 lists the runs.

Step 5, the unique transition. Put $I = [0.66881285, 0.66881287]$, which contains $[z_-, z_+]$. Differentiating the fixed-point equation gives inhomogeneous integral inequalities for $|\partial_t \phi_a(t)|$ and $|\partial_a \phi_a(t)|$. The same continuity argument as in Step 1 propagates the bounds $2t^{-8}$ and $4t^{-7}$, respectively, from $[6, 10]$ to $[6, \infty)$. The latter justifies differentiation under the Fourier integral, with tail error at most $1/(\pi 10^4)$ beyond $t = 10$. The script `certify_transition.py` certifies, uniformly for $a \in \operatorname{int} I$,

$$0.68 < \partial_a g_a''(0) < 0.70, \qquad -0.09 < g_a^{(4)}(0) < -0.07.$$

range of a	method	pieces	parameters	time
$[\frac{1}{2}, 0.668]$	Steps 1 to 3	40	$\vartheta = \frac{1}{8}$, $T_2 \in [3, 10]$	2.1 h
$[0.668, 0.6712]$	Steps 1 and 2	8	$\vartheta = \frac{1}{32}$, $\rho \le \frac{1}{100}$, $T_2 \in [4, 10]$	13 min
$[0.6705, 1]$	Step 4	115	$k \le 56$	25 min

Table 4: The runs of the certificate. The times are summed over the processes of each run.

At finite frequencies we use parameter Taylor models and differentiate their Cauchy remainder bounds. Section C.4 gives the argument and constants.

Proof of Theorem 3.5. The 40 intervals of the first run cover $[\frac{1}{2}, 0.668]$. On each, the scaled enclosure from Step 3 is negative, so $g_a''(0) < 0$ for $a \in (\frac{1}{2}, 0.668]$. The 8 intervals of the second run cover $[0.668, 0.6712]$. On the 2 intervals that cover $[0.668, 0.6688]$ the enclosure is negative, and on the 5 that cover $[0.6692, 0.6712]$ it is positive. On $[0.6688, 0.6692]$ we have $\eta < 1.1 \cdot 10^{-15}$. There the enclosure is negative on closed pieces that end at z_- and positive on closed pieces that start at z_+, and only (z_-, z_+) is undecided. The script prints these signs and checks this arrangement. So $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. The 115 boxes of the third run cover $[0.6705, 1]$, and on each the first sum of Lemma B.1 is negative.

By Step 1, g_a is even with two continuous derivatives, so $g_a'(0) = 0$. If $g_a''(0) < 0$, then $g_a' < 0$ on some $(0, \delta)$, which rules out case (b), where g_a is nondecreasing on $[0, \kappa]$. If $g_a''(0) > 0$, then $g_a' > 0$ on some $(0, \delta)$, which rules out case (a). Lemma B.1 gives case (b) on $[0.6705, 1)$. By Step 2, $a \mapsto g_a''(0)$ is continuous on $[0.6688, 0.6692]$, so it has a zero in (z_-, z_+). Step 5 makes it strictly increasing throughout that bracket, so the zero a_* is unique. The signs already proved exclude other zeros in $(\frac{1}{2}, 0.6712]$ and give the stated shape on either side of a_*. At a_*, evenness and Step 5 give $g_{a_*}'(x) = g_{a_*}^{(4)}(0)x^3/6 + o(x^3) < 0$ for small positive x. This excludes case (b), so the critical density is unimodal too. □

C Proofs for the threshold certificate

We prove the certificate bounds used in Appendix B, keeping its notation and the smoothing inequality (B.1). Together these bounds prove Theorem 3.5. We write $\Phi_N(t) = \sum_{n \le N} c_n (it)^n$ for the partial sums of the moment series.

C.1 Bounds at a fixed memory

Lemma C.1 (moment bounds)**.** *Let $N \ge 3$.*

(a) Put $A = ((N+1)a - 1)/N$. If $m_j \le AB^j$ for $1 \le j \le N$, then $m_n \le AB^n$ for all $n \ge 1$.

(b) Let $S \subset \mathbb{C}$ be compact, with $\frac{1}{2} < \alpha \le \operatorname{Re} z$, $|\operatorname{Im} z| \le \rho$ and $|z| \le 1$ for $z \in S$. Define $m_n(z)$ by the recursion with z in place of a, and $c_n(z) = m_n(z)/\Gamma(1 + nz)$. Then every m_n is analytic near S. Put $A = ((N+1)\alpha - 1)/N$. If $|m_j| \le AB^j$ on S for $1 \le j \le N$, then for all $n \ge 1$ and $z \in S$

$$|m_n(z)| \le AB^n, \qquad |c_n(z)| \le \frac{\tilde{A}\tilde{B}^n}{\Gamma(1 + n\alpha)}, \qquad \tilde{A} = Ae^{\rho^2/(2\alpha^2)}, \quad \tilde{B} = Be^{\rho^2/(2\alpha)}.$$

Proof. For (a) we use induction on $n > N$. As $\theta_j \le 1$ and $na - \theta_n \ge na - 1 > 0$, the bound for smaller indices gives $m_n \le (n-1)A^2B^n/(na-1)$. The ratio $(na-1)/(n-1)$ increases in n, since its derivative is $(1-a)/(n-1)^2$, and it equals A at $n = N+1$. So $m_n \le AB^n$.

For (b), $\mathrm{Re}(nz-\theta_n)\ge n\alpha-1>0$ on S, because $\alpha\le 1$. So m_n is analytic near S, and the induction of (a) goes through with $|\theta_j|\le 1$. For $x>0$ and real y,

$$\Big|\frac{\Gamma(x)}{\Gamma(x+iy)}\Big|^2=\prod_{k\ge 0}\Big(1+\frac{y^2}{(x+k)^2}\Big)\le\exp\Big(\frac{y^2}{x^2}+\frac{y^2}{x}\Big)$$

by [45, 5.8.3]. We take $x=1+n\,\mathrm{Re}\,z\ge 1+n\alpha$ and $|y|\le n\rho$. Then $y^2/x^2\le\rho^2/\alpha^2$ and $y^2/x\le n\rho^2/\alpha$. Finally $\Gamma(1+n\,\mathrm{Re}\,z)\ge\Gamma(1+n\alpha)$, as Γ increases on $[1.47,\infty)$. □

Lemma C.2 (truncation)**.** *Let $0<\alpha\le 1$, and suppose that $|c_n|\le\tilde A\tilde B^n/\Gamma(1+n\alpha)$ for $n>N$. Let $t>0$, $0\le k\le N$ and $n_0=N+1$, and put*

$$u=\frac{\tilde A\tilde B^{n_0}n_0^k t^{n_0-k}}{\Gamma(1+\alpha n_0)},\qquad r=\tilde Bt\Big(1+\frac{1}{n_0}\Big)^k\frac{\Gamma(1+\alpha n_0)}{\Gamma(1+\alpha n_0+\alpha)}.$$

If $r<1$, then $\big|\frac{d^k}{dt^k}\sum_{n>N}c_n(it)^n\big|\le\tau_k(t):=u/(1-r)$, and τ_k increases on $(0,t]$.

Proof. The kth derivative of $(it)^n$ has modulus at most n^kt^{n-k}. The ratio of consecutive terms of the majorant $\tilde A\tilde B^nn^kt^{n-k}/\Gamma(1+\alpha n)$ is $\tilde Bt(1+1/n)^k\Gamma(1+\alpha n)/\Gamma(1+\alpha n+\alpha)$. It decreases in n, because $\log\Gamma$ is convex. So the majorant is at most a geometric series with first term u and ratio r. Both u and r increase in t. □

Lemma C.3 (frequency cells)**.** *Let ν be even, $t_c>0$ and $|\delta|\le h$. Then*

$$|\phi(t_c+\delta)|\le\sum_{k<\nu}|\phi^{(k)}(t_c)|h^k/k!+c_\nu h^\nu.$$

Proof. This is Taylor's formula, with $|\phi^{(\nu)}|\le\mathbb{E}|\mathcal{L}_a|^\nu=\mathbb{E}\mathcal{L}_a^\nu=\nu!\,c_\nu$. □

Lemma C.4 (Fourier tail)**.** *Let $0<T_2<T_3$ with $T_3\ge 2^aT_2$, and let $\beta>3$ with $a\beta>1$. Cover $[0,T_3]$ by cells $[l_j,r_j]$ with $l_0=0$ and with T_2 among the end points, and let $s_j\ge\sup_{[l_j,r_j]}|\phi|$. Let C be the largest $s_jr_j^\beta$ over the cells that meet $[T_2,T_3]$, and put*

$$R=T_2^{-1/a}\sum_{r_j\le T_2}s_j\Big(1-\frac12\Big(\frac{r_j}{T_2}\Big)^{1/a}\Big)^{-a\beta}\big(r_j^{1/a}-l_j^{1/a}\big)+\frac{2^{a\beta}CT_2^{-\beta}}{a\beta-1}.$$

If $R<1$, then $|\phi(t)|\le Ct^{-\beta}$ for every $t\ge T_3$.

Proof. Let $b=s_j$ on the cells in $[0,T_3)$ and $b(t)=Ct^{-\beta}$ for $t\ge T_3$. On a cell that meets $[T_2,T_3]$ we have $s_j\le Cr_j^{-\beta}\le Ct^{-\beta}$. So $b(t)\le Ct^{-\beta}$ for $t\ge T_2$, and $|\phi|\le b$ on $[0,T_3]$. Let $J(t)=\int_0^1 b(tv^a)\,b(t(1-v)^a)\,dv$.

We show that $J(t)\le Rb(t)$ for $t\ge T_3$. By symmetry $J(t)$ is twice the integral over $v\le\frac12$. There $t(1-v)^a\ge 2^{-a}T_3\ge T_2$, so $b(t(1-v)^a)\le b(t)(1-v)^{-a\beta}$. Hence $J(t)/b(t)$ is at most $2\int_0^{1/2}b(tv^a)(1-v)^{-a\beta}\,dv$. Put $v_0=(T_2/t)^{1/a}\le\frac12$. On $v<v_0$ we substitute $u=tv^a$, and this part becomes

$$\frac2a t^{-1/a}\int_0^{T_2}s(u)\big(1-(u/t)^{1/a}\big)^{-a\beta}u^{1/a-1}\,du.$$

Here $t^{-1/a}\le\frac12T_2^{-1/a}$ and $(u/t)^{1/a}\le\frac12(u/T_2)^{1/a}$. The weight increases in u, so on each cell we take it at r_j. As $\int_l^r u^{1/a-1}du=a(r^{1/a}-l^{1/a})$, this part is at most the first term of R. On $v\ge v_0$ we use $b(tv^a)\le Ct^{-\beta}v^{-a\beta}$ and $(1-v)^{-a\beta}\le 2^{a\beta}$. This part is at most $2^{1+a\beta}T_2^{-\beta}(T_2/t)^{1/a}C/(a\beta-1)$, which is at most the second term of R.

Let ς be the supremum of the $\tau\ge T_3$ with $|\phi|\le b$ on $[0,\tau]$, and suppose that $\varsigma<\infty$. By continuity $|\phi|\le b$ on $[0,\varsigma]$. The arguments in (B.1) at $t=\varsigma$ lie in $[0,\varsigma]$, so $|\phi(\varsigma)|\le J(\varsigma)\le Rb(\varsigma)<b(\varsigma)$. As $|\phi|$ and b are continuous on $[T_3,\infty)$, we get $|\phi|<b$ near ς, against the choice of ς. □

This is an explicit form of Step 4 in the proof of [11, Theorem 2.5], with computed constants.

Lemma C.5 (the second derivative)**.** *If $|\phi(t)| \le Ct^{-\beta}$ for $t \ge T_3$, with $\beta > 3$, then g_a has two continuous derivatives and*

$$g_a''(0) = -\frac{1}{\pi}\Big(\sum_{\substack{n \le N \\ n \text{ even}}} (-1)^{n/2} c_n \frac{T_3^{n+3}}{n+3} + E \Big), \qquad |E| \le \frac{T_3^3}{3}\tau_0(T_3) + \frac{CT_3^{3-\beta}}{\beta - 3}.$$

Proof. As $t^2 \operatorname{Re}\phi(t)$ is integrable, Fourier inversion gives $g_a(x) = \frac{1}{2\pi}\int e^{-itx} \operatorname{Re}\phi(t)\,dt$, and dominated convergence allows two derivatives under the integral [43, Section 3.3]. So $g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\phi(t)\,dt$. On $[0, T_3]$ we replace ϕ by Φ_N, whose real part is $\sum_{n \text{ even}} (-1)^{n/2} c_n t^n$, and integrate term by term. By Lemma C.2 the error is at most $\int_0^{T_3} t^2 \tau_0(t)\,dt \le \frac{1}{3} T_3^3 \tau_0(T_3)$. The integral over $t > T_3$ is at most $CT_3^{3-\beta}/(\beta - 3)$. □

C.2 Bounds on memory intervals

Direct interval evaluation loses too much to cancellation. Near the threshold the series for ϕ cancels terms of size about 10^{60}, requiring intervals in a narrower than about 10^{-65}. We retain this cancellation by expanding in a around one exact point.

Lemma C.6 (Cauchy remainder)**.** *Let h be analytic near the closed disc $|z - a_0| \le \rho$, with $|h| \le M$ there and Taylor coefficients h_i at a_0. If $0 < \vartheta < 1$ and $|a - a_0| \le \vartheta\rho$, then $|h(a) - \sum_{i \le K} h_i (a - a_0)^i| \le M\vartheta^{K+1}/(1 - \vartheta)$.*

Proof. Cauchy's estimate gives $|h_i| \le M\rho^{-i}$. Sum the geometric tail. □

Let $I = [a_0 - w, a_0 + w]$, $0 < \vartheta < 1$ and $\rho = w/\vartheta$, and let S be the square $a_0 + [-\rho, \rho] + i[-\rho, \rho]$, which contains the disc of Lemma C.6. Suppose that $\alpha = a_0 - \rho > \frac{1}{2}$, that $|z| \le 1$ on S, and that $\tilde{A}, \tilde{B}$ are as in Lemma C.1(b). Let τ_k be the bound of Lemma C.2 with these constants. It holds for every real $a \in I$. For a cell with center t_c write $\partial_t^k \Phi_N(z, t_c) = E_k(z) + iO_k(z)$, where E_k is the sum over even n and iO_k the sum over odd n. Both E_k and O_k are real for real z. Let $E_{k,i}$ and $O_{k,i}$ be their Taylor coefficients at a_0, and put $M_k = \sum_{n \le N} \sup_S |c_n|\, n(n-1)\cdots(n-k+1)\, t_c^{n-k}$.

Proposition C.7 (enclosure on an interval)**.** *(a) For every $a \in I$ and every t in the cell with center t_c and half-width h,*

$$|\phi_a(t)| \le \sum_{k<12} \Big(\sum_{i \le K} w^i \sqrt{E_{k,i}^2 + O_{k,i}^2} + \frac{M_k \vartheta^{K+1}}{1 - \vartheta} + \tau_k(t_c) \Big) \frac{h^k}{k!} + h^{12} \sup_S |c_{12}|.$$

(b) Let T_2, $T_3 \ge 2^{a_0 + w} T_2$, β and the cells be as in Lemma C.4, with s_j the bound of (a), and suppose that $R < 1$ for every $a \in I$. Put

$$F(z) = \sum_{n \le N,\, n \text{ even}} (-1)^{n/2} c_n(z) T_3^{n+3}/(n+3).$$

Let F_i be its Taylor coefficients at a_0, and let M_F be the same sum with $\sup_S |c_n|$ in place of $(-1)^{n/2} c_n(z)$. Then for every $a \in I$

$$\Big| g_a''(0) + \frac{1}{\pi} \sum_{i \le K} F_i (a - a_0)^i \Big| \le \eta = \frac{1}{\pi}\Big(\frac{M_F \vartheta^{K+1}}{1 - \vartheta} + \frac{T_3^3}{3}\tau_0(T_3) + \frac{CT_3^{3-\beta}}{\beta - 3} \Big),$$

and $a \mapsto g_a''(0)$ is continuous on I.

Proof. For (a) we apply Lemma C.3 with $\nu = 12$, and Lemma C.2 to replace $\phi_a^{(k)}$ by $\partial_t^k \Phi_N$. For real a we have $|\partial_t^k \Phi_N(a, t_c)| = |(E_k(a), O_k(a))|$. Lemma C.6 for E_k and O_k, with $\sup|E_k| + \sup|O_k| \le M_k$ on S, and the triangle inequality in $\mathbb{R}^2$ give the bracket. Also $c_{12}(a) \le \sup_S |c_{12}|$.

For (b), $T_3 \ge 2^a T_2$ for every $a \in I$, so Lemma C.4 applies at each a. Then Lemma C.5 and Lemma C.6 for F give the enclosure. By Lemma C.1(b) the series for $\phi_a(t)$ converges uniformly for $a \in I$ and t in compact sets. So $\phi_a(t)$ is continuous in a. Moreover $t^2|\phi_a(t)|$ is at most t^2 on $[0, T_3]$ and $Ct^{2-\beta}$ beyond, uniformly in $a \in I$. Dominated convergence gives the continuity. □

Near $a = \frac{1}{2}$ the bounds of Lemma C.1 degenerate, since $m_2 = a/(2a-1)$. We rescale.

Lemma C.8 (the scaled variable)**.** *Put $y = 2a - 1$ and $\mu_n = m_n y^{\lfloor n/2 \rfloor}$.*

(a) We have $\mu_0 = \mu_1 = 1$, $\mu_2 = a$, and for $n \ge 3$

$$(na - \theta_n)\mu_n = \sum_{j=1}^{n-1} \theta_j y^{e_{nj}} \mu_j \mu_{n-j},$$

where $e_{nj} = 1$ if n is even and j is odd, and $e_{nj} = 0$ otherwise. Here $na - \theta_n$ is $(n-1)a$ for odd n and $na - 1$ for even n. So every μ_n is a rational function of a with no pole in $\operatorname{Re} a > \frac{1}{4}$.

(b) Let $\sigma^2 = \mathbb{E}\mathcal{L}_a^2$, $X = \mathcal{L}_a/\sigma$ and $\hat{c}_n = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor}/\Gamma(1+an)$. Then $\sigma^2 = 1/(y\Gamma(2a))$,

$$\mathbb{E}e^{iuX} = \sum_{n \text{ even}} \hat{c}_n (iu)^n + \sqrt{y\Gamma(2a)} \sum_{n \text{ odd}} \hat{c}_n (iu)^n, \qquad g_a''(0) = -\frac{1}{\pi\sigma^3} \int_0^\infty u^2 \operatorname{Re} \mathbb{E}e^{iuX}\, du.$$

Proof. Multiply the recursion for m_n by $y^{\lfloor n/2 \rfloor}$. The exponent $\lfloor n/2 \rfloor - \lfloor j/2 \rfloor - \lfloor (n-j)/2 \rfloor$ is 1 when n is even and j is odd, and 0 otherwise. Also $m_2 = a/y$. This proves (a), as the zeros of the factors are $a = 0$ and $a = 1/n \le \frac{1}{4}$. For (b), $\sigma^2 = 2m_2/\Gamma(1+2a) = 1/(y\Gamma(2a))$ and $\mathbb{E}X^n/n! = c_n \sigma^{-n} = \mu_n \Gamma(2a)^{\lfloor n/2 \rfloor} (y\Gamma(2a))^{(n \bmod 2)/2}/\Gamma(1+an)$. The formula for $g_a''(0)$ follows from the substitution $t = u/\sigma$ in Lemma C.5. □

The fixed-point equation is linear, so X satisfies it too, and (B.1) and Lemmas C.3 to C.5 hold for X. The proof of Lemma C.1(b) carries over to μ_n for $n \ge 3$ and $\alpha > \frac{1}{4}$, with two changes. The constant A becomes $((N+1)\alpha - 1)/(N\kappa)$, where κ is the largest value of $\max(1, |z|)\max(1, |2z-1|)$ on S. And $\tilde{B}$ gains the factor $\max(1, \sup_S |\Gamma(2z)|)^{1/2}$. For real a the odd part carries the factor $\sqrt{y\Gamma(2a)}$. In the cell bounds we treat the even and odd parts as separate analytic functions. We multiply the odd part by $f = \sup_I \sqrt{y\Gamma(2a)}$ only for real a, and the remainders by $\max(1, f)$. The real part of $\mathbb{E}e^{iuX}$ involves only the even part. So the disc may cross $a = \frac{1}{2}$, and the cover reaches $a = \frac{1}{2}$. The sign of $g_a''(0)$ is the opposite of the sign of the integral, since $\sigma^3 > 0$ and the Fourier formula has a minus sign.

C.3 Checking the parameter cover

The script accepts an interval of a only when $G \pm \eta$ has one sign on each piece of a subdivision of it. In the 7 scaled runs every piece of every interval is negative, so $g_a''(0) < 0$ on $(\frac{1}{2}, 0.668]$. The zero run has 8 intervals that cover $[0.668, 0.6712]$. On the 2 intervals that cover $[0.668, 0.6688]$ every piece is negative. On the 5 intervals that cover $[0.6692, 0.6712]$ every piece is positive. On $[0.6688, 0.6692]$ the pieces up to z_- are negative, the pieces from z_+ on are positive, and only (z_-, z_+) is undecided. Proposition C.7(b) turns these signs into $g_a''(0) < 0$ on $(\frac{1}{2}, z_-]$ and $g_a''(0) > 0$ on $[z_+, 0.6712]$. To check the sign

classification in Theorem 3.5, the script records each accepted interval with its sign and exact endpoints. Its mode `zero` fails unless the signs are $-$ before the undecided piece and $+$ after it. Its mode `check` requires the same of the stored records, together with the sign $-$ on every interval of the scaled runs.

C.4 Uniqueness of the transition

To prove uniqueness inside the bracket, we differentiate the smoothing equation and bound the derivative's tail. Differentiating the error bound in Lemma C.5 would not bound the derivative of the error.

Lemma C.9 (differentiated smoothing bound)**.** *Fix a compact memory interval $I \subset (1/2, 1)$. Suppose that the cell certificate of Lemma C.4 gives, uniformly on I, $f(t) := |\phi_a(t)| \le Ct^{-\beta}$ for $t \ge T_2$, and*

$$H_\beta(t) := 2\int_0^{1/2} f(tv^a)(1-v)^{-a\beta}\,dv \le R < 1 \quad (t \ge T), \qquad T \ge 2^a T_2.$$

Let $\beta > 4$, $\gamma = \beta - 1$, $a\gamma > 1$, and $u = |\partial_t \phi_a|$, $w = |\partial_a \phi_a|$. Suppose that $u \le M$ everywhere, $w \le M_a$ on $[0, T_2]$, and cells give $u(t) < D_1 t^{-\beta}$ and $w(t) < D_a t^{-\gamma}$ on $[T_2, T]$. Put

$$A_1 = \frac{MC2^{a(\beta-1)}}{a+1}, \qquad J = 2C2^{a\beta}T^{\gamma-\beta-1/a},$$
$$\kappa = R + \frac{JT_2^{1/a-\gamma}}{a\gamma - 1}, \qquad A_0 = JM_aT_2^{1/a},$$
$$A_S = MC2^{a(\beta-1)}\left\{\frac{\log 2}{a+1} + \frac{1}{(a+1)^2}\right\} + D_1(\log 2)R + CR/T.$$

If $A_1 + RD_1 < D_1$ and $A_0 + A_S + \kappa D_a < D_a$, with $\kappa < 1$, then the two derivative bounds hold for every $t \ge T_2$. Consequently, in the interior of I,

$$\frac{d}{da}g_a''(0) = -\frac{1}{\pi}\int_0^\infty t^2 \operatorname{Re}\partial_a\phi_a(t)\,dt, \qquad \left|\frac{1}{\pi}\int_T^\infty t^2 \operatorname{Re}\partial_a\phi_a(t)\,dt\right| \le \frac{D_aT^{3-\gamma}}{\pi(\gamma-3)}. \tag{C.1}$$

Proof. The complex majorant in Lemma C.1(b) makes the moment series locally uniformly convergent in the frequency and memory. Cauchy's estimate on a smaller parameter disc gives the same conclusion for its memory derivative and every fixed frequency derivative. Thus these are actual derivatives of ϕ_a. The smoothing equation, with $p = (1+a)/2$, is

$$\phi_a(t) = \int_0^1 \phi_a(tv^a)\{p\phi_a(t(1-v)^a) + (1-p)\phi_a(-t(1-v)^a)\}\,dv.$$

On compact frequency sets, its differentiation under the integral is justified by local boundedness of these derivatives and boundedness of $v^{\inf I}|\log v|$. Conjugate symmetry gives even moduli. Writing $x = tv^a$ and $y = t(1-v)^a$, we obtain

$$u(t) \le \int_0^1 \{v^a u(x)f(y) + (1-v)^a f(x)u(y)\}\,dv,$$
$$w(t) \le \int_0^1 \{w(x)f(y) + f(x)w(y)\}\,dv + S(t),$$
$$S(t) \le \int_0^1 \{x|\log v|u(x)f(y) + y|\log(1-v)|f(x)u(y) + f(x)f(y)\}\,dv.$$

The final term includes the derivative of p. Its coefficient is one because $p' = 1/2$ and $|\phi_a(y) - \phi_a(-y)| \le 2f(y)$.

Split each integral at $1/2$ and reflect its second half. For the frequency derivative, the smaller-argument term is at most $A_1 t^{-\beta}$, by $u \le M$ and

$$2\int_0^{1/2} v^a (1-v)^{-a\beta}\, dv \le \frac{2^{a(\beta-1)}}{a+1}.$$

The larger-argument term is at most $RD_1 t^{-\beta}$. For the memory derivative, split the smaller-argument term again at $v_0 = (T_2/t)^{1/a}$. Its bound is

$$2C2^{a\beta} t^{-\beta-1/a} \left\{ M_a T_2^{1/a} + \frac{D_a T_2^{1/a-\gamma}}{a\gamma - 1} \right\}.$$

The larger-argument term is at most $RD_a t^{-\gamma}$, since $H_\gamma \le H_\beta$. As $\gamma - \beta - 1/a < 0$ and $t \ge T$, these two terms are bounded by $(A_0 + \kappa D_a) t^{-\gamma}$. In S, the smaller logarithmic term uses

$$\int_0^{1/2} v^a (-\log v)\, dv = 2^{-a-1} \left\{ \frac{\log 2}{a+1} + \frac{1}{(a+1)^2} \right\};$$

the larger uses $-\log(1-v) \le \log 2$ and $H_\beta \le R$. The mixing-weight term is at most $CRt^{-\beta}$. Thus $S(t) \le A_S t^{-\gamma}$.

To extend the bounds beyond the cells, maximize the continuous weighted derivative on any $[T_2, U]$. If its maximum lies in $[T_2, T]$, use the cell bound. Otherwise repeat the preceding inequalities with that maximum in place of the proposed constant, and rearrange the strict contraction inequality. This first proves the frequency bound, then the memory bound, independently of U. The integrable uniform bound $t^2 w(t) \le D_a t^{2-\gamma}$ now permits differentiation of the Fourier formula for $g_a''(0)$ on the interior of I and gives (C.1). □

The computation uses

$$I = [0.66881285, 0.66881287], \quad T_2 = 6, \quad T = 10, \quad \beta = 8, \quad \gamma = 7,$$

and the conservative constants

$$C = 17/1000, \quad R = 83/1000, \quad M = 2, \quad D_1 = 2, \quad M_a = D_a = 4.$$

It checks $A_1 < 1$, $\kappa < 1/10$ and $A_0 + A_S < 2$, so both contractions have strict slack. The new tail is at most $1/(10000\pi) < 1/30000$. The certified zero bracket lies strictly inside I.

For the finite-frequency integral, let w_0 be the halfwidth of I, $\rho > w_0$ the Cauchy radius and $\vartheta = w_0/\rho$. Differentiating the finite Taylor polynomial in memory leaves the coefficient remainder factor

$$\frac{1}{\rho} \sum_{i>K} i\vartheta^{i-1} = \frac{\vartheta^K \{(K+1) - K\vartheta\}}{\rho(1-\vartheta)^2}.$$

The omitted moment series has derivative bound $\tau_k(t)/(\rho - w_0)$ by Cauchy's estimate. On each frequency cell we use a Taylor expansion of order 48, with its remainder bounded by the complex coefficient majorant. (At complex memory we cannot use a probability moment.) The signed interval checks give

$$0.68 < \frac{d}{da} g_a''(0) < 0.70, \qquad -0.09 < g_a^{(4)}(0) < -0.07 \quad (a \in \operatorname{int} I). \tag{C.2}$$

The second enclosure uses the same Fourier bound with weight t^4, whose tail is $CT^{-3}/(3\pi)$. The file `certify_transition.py` performs these checks, including the

initial derivative-cell bounds. It was rerun with 1100 moments and 1600-bit precision under `python -O`. Every check uses `require`, so optimization cannot remove it.

Continuity and the sign bracket give a zero a_* of $g_a''(0)$. The first inequality in (C.2) makes it unique within I. At a_*, evenness and the second inequality give

$$g'_{a_*}(x) = \tfrac{1}{6} g^{(4)}_{a_*}(0) x^3 + o(x^3) < 0 \quad \text{for all sufficiently small } x > 0.$$

The strict shape dichotomy therefore puts a_* in the unimodal branch. The signs inside I and the shape certificate outside the bracket now prove the unique transition in Theorem 3.5. Central curvature could have further zeros in the bimodal region without changing this conclusion.

The critical constants. For Table 1 we apply the same estimates on $I_* = [z_- - 10^{-14}, z_- + 10^{-14}]$, where $z_- = 1838418790980431/2748779069440000$ is the lower endpoint of the certified threshold bracket. The entire bracket lies inside I_*. Take

$$T_2 = 10, \quad T = 16, \quad \beta = 12, \quad \gamma = 11,$$
$$C = 1/1000, \quad R = 1/20, \quad M = D_1 = 2, \quad M_a = D_a = 4.$$

The base Fourier bound, initial derivative cells and both strict contraction inequalities hold uniformly on I_*. We use 2000 moments, 2600-bit ball arithmetic, Cauchy ratio $1/10000$, frequency halfwidth $1/32$ and frequency Taylor order 48. The omitted tails for C_*, B_* and H_* are bounded respectively by

$$\frac{D_a T^{3-\gamma}}{\pi(\gamma-3)}, \qquad \frac{C T^{5-\beta}}{\pi(\beta-5)}, \qquad \frac{C T^{2-\beta}}{\pi(\beta-2)}.$$

The finite-frequency integrals use the same parameter and moment remainders as above, with weights t^2, t^4 and t. Since the bounds hold on the whole interval, they include a_* even though its exact value is unknown. Outward rounding gives the table's two derived coefficients. The record and rerun command are in Appendix I.

D Urn bounds for the central mass

We prove Theorem 4.5. Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 4.4(a) we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1 - r_s(b))F(b) + r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = k_c\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1, and for $1 \leq b \leq s-1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq k_c,\ 1 \leq s - b \leq k_c\}$. At the final time, $\mathcal{S}_N = \{k_c\}$.

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b + c$ and $s - b + c$ for the two colors, that is, with c extra balls of each color. For $b + c > 0$ and $s - b + c > 0$ let $\Phi_s^c(b) = \binom{N-s}{k_c - b} \operatorname{Beta}(k_c + c, k_c + c)/\operatorname{Beta}(b + c, s - b + c)$, where Beta is the beta function and the binomial coefficient is 0 unless $0 \leq k_c - b \leq N - s$. Theorem 4.5(a) will follow from Lemma D.1(d), and (b) from Proposition D.4.

Lemma D.1. *Let $c \geq 0$, $1 \leq s \leq N$, $b \in \mathbb{Z}$, $b + c > 0$ and $s - b + c > 0$.*

(a) The number $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at k_c at time N. So $\Phi_N^c(b) = \mathbf{1}\{b = k_c\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1 - r_s^c(b))\Phi_{s+1}^c(b)$ for $s < N$.

(b) If $s < N$ and $\Phi_s^c(b) > 0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{k_c - b}{N - s}$.

(c) If $0 < \varepsilon < \frac{1}{2}$ *and* $\alpha = \frac{\varepsilon}{1-2\varepsilon}$*, then* $r_s(b) = r_s^{\alpha s}(b)$ *for* $0 \le b \le s$*.*

(d) If $0 < \varepsilon < \frac{1}{2}$ *and* $0 \le c \le \alpha s$*, then* $V_s(b) \ge \Phi_s^c(b)$ *for* $1 \le b \le s-1$*. In particular* $V_{t+1}(1) \ge \Phi_{t+1}^0(1) = \pi_t$ *for* $1 \le t \le k_c$*.*

Proof. (a) Let $x = b + c$, $y = s - b + c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. As in the proof of Lemma 4.3, each order of the $N - s$ draws with $k_c - b$ draws of the first color has probability $(x)_{k_c-b}(y)_{N-s-k_c+b}/(x+y)_{N-s} = \mathrm{Beta}(k_c + c, k_c + c)/\mathrm{Beta}(x, y)$, and there are $\binom{N-s}{k_c-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{k_c-b-1}/\binom{N-s}{k_c-b} = \frac{k_c-b}{N-s}$ and $\mathrm{Beta}(x, y)/\mathrm{Beta}(x+1, y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1 + 2\alpha = 1/(1-2\varepsilon)$, we have $\frac{b+\alpha s}{s+2\alpha s} = (1-2\varepsilon)(\frac{b}{s} + \alpha) = r_s(b)$.

(d) We check the hypotheses of Lemma D.2(b) below for $F_u = \Phi_u^c$ on $\{1, \dots, u-1\}$, where $s \le u \le N$. The comparison has a simple interpretation. At time u, M has $\alpha u \ge c$ extra balls, which pull it toward $\frac{u}{2}$ at least as strongly as in the Pólya(c) chain. Moving toward the middle raises the chance of ending at k_c.

By (a), $F_N(b) = \mathbf{1}\{b = k_c\}$, and F_u vanishes off $\mathcal{S}_u$ because the binomial coefficient does. Let $s \le u < N$ and $b \in \mathcal{S}_u$. Subtracting the recursion in (a) from the definition of $\mathcal{L}_u$ gives

$$\mathcal{L}_u F_{u+1}(b) - F_u(b) = \big(r_u(b) - r_u^c(b)\big)\big(F_{u+1}(b+1) - F_{u+1}(b)\big).$$

We show that the two factors on the right never have opposite signs. For the first factor, $r_u^c(b) = \frac{1}{2} - (\frac{u}{2} - b)/(u + 2c)$, and by (c) the same formula with αu in place of c gives $r_u(b)$. So the first factor is $\frac{u}{2} - b$ times $\frac{1}{u+2c} - \frac{1}{u+2\alpha u}$. The second of these is nonnegative because $c \le \alpha s \le \alpha u$. Hence the first factor has the sign of $\frac{u}{2} - b$ or vanishes. For the second factor there are three cases. If $b = k_c$ then $b \ge \frac{u}{2}$ and $F_{u+1}(b+1) = 0$, so the factor is ≤ 0. If $u - b = k_c$ then $b \le \frac{u}{2}$ and $F_{u+1}(b) = 0$, so the factor is ≥ 0. Otherwise both values are positive, and the formula for Φ gives $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(k_c-b)(u-b+c)}{(k_c-u+b)(b+c)}$. With $b = \frac{u}{2} - d$ the numerator minus the denominator is $2d(k_c + c)$, so the ratio is ≥ 1 if $b \le \frac{u}{2}$ and ≤ 1 if $b \ge \frac{u}{2}$. In all three cases the second factor has the sign of $\frac{u}{2} - b$ or vanishes. Hence $\mathcal{L}_u F_{u+1} \ge F_u$ on $\mathcal{S}_u$, and Lemma D.2(b) with $s_0 = s$ gives $V_s \ge F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(k_c) = \pi_t$ by (a) and Lemma 4.3. □

Lemma D.2. *Let* $1 \le s_0 \le N$*, and for* $s_0 \le s \le N$ *let* $F_s \ge 0$ *be a function on* $\{1, \dots, s-1\}$*.*

(a) If $F_N(k_c) \ge 1$ *and* $\mathcal{L}_s F_{s+1} \le K_s F_s$ *on* $\mathcal{S}_s$ *for* $s_0 \le s < N$*, with constants* $K_s \ge 0$*, then* $V_s \le \big(\prod_{u=s}^{N-1} K_u\big) F_s$ *on* $\mathcal{S}_s$*.*

(b) If $F_N(b) \le \mathbf{1}\{b = k_c\}$*, each* F_s *vanishes off* $\mathcal{S}_s$*, and* $\mathcal{L}_s F_{s+1} \ge F_s$ *on* $\mathcal{S}_s$ *for* $s_0 \le s < N$*, then* $V_s \ge F_s$*.*

Proof. We use backward induction on s, from $V_N(b) = \mathbf{1}\{b = k_c\}$ and $V_s = \mathcal{L}_s V_{s+1}$. The operator $\mathcal{L}_s$ averages two values, so it preserves inequalities. In (a), suppose that $V_{s+1} \le \prod_{u>s} K_u\, F_{s+1}$ on $\mathcal{S}_{s+1}$. This also holds off $\mathcal{S}_{s+1}$, because there $V_{s+1} = 0$ and $F_{s+1} \ge 0$. So on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \le \prod_{u>s} K_u\, \mathcal{L}_s F_{s+1} \le \prod_{u \ge s} K_u\, F_s$. In (b), suppose that $V_{s+1} \ge F_{s+1}$ on $\{1, \dots, s\}$. Then $V_s = \mathcal{L}_s V_{s+1} \ge \mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$, and $F_s = 0 \le V_s$ off $\mathcal{S}_s$. □

Choosing the comparison urn. We compare M with an urn whose number of extra balls, $\zeta(s+10)$ of each color at time s, grows more than twice as fast as the αs of M. Its hitting probability $\Phi_s^{\zeta_s}$ is explicit. One step of M multiplies this hitting probability by Δ_s. Proposition D.4 bounds $\log \Delta_s \le \zeta/s$ up to a smaller term. Two effects enter Δ_s. Raising the extra balls from ζ_s to ζ_{s+1} changes Φ, and we bound this through the derivative of Φ_s^τ in τ (Lemma D.3). Also M pulls toward the middle less than the comparison urn, which helps. The product of the factors is about $e^{\zeta H_N}$. Let ψ be the digamma function.

Lemma D.3. *Let $2 \le s \le N$, $b \in \mathcal{S}_s$ and $\tau \ge 0$. Put $X = b + \tau$, $Y = s - b + \tau$, $\Sigma = X + Y$ and $z = (s-2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1-z) + \frac{1+z}{(1-z)\Sigma} + \frac{2(1+z)}{3\Sigma^2(1-z)^2}.$$

Proof. We differentiate the beta functions and bound the digamma function by the first terms of its expansion. Only the two beta functions in $\Phi_s^\tau(b)$ depend on τ, and $\log \mathrm{Beta}(x,y) = \log\Gamma(x) + \log\Gamma(y) - \log\Gamma(x+y)$. So the derivative is $2\psi(\Sigma) - \psi(X) - \psi(Y)$ minus the difference $2\psi(N+2\tau) - 2\psi(k_c + \tau)$. The duplication formula [45, 5.5.5] gives $2\psi(2y) = \psi(y) + \psi(y + \frac12) + 2\log 2$, and ψ is increasing. With $y = k_c + \tau$ this shows that the difference we subtract is more than $2\log 2$. For $x > 0$, [45, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2\int_0^\infty \frac{t\,dt}{(t^2+x^2)(e^{2\pi t}-1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t\,dt}{e^{2\pi t}-1} = \frac{1}{24}$. So $\psi(\Sigma) < \log\Sigma - \frac{1}{2\Sigma}$, and $\psi(x) > \log x - \frac{1}{2x} - \frac{1}{12x^2}$ for $x = X$ and $x = Y$. Hence the derivative is less than $\log\frac{\Sigma^2}{4XY} + (\frac{1}{2X} + \frac{1}{2Y} - \frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2} + \frac{1}{Y^2})$. Now $X - Y = 2b - s$, so $4XY = \Sigma^2 - (X-Y)^2 = \Sigma^2(1-z)$ and $X^2 + Y^2 = \Sigma^2(1+z)/2$. So the three terms are $-\log(1-z)$, $\frac{X^2+Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$ and $\frac{X^2+Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$. □

Proposition D.4. *Let $0 < \varepsilon < \frac14$, $\zeta = \frac{2\varepsilon}{1-4\varepsilon}$, $\zeta_s = \zeta(s+10)$ and $\Sigma_s = s + 2\zeta_s$. For $1 \le s \le N-1$ and $b \in \mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s \Phi_{s+1}^{\zeta_{s+1}}(b) \big/ \Phi_s^{\zeta_s}(b)$.*

(a) Let $c = \zeta_s$, $c' = \zeta_{s+1}$, $d = \frac{s}{2} - b$, $u = \frac{d}{s+2c'}$, $v = \frac{d}{N-s}$ and $\rho = \frac{s+2c'}{s(1+2\alpha)}$. Then $|u| < \frac12$ and

$$\Delta_s(b) = \mathcal{J}\,\frac{\Phi_s^{c'}(b)}{\Phi_s^{c}(b)}, \qquad \mathcal{J} = 1 - (\rho-1)\frac{u(u+v)}{\frac14 - u^2}.$$

(b) The factor satisfies $\log\Delta_s(b) \le \frac{\zeta}{\Sigma_s} + \frac{2\zeta}{3\Sigma_s^2}$.

(c) For $1 \le t \le k_c$, $V_{t+1}(1) \le \pi_t \exp\big(\zeta\big[H_{N-1} - H_t + \frac{2}{3t}\big] + \zeta_{t+1}\big[\log\frac{(t+1)^2}{4t} + \frac23\big]\big)$.

(d) The center satisfies $C_N \le U_N := \varepsilon c_1 e^{\zeta H_{N-1}} \sum_{t=1}^{k_c} w_t (1-\varepsilon)^{t-1} e^{\zeta \mathcal{G}(t)}$, where $\mathcal{G}(t) = (t+11)\big(\log\frac{(t+1)^2}{4t} + \frac23\big) - H_t + \frac{2}{3t}$.

The slope ζ solves $2(\zeta-\alpha)/(1+2\alpha) = \zeta$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c'.

Proof. (a) Write $r' = r_s^{c'}(b)$, $\bar r = \bar r_s(b)$ and $r = r_s(b)$. By Lemma D.1(b) with c', and the recursion in Lemma D.1(a), we have $r'\Phi_{s+1}^{c'}(b+1) = \bar r\,\Phi_s^{c'}(b)$ and $(1-r')\Phi_{s+1}^{c'}(b) = (1-\bar r)\Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r)(1-\bar r)}{1-r'} + \frac{r\bar r}{r'}$. Here $r' = \frac12 - u$, $\bar r = \frac12 + v$ and $r = \frac12 - \rho u$, since $\frac12 - r = (1-2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $r'(1-r') = \frac14 - u^2$ the numerator is $f(u,v) + f(-u,-v)$ with $f(u,v) = (\frac12 + \rho u)(\frac12 - v)(\frac12 - u)$, and this equals $\frac14 - \rho u^2 - (\rho-1)uv$. Also $|d| \le \frac{s}{2} - 1$, so $|u| < \frac12$.

(b) *Step 1. The factor $\mathcal{J}$.* Write $c = \zeta_s$, $c' = c + \zeta$ and $\Sigma = \Sigma_s$. Since $c' - \alpha s = (\zeta - \alpha)s + 11\zeta$, the choice of ζ gives $\rho - 1 = \zeta + \nu_s$ with $\nu_s = \frac{22\zeta}{s(1+2\alpha)}$. As u and v have the sign of d, $u(u+v) \ge u^2$. So $\mathcal{J} \le 1 - (\zeta+\nu_s)W'$ with $z' = 4u^2$ and $W' = \frac{z'}{1-z'}$. Also $\mathcal{J} > 0$ by (a), since $0 < r < 1$ and $\bar r$, $1 - \bar r$ are not both 0. So $\log\mathcal{J} \le -(\zeta+\nu_s)W'$.

Step 2. The ratio $\Phi_s^{c'}/\Phi_s^c$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau \in [c, c']$ of the derivative in Lemma D.3. Its bound increases in z and decreases in Σ, while z decreases and Σ increases in τ. So we may take the bound at $\tau = c$, where $X + Y = \Sigma$, and multiply it by the length ζ of the interval. Put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1 + W$, $\frac{1+z}{1-z} = 1 + 2W$ and $-\log(1-z) \le W$, so

$$\log\Delta_s(b) \le \zeta\Big[W + \frac{1+2W}{\Sigma} + \frac{2(1+2W)(1+W)}{3\Sigma^2}\Big] - (\zeta+\nu_s)W'. \tag{D.1}$$

Step 3. The remaining terms. The terms of (D.1) without W and W' are $\frac{\zeta}{\Sigma}+\frac{2\zeta}{3\Sigma^2}$, which is the bound we want. Since $(1+2W)(1+W)=1+3W+2W^2$, the other terms are

$$\zeta(W-W')+\frac{2\zeta W}{\Sigma}+\frac{2\zeta W}{\Sigma^2}+\frac{4\zeta W^2}{3\Sigma^2}-\nu_s W',$$

and we must show that they add up to at most 0. If $W=0$ then $d=0$, so $W'=0$ and there is nothing to do. So let $W>0$. We bound the four positive terms by multiples of $\frac{\zeta}{s}W$. Their sum is smaller than the magnitude of the negative term (the reason for the shift 10 in ζ_s), also a multiple of $\frac{\zeta}{s}W$. We need three facts, with $X=b+c$ and $Y=s-b+c$ as in Step 2.

(i) We have $1+W\le\frac{\Sigma}{2(1+c)}$. Indeed $X,Y\ge 1+c$, since $b\ge 1$ and $s-b\ge 1$, and the larger of X and Y is at least $\frac{\Sigma}{2}$. So $XY\ge(1+c)\Sigma/2$, and $1+W=\frac{1}{1-z}=\frac{\Sigma^2}{4XY}$.

(ii) We have $W-W'\le\frac{2\zeta}{1+c}W$, and so $W'\ge\frac{9}{11}W$. Indeed z and z' have the same numerator $(s-2b)^2$, and their denominators are Σ^2 and $(s+2c')^2=(\Sigma+2\zeta)^2$. So $z'=\lambda z$ with $\lambda=(\frac{\Sigma}{\Sigma+2\zeta})^2\ge 1-\frac{4\zeta}{\Sigma}$. Then

$$W-W'=\frac{(1-\lambda)z}{(1-z)(1-\lambda z)}\le(1-\lambda)W(1+W)\le\frac{4\zeta}{\Sigma}\,W\,\frac{\Sigma}{2(1+c)}=\frac{2\zeta}{1+c}W,$$

where we used $1-\lambda z\ge 1-z$ and then (i). Since $c=\zeta(s+10)\ge 11\zeta$, we have $\frac{2\zeta}{1+c}\le\frac{2}{11}$.

(iii) We have $\Sigma\ge s$, $1+c>\zeta s$ and $\Sigma^2\ge 3s$. The first two are clear. For the third, $W>0$ means $s\ne 2b$, and with $b\ge 1$ and $s-b\ge 1$ this forces $s\ge 3$.

Step 4. The terms with W. We bound the four positive terms one by one.

$$\begin{aligned}
\zeta(W-W')&\le\frac{2\zeta^2}{1+c}W\le\frac{2\zeta}{s}W && \text{by (ii) and } 1+c>\zeta s,\\
\frac{2\zeta W}{\Sigma}&\le\frac{2\zeta}{s}W && \text{by } \Sigma\ge s,\\
\frac{2\zeta W}{\Sigma^2}&\le\frac{2\zeta}{3s}W && \text{by } \Sigma^2\ge 3s,\\
\frac{4\zeta W^2}{3\Sigma^2}&\le\frac{4\zeta W(1+W)}{3\Sigma^2}\le\frac{2\zeta W}{3\Sigma(1+c)}\le\frac{2\zeta}{3s}W && \text{by (i) and } \Sigma\ge s.
\end{aligned}$$

The four bounds add up to $\frac{16\zeta}{3s}W$. For the negative term, (ii) gives $-\nu_s W'\le-\frac{9}{11}\nu_s W=-\frac{18\zeta}{s(1+2\alpha)}W$. So

$$\log\Delta_s(b)\le\frac{\zeta}{\Sigma}+\frac{2\zeta}{3\Sigma^2}+W\Big[\frac{16\zeta}{3s}-\frac{18\zeta}{s(1+2\alpha)}\Big],$$

and the bracket is negative because $1+2\alpha<2$ for $\varepsilon<\frac14$, so $18/(1+2\alpha)>9>16/3$.

(c) We first compare $V_{t+1}(1)$ with $\Phi_{t+1}^{\zeta_{t+1}}(1)$, and then this with $\pi_t=\Phi_{t+1}^{0}(1)$. For the first step apply Lemma D.2(a) with $s_0=t+1$, $F_s=\Phi_s^{\zeta_s}$ and $K_s=\exp(\frac{\zeta}{s}+\frac{2\zeta}{3s^2})$. This is allowed, since $F_N(k_{\mathrm c})=1$, $F_s>0$ on $\mathcal{S}_s$, and $\mathcal{L}_sF_{s+1}=\Delta_sF_s\le K_sF_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s\ge s$. The product of the K_s over $t<s<N$ is the exponential of $\zeta(H_{N-1}-H_t)+\frac{2\zeta}{3}\sum_{s>t}s^{-2}$, and $\sum_{s>t}s^{-2}\le\frac1t$. Since $1\in\mathcal{S}_{t+1}$, we get $V_{t+1}(1)\le e^{\zeta[H_{N-1}-H_t+2/(3t)]}\,\Phi_{t+1}^{\zeta_{t+1}}(1)$.

For the second step, $\log(\Phi_{t+1}^{\zeta_{t+1}}(1)/\pi_t)$ is the integral over $\tau\in[0,\zeta_{t+1}]$ of the derivative in Lemma D.3 with $s=t+1$ and $b=1$. As in Step 2 of (b), the bound of that lemma is largest at $\tau=0$. There $X=1$, $Y=t$ and $\frac{1}{1-z}=\frac{(t+1)^2}{4t}$, and the bound is $\log\frac{(t+1)^2}{4t}+\frac{t^2+1}{2t(t+1)}+\frac{t^2+1}{12t^2}\le\log\frac{(t+1)^2}{4t}+\frac23$, because the two fractions are at most $\frac12$ and $\frac16$ for $t\ge1$. Multiplying by the length ζ_{t+1} of the interval gives (c).

(d) Insert (c) into Proposition 4.4(a), and use $\zeta_{t+1}=\zeta(t+11)$ and $\pi_t=c_1w_t$. □

Proof of Theorem 4.5. (a) By Proposition 4.4(a) and Lemma D.1(d), $C_N \geq \varepsilon c_1 \sum_t w_t (1-\varepsilon)^{t-1}$. Since $u \mapsto (1-\varepsilon)^u$ is convex, Jensen's inequality and $\sum_t w_t (t-1) = 2 - \frac{6}{k_c+2}$ give the stronger bound

$$C_N \geq \frac{4\varepsilon}{N+2}(1-\varepsilon)^{\,2-6/(k_c+2)} \geq \frac{4\varepsilon}{N+2}(1-2\varepsilon). \tag{D.2}$$

The last step uses $0 \leq 2 - 6/(k_c+2) \leq 2$ and $(1-\varepsilon)^2 \geq 1-2\varepsilon$.

(b) *Step 1. The constants.* Let $0 < \varepsilon(1+\log N) \leq \frac{1}{100}$. Since $\log N \geq 0$, we have $\varepsilon \leq \frac{1}{100}$. So $1 - 4\varepsilon \geq 0.96$ and $\zeta \leq 2\varepsilon/0.96 \leq 2.1\varepsilon$. Also $\zeta \leq 2\varepsilon(1+8\varepsilon)$, because $(1+8\varepsilon)(1-4\varepsilon) = 1 + 4\varepsilon(1-8\varepsilon) \geq 1$. Since $\log \frac{(t+1)^2}{4t} \geq 0$ and $\frac{2}{3}(t+11) > 1 + \log t \geq H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \leq \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \geq \frac{2}{3t}$, also $\mathcal{G}(t) \leq (t+11)\log(t+1) \leq 12t\log(t+1)$. So $\zeta\mathcal{G}(t) \leq 25.2\,\varepsilon t \log(t+1)$, and this is at most $\frac{t}{2}\log 2$ for $t \leq k_c$, since then $25.2\,\varepsilon\log(t+1) \leq 25.2\,\varepsilon \log N \leq 0.252 < \frac{1}{2}\log 2$.

Step 2. The sum over t. Since $e^y - 1 \leq y e^y$ for $y \geq 0$, Step 1 gives $e^{\zeta\mathcal{G}(t)} - 1 \leq \zeta\mathcal{G}(t) 2^{t/2}$. By Proposition D.4(d), $(1-\varepsilon)^{t-1} \leq 1$, $\sum_t w_t = 1$ and $w_t \leq 2t2^{-t}$ (Proposition 4.4(c)),

$$C_N \leq \varepsilon c_1 e^{\zeta H_{N-1}} \Big(1 + 50.4\,\varepsilon \sum_{t \geq 1} t^2 \log(t+1) 2^{-t/2}\Big),$$

The ratio of consecutive terms in the series is

$$(1+\tfrac{1}{t})^2 \frac{\log(t+2)}{\log(t+1)} 2^{-1/2}.$$

It decreases in t and is below $\frac{3}{4}$ at $t = 60$. So the terms with $t > 60$ add up to less than four times the term at $t = 61$, and this is below 10^{-4}. The first 60 terms add up to 102.5864 to four decimals (enclosed in ball arithmetic by `certify_elephant.py`), so the series is below 102.59.

Step 3. The factor $e^{\zeta H_{N-1}}$. Finally $H_{N-1} \leq 1 + \log N$, so $\zeta H_{N-1} \leq 2\varepsilon\log N + 2\varepsilon + 16\varepsilon^2(1+\log N) \leq 2\varepsilon\log N + 3\varepsilon$. With $1 + y \leq e^y$ this proves (b), since $3 + 50.4 \cdot 102.59 < 5200$. Any upper bound for the series below $(5200-3)/50.4$, which exceeds 103.1, would give the same constant. □

E Family sizes and a bound on the mode

The following family estimates give the mode bound in Corollary 5.11. We keep the conventions of Section 5 and prove the estimates, with explicit constants, in Appendix G. Since M is independent of X_1, $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 5.3. The size $|\mathcal{T}_v|$ of the family has the law β_v of Lemma 4.3. Let $\bar\beta_v(s) = \mathbb{P}(|\mathcal{T}_v| \geq s)$, with $\bar\beta_v(s) = 1$ for $s \leq 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the tail of β_v telescopes. So, for $1 \leq s \leq N-1$,

$$\begin{aligned} \bar\beta_v(s) &= \binom{N-s}{v-1} \Big/ \binom{N-1}{v-1} = \prod_{i=1}^{v-1} \Big(1 - \frac{s-1}{N-i}\Big) \leq e^{-(v-1)(s-1)/(N-1)}, \\ \beta_v(s) &= \frac{v-1}{N-s}\,\bar\beta_v(s). \end{aligned} \tag{E.1}$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \qquad \frac{\bar\beta_v(s-1)}{\bar\beta_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \tag{E.2}$$

Lemma E.1 (mass between two points)**.** *Let $0 \le j \le b \le N$. If $m_{\max} \le j$, then $(b-j+1)f_N(b) \le \mathbb{P}(j \le M_N \le b)$. If $b \le m_{\max}$, then $(b-j+1)f_N(j) \le \mathbb{P}(j \le M_N \le b)$.*

Lemma E.2 (disagreement tail)**.** *Let $0 < \varepsilon < 1$, $k_0 \ge 1$ and $t \ge k_0$. For every integer $z \ge 1$,*

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \frac{\varepsilon t H_z}{z}, \qquad \mathbb{E}M_t^{[k_0]} \le \varepsilon t \log \frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \ge z) \le xH_z/z$, and $\mathbb{P}(Z_s^{(j)} \ge z) \le xH_z/z$ for $j-1 \le s \le N$.

Lemma E.3 (the first flipped family)**.** *For $1 \le d \le N-1$,*

$$\frac{x}{d(d+1)}\Big(1 - \frac{2x(N-d-1)}{N(d+2)}\Big) \le \mathbb{P}\big(|\mathcal{T}_V| = d\big) \le \frac{x}{d(d+1)}, \qquad \mathbb{P}\big(|\mathcal{T}_V| \ge d\big) \le x\Big(\frac{1}{d} - \frac{1}{N}\Big).$$

Proposition E.4 (local lower bound)**.** *Let $N \ge 64$, $0 < \varepsilon < 1$ and $2 \le m \le N/4$. Then $f_N(m) \ge \frac{x}{m(m+1)}(1 - e_1'(m)) \ge \frac{x}{m(m+1)}(1 - e_1(m))$, where*

$$e_1'(m) = 2\varepsilon H_{m+1} + \frac{xH_m}{m} + \frac{2x}{m+2} + 8\varepsilon \log N + 20me^{-m/9}$$

and

$$e_1(m) = 20\Big(\frac{x\log(m+1)}{m} + \varepsilon \log N + me^{-m/9}\Big).$$

How the first family gives the lower bound. We use the formula of Lemma 5.3. Given the time j of the first flip, the count M_N is $d - Y_d + Z_{N-d}^{(j)}$, where d is the size of the first flipped family. As d grows by one this quantity moves by at most one, so it hits m at some d close to m unless Y or Z is large. (With no further flips, $Y = Z = 0$, and the count equals d, reaching m at $d = m$.) The law β_j of d is nearly flat near m, and this gives almost the full weight $\beta_j(m)$. Then we sum over j. The error e_1 collects the chance that Y or Z is large, the small slope of β_j, and the first flips that come too late to be treated this way.

Lemma E.5 (mode bound)**.** *Let $N \ge 64$ and $x \ge 1$, and put $\bar{c} = \frac{1}{40}$ and $A_0 = 15400$. If $\varepsilon \log N \le \bar{c}$ and $804x\log(x+2) \le N$, then $m_{\max} \le A_0 x \log^2(x+1)$. In particular this holds if $\varepsilon \log N \le \bar{c}$ and $A_0 x \log(x+2) \le N/4$.*

F Poisson approximation and the Landau limit

We prove Theorems 5.8 and 5.9, Lemma 5.10 and Corollary 5.11. We keep the notation of Section 5 and of Appendix E. So $\mathcal{T}$ is the tree of Lemma 5.1, $\mathcal{T}_v$ is the family of v, and μ_v is the mark of v. By Lemma 4.3 and the discussion after Lemma 5.1, $|\mathcal{T}_v|$ has the law β_v, with mean N/v, and $\sum_{v=2}^N \beta_v(d) = N/(d(d+1))$ for $1 \le d \le N-1$. Let

$$\Psi_N(\theta) = \sum_{d=1}^{N-1} \frac{e^{i\theta d} - 1}{d(d+1)}, \qquad \Psi_\infty(\theta) = \sum_{d \ge 1} \frac{e^{i\theta d} - 1}{d(d+1)}.$$

The law CP_N has the characteristic function $\widehat{\mathrm{CP}}_N(\theta) = e^{x\Psi_N(\theta)}$.

F.1 Poisson approximation

We use $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_k |P(k) - Q(k)|$. The case $N = 1$ is immediate. Assume $N \ge 2$ and $0 < \varepsilon < 1$. Put $Z_+ = \sum_{v=2}^N \mu_v |\mathcal{T}_v|$. Unless two marked vertices are strictly nested, $M_N = Z_+$. The family means therefore give

$$\mathbb{P}(M_N \ne Z_+) \le \varepsilon^2 \sum_{v=2}^N (N/v - 1) = \varepsilon^2(NH_N - 2N + 1).$$

Given the tree, replace the independent Bernoulli marks by independent Poisson variables of mean ε. The scalar total variation is $\varepsilon(1-e^{-\varepsilon}) \le \varepsilon^2$, since the only positive Bernoulli-minus-Poisson mass is at 1. Coupling the marks one at a time changes the weighted sum's law by at most $(N-1)\varepsilon^2$. The resulting conditional law is $\mathrm{CP}(\Lambda)$, where $\Lambda = \varepsilon \sum_{v=2}^{N} \delta_{|\mathcal{T}_v|}$ has deterministic mass $\lambda = (N-1)\varepsilon$. By the family identity, $\mathbb{E}\Lambda\{d\} = x/[d(d+1)]$, so $\mathrm{CP}(\mathbb{E}\Lambda) = \mathrm{CP}_N$.

Use the L1 norm on signed measures and convolution $*$, for which $\|\sigma * \tau\|_1 \le \|\sigma\|_1 \|\tau\|_1$. On intensities of mass λ, the CP map and its derivatives in a zero-mass direction D are

$$F(\nu) = e^{-\lambda} \sum_{r \ge 0} \frac{\nu^{*r}}{r!}, \qquad F'(\nu)[D] = F(\nu) * D, \qquad F''(\nu)[D, D] = F(\nu) * D * D,$$

where $\nu^{*0} = \delta_0$. The series and derivatives converge in L1. Let $\mathcal{F}_j = \sigma(U_1, \dots, U_j)$, $\Lambda_j = \mathbb{E}(\Lambda \mid \mathcal{F}_j)$ and $D_j = \Lambda_j - \Lambda_{j-1}$. Each Λ_j is nonnegative of mass λ. Along the segment between consecutive intensities F has norm 1, so Taylor's remainder has norm at most $\|D_j\|_1^2/2$. The linear term has conditional mean zero given $\mathcal{F}_{j-1}$. Telescoping expectations and dividing by two gives

$$\mathrm{TV}(\mathbb{E}F(\Lambda), F(\mathbb{E}\Lambda)) \le \frac{1}{4} \sum_{j=1}^{N-1} \mathbb{E}\|D_j\|_1^2.$$

If $\Lambda^{(j)}$ resamples only U_j, independence gives $D_j = \mathbb{E}(\Lambda - \Lambda^{(j)} \mid \mathcal{F}_j)$. Conditional Jensen thus bounds its squared norm in expectation by that of $\Lambda - \Lambda^{(j)}$. Moving the subtree of $j+1$ changes family sizes only along the old and new parent paths, excluding the root. Each altered atom contributes at most 2ε in L1. Writing dp for depth, and using $(u+v)^2 \le 2(u^2+v^2)$, gives

$$\mathbb{E}\|\Lambda - \Lambda^{(j)}\|_1^2 \le 16\varepsilon^2 \mathbb{E}\mathrm{dp}(U_j)^2 \le 16\varepsilon^2 (H_{j-1}^2 + H_{j-1}).$$

For the last inequality, the depth pgf satisfies $g_1(s) = 1$ and $j g_{j+1}(s) = s \sum_{v \le j} g_v(s)$. Subtraction gives $g_v(s) = \prod_{i=1}^{v-1} (i - 1 + s)/i$. Thus the depth has the law of a sum of independent Bernoulli$(1/i)$ variables, and its second moment is at most $H_{v-1}^2 + H_{v-1}$. Average over the independent uniform parent $U_j \in \{1, \dots, j\}$.

Adding the three errors proves the stronger bound

$$\mathrm{TV}(\mathcal{L}(M_N), \mathrm{CP}_N) \le \varepsilon^2 \left\{ N H_N - N + 4 \sum_{j=0}^{N-2} (H_j^2 + H_j) \right\} \le \delta_\Sigma.$$

Finally $H_N \le 1 + \log N$ and $H_j \le H_N$ give $\delta_\Sigma \le (x^2/N)\{\log N + 2 + 4(H_N^2 + H_N)\} \le \delta_B$. Every singleton discrepancy is at most total variation. □

F.2 The local limit and its error

We write $\Psi_{\mathrm{L}}(t) = -it \log|t| - \frac{\pi}{2}|t|$ for real $t \ne 0$, and $\Psi_{\mathrm{L}}(0) = 0$. For $t > 0$ this is the value of $u \log u$ at $u = -it$, with the principal logarithm, and $\Psi_{\mathrm{L}}(-t) = \overline{\Psi_{\mathrm{L}}(t)}$. We need three facts about Ψ_N and Ψ_∞.

Fact 1. For real θ, $|\Psi_N(\theta) - \Psi_\infty(\theta)| \le \sum_{d \ge N} \frac{2}{d(d+1)} = \frac{2}{N}$.

Fact 2. For $0 < |\theta| \le \pi$ we have $\Psi_\infty(\theta) = -i\theta \log(-i\theta) + R(\theta)$ with $|R(\theta)| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big)$. The reason is that Ψ_∞ has a closed form in which θ enters only through $1 - e^{i\theta}$, and $1 - e^{i\theta}$ is close to $-i\theta$ for small θ. The remainder R measures the error in this replacement.

We start with the closed form. For $0 < |s| \le 1$, $s \ne 1$,

$$\sum_{d\ge 1} \frac{s^d - 1}{d(d+1)} = \chi(s) := \frac{(1-s)\log(1-s)}{s},$$

because $\frac{1}{d(d+1)} = \frac{1}{d} - \frac{1}{d+1}$, $\sum_d s^d/d = -\log(1-s)$, $\sum_d s^d/(d+1) = (-\log(1-s) - s)/s$ and $\sum_d \frac{1}{d(d+1)} = 1$. So $\Psi_\infty(\theta) = \chi(e^{i\theta})$.

Next we compare $1 - e^{i\theta}$ with $-i\theta$. Write $1 - e^{i\theta} = (-i\theta)\,\mathsf{u}$ with $\mathsf{u} = \operatorname{sinc}(\theta/2)\, e^{i\theta/2}$, where $\operatorname{sinc} y = \sin y/y$ is positive for $|y| \le \frac{\pi}{2}$. As θ tends to zero, the factor u tends to one. The argument of $-i\theta$ is $\mp\frac{\pi}{2}$ for $\pm\theta > 0$ and that of u is $\theta/2$. They add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\log(1 - e^{i\theta}) = \log(-i\theta) + \log \mathsf{u}$ for principal logarithms.

The two identities give $\chi(e^{i\theta}) = (-i\theta)\,\mathsf{u}e^{-i\theta}\big(\log(-i\theta) + \log \mathsf{u}\big)$. Subtracting $-i\theta \log(-i\theta)$ gives

$$R(\theta) = -i\theta \log(-i\theta)\big(\mathsf{u}e^{-i\theta} - 1\big) - i\theta\, \mathsf{u}e^{-i\theta} \log \mathsf{u}.$$

It remains to bound the three quantities in this formula. We show that

$$|\mathsf{u}e^{-i\theta} - 1| \le 0.631\,|\theta|, \qquad |\log \mathsf{u}| \le 0.53\,|\theta|, \qquad |\log(-i\theta)| \le |\log|\theta|| + \frac{\pi}{2}.$$

For the first, put $y = \theta/2$. Then $|\mathsf{u}e^{-i\theta} - 1| = |\operatorname{sinc} y - e^{iy}| \le (1 - \operatorname{sinc} y) + |1 - e^{iy}| \le \frac{y^2}{6} + |y| \le (\frac{1}{2} + \frac{\pi}{24})|\theta|$, where the last step uses $|\theta| \le \pi$, and $\frac{1}{2} + \frac{\pi}{24} \le 0.631$. For the second, $|\log \mathsf{u}|^2 = \log^2 \operatorname{sinc}(\theta/2) + \theta^2/4$, and $-\log \operatorname{sinc} y = \sum_{k\ge 1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log \operatorname{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and its value at $|\theta| = \pi$ gives $|\log \mathsf{u}| \le \big(\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2}\big)^{1/2}|\theta| \le 0.53\,|\theta|$. The third holds because $-i\theta$ has modulus $|\theta|$ and argument $\pm\frac{\pi}{2}$. Since $|\mathsf{u}e^{-i\theta}| \le 1$, the formula for R now gives

$$|R(\theta)| \le |\theta|\Big(|\log|\theta|| + \frac{\pi}{2}\Big) \cdot 0.631\,|\theta| + |\theta| \cdot 0.53\,|\theta| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big),$$

as $0.631 \cdot \frac{\pi}{2} + 0.53 \le 1.53$.

Fact 3. For $|\theta| \le \pi$, $-\operatorname{Re}\Psi_\infty(\theta) \ge (2/5)|\theta|$. For $0 < \theta \le 1/10$, Fact 2 gives

$$-\operatorname{Re}\Psi_\infty(\theta)/\theta \ge \pi/2 - \theta\{0.631\log(1/\theta) + 1.53\} > 2/5.$$

The subtracted expression increases on this interval. For $1/10 \le \theta \le \pi$ we use the exact identity

$$-\operatorname{Re}\Psi_\infty(\theta) = (1 - \cos\theta)\log(2\sin(\theta/2)) + \tfrac{1}{2}(\pi - \theta)\sin\theta.$$

Ball arithmetic on 305 closed intervals of width $1/100$, starting at $1/10$, proves that this exceeds $2\theta/5$. The intervals reach $3.15 > \pi$. Symmetry covers negative θ, and equality holds at zero. Appendix I gives the record.

The function $e^{\Psi_{\mathrm{L}}}$ is a characteristic function of class L. Let S_x have the law CP_∞ with parameter x, namely the law of $\sum_{d\ge 1} d\,\mathcal{N}_d$. Its characteristic function is $e^{x\Psi_\infty(\theta)}$. By Fact 2, for $t \ne 0$ and $x \ge \max(1, |t|/\pi)$,

$$x\Psi_\infty(t/x) - it\log x = \Psi_{\mathrm{L}}(t) + xR(t/x), \qquad |xR(t/x)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big). \tag{F.1}$$

So the characteristic function of $(S_x - x\log x)/x$ tends to $e^{\Psi_{\mathrm{L}}(t)}$ as $x \to \infty$, for every real t. The limit is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a probability law. It is integrable, so this law has the bounded continuous density f_L of Theorem 5.9. For $0 < c < 1$ and real t we have $\Psi_{\mathrm{L}}(ct) = c\Psi_{\mathrm{L}}(t) - ict\log c$, and hence

$$e^{\Psi_{\mathrm{L}}(t)} = e^{\Psi_{\mathrm{L}}(ct)} \cdot e^{\Psi_{\mathrm{L}}((1-c)t)}\, e^{it[(1-c)\log(1-c) + c\log c]}.$$

The last two factors form the characteristic function of a scaled and shifted copy of the law. So the law is of class L (self-decomposable), and f_L is unimodal by [38].

Proof of (a). By Fourier inversion on $\mathbb{Z}$ [42] and the substitution $\theta = t/x$,

$$x\,\mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)}\,dt, \qquad A(t) = x\Psi_N(t/x) - it\log x.$$

Let $|t| \le \pi x$. By (F.1) and Fact 1,

$$|A(t) - \Psi_{\mathrm{L}}(t)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N},$$

and by Facts 1 and 3, $\max(\mathrm{Re}\,A(t), \mathrm{Re}\,\Psi_{\mathrm{L}}(t)) \le -\frac{2}{5}|t| + \frac{2x}{N}$. Now $|e^u - e^v| \le |u - v|\,e^{\max(\mathrm{Re}\,u, \mathrm{Re}\,v)}$, and $\frac{2x}{N} \le \frac{1}{2}$ because $N \ge 4x$. So for $|t| \le \pi x$

$$|e^{A(t)} - e^{\Psi_{\mathrm{L}}(t)}| \le e^{1/2} e^{-2|t|/5}\Big[\frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N}\Big].$$

The tangent bound for $\log s$ at 8, together with $s|\log s| \le 1/e$ for $0 < s < 1$, gives

$$s^2|\log s| \le s^3/8 + (\log 8 - 1)s^2 + s/e \qquad (s \ge 0).$$

Integrate the preceding estimate with $\rho = 2/5$ and $\int_0^\infty t^k e^{-\rho t}\,dt = k!/\rho^{k+1}$. The coefficient of $\log x/x$ is

$$\frac{e^{1/2}}{\pi}\frac{2(0.631)}{\rho^3} < 10.4,$$

and that of x/N is $2e^{1/2}/(\pi\rho) < 2.7$. The omitted Landau tail is $2e^{-\pi^2 x/2}/\pi^2 \le 4/(\pi^4 x)$. Including it, the remaining coefficient of $1/x$ is at most

$$\frac{e^{1/2}}{\pi}\left[0.631\left\{\frac{6}{8\rho^4} + \frac{2(\log 8 - 1)}{\rho^3} + \frac{1}{e\rho^2}\right\} + \frac{2(1.53)}{\rho^3}\right] + \frac{4}{\pi^4} < 47.$$

These strict numerical inequalities are enclosed in the same record. Fourier inversion therefore proves (a), including negative integers m, where $\mathrm{CP}_N(m) = 0$.

Proof of (b). Since $\varepsilon \le \frac{1}{4}$, we have $N \ge 4x$. By Theorem 5.8, $|xf_N(m) - x\,\mathrm{CP}_N(m)| \le x\delta_\Sigma \le x\delta_B$. Add (a).

Proof of (c). Drop the index n and let S_x have law CP_∞. Adding the independent Poisson jumps of sizes $d \ge N$ couples CP_N to S_x at cost at most $1 - e^{-x\sum_{d\ge N} 1/(d(d+1))} = 1 - e^{-x/N}$. Thus

$$\mathrm{TV}(\mathcal{L}(M_N), \mathcal{L}(S_x)) \le \delta_\Sigma + 1 - e^{-x/N} \longrightarrow 0$$

under the weaker condition $x^2\log^2 N/N \to 0$. Equation (F.1) and Lévy's continuity theorem give $(S_x - x\log x)/x \Rightarrow f_L$ as $x \to \infty$. Total variation contracts under this affine map, so the same weak limit holds for M_N. At the crossing $x_N \sim 2\log N$, and both conditions hold. □

F.3 The Landau mode and its curvature

Step 1. A real integral. We claim that for every real λ and $k \ge 0$,

$$f_L^{(k)}(\lambda) = \frac{1}{\pi}\int_0^\infty (-s)^k\, e^{-s\log s - \lambda s}\sin(\pi s)\,ds. \tag{F.2}$$

For $k = 0$ this is [29, Eq. (4.3)], and we include a short proof. Since $\Psi_{\mathrm{L}}(-t) = \overline{\Psi_{\mathrm{L}}(t)}$, the definition of f_L gives $f_L(\lambda) = \frac{1}{\pi}\,\mathrm{Re}\int_0^\infty \Phi(-it)\,dt$ with $\Phi(u) = \exp(u\log u + \lambda u)$. The function Φ is holomorphic off the negative real axis. Put $u = \rho e^{i\varphi}$ with $-\pi < \varphi \le -\frac{\pi}{2}$.

Then $\mathrm{Re}(u\log u)=\rho(\log\rho\cos\varphi-\varphi\sin\varphi)$, where $\cos\varphi\le 0$ and $\varphi\sin\varphi\ge 0$. For $\log\rho\ge|\lambda|+1$ this gives $\mathrm{Re}(u\log u+\lambda u)\le-\rho(|\cos\varphi|+\varphi\sin\varphi)\le-\rho$, because $|\cos\varphi|+\varphi\sin\varphi\ge 1$ on $[-\pi,-\frac{\pi}{2}]$ (for $\psi=\varphi+\pi\in[0,\frac{\pi}{2}]$ it equals $\cos\psi+(\pi-\psi)\sin\psi\ge\cos\psi+\sin\psi\ge 1$). Also Φ is bounded near 0. So by Cauchy's theorem we may turn the ray $u=-it$ to the lower side of the negative real axis, where $u=-s$ and $\log u=\log s-i\pi$. There $u\log u=-s\log s+i\pi s$, and $dt=i\,du=-i\,ds$. Hence

$$f_L(\lambda)=\frac{1}{\pi}\,\mathrm{Re}\Big[-i\int_0^\infty e^{-s\log s-\lambda s}\,e^{i\pi s}\,ds\Big]=\frac{1}{\pi}\int_0^\infty e^{-s\log s-\lambda s}\sin(\pi s)\,ds.$$

The integrand decays faster than every exponential, so we may differentiate under the integral sign. This proves (F.2). Also f_L'' is bounded on the real line, because $t^2e^{\Psi_{\mathrm{L}}(t)}$ is integrable and we may differentiate the Fourier formula for f_L twice.

Step 2. Certified enclosures. Let $-0.223\le\lambda\le 0$ and $k\in\{1,2\}$, and split the integral in (F.2) at $\delta=2^{-20}$ and at 30. On $[0,\delta]$ we have $-s\log s\le 1/e$, $-\lambda s\le 0.223\,s$ and $|\sin\pi s|\le\pi s$, so this part is at most $\pi e^{1/e+0.223\delta}\delta^{k+2}/(k+2)$ in absolute value. On $[30,\infty)$ we have $s\log s+\lambda s\ge cs$ with $c=\log 30-0.223>3$, so this part is at most $\int_{30}^\infty s^2e^{-cs}\,ds\le 30^2e^{-30c}$. Both are below 10^{-17}. On $[\delta,30]$ the integrand is holomorphic in a neighborhood of the path, and we enclose the integral with the rigorous integration algorithm of [39], in ball arithmetic with 128 bits. The parameter λ enters as a ball, so one run encloses the integral for every λ in an interval. The script `certify_elephant.py` does this and reports

$$f_L'(-0.2229)\in[9.232\cdot10^{-6},9.234\cdot10^{-6}],\qquad f_L'(-0.2227)\in[-6.548\cdot10^{-6},-6.546\cdot10^{-6}],$$

and $f_L''(\lambda)\in[-0.0792,-0.0786]$ for every $\lambda\in[-0.2229,-0.2227]$. In the same way it finds that f_L' is positive at -0.22278299 and negative at -0.22278297. The two enclosures are $6.8984026\cdot10^{-10}$ and $-8.8809315\cdot10^{-10}$ with radii below 10^{-17}, and these are the smallest margins in the proof.

Step 3. The mode. Write $[u_-,u_+]=[-0.2229,-0.2227]$. By Step 2, f_L' is strictly decreasing on $[u_-,u_+]$ with $f_L'(u_-)>0>f_L'(u_+)$. So f_L' has exactly one zero λ_{L} in $[u_-,u_+]$, and f_L is strictly increasing on $[u_-,\lambda_{\mathrm{L}}]$ and strictly decreasing on $[\lambda_{\mathrm{L}},u_+]$. Since f_L is unimodal, there is a point λ_* such that f_L is nondecreasing on $(-\infty,\lambda_*]$ and nonincreasing on $[\lambda_*,\infty)$. If $\lambda_*>\lambda_{\mathrm{L}}$, this would contradict the strict decrease just to the right of λ_{L}. The strict increase on the left likewise rules out $\lambda_*<\lambda_{\mathrm{L}}$. So $\lambda_*=\lambda_{\mathrm{L}}$, and f_L is nondecreasing on $(-\infty,\lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}},\infty)$. By Step 2 again, $\lambda_{\mathrm{L}}\in(-0.22278299,-0.22278297)$. This proves (a) and (b).

Step 4. The global gap. Write $v=-0.22278298$ and $r_0=10^{-8}$, so $|\lambda_{\mathrm{L}}-v|<r_0$. Fourier inversion gives

$$\|f_L'\|_\infty\le 4/\pi^3,\qquad\|f_L^{(3)}\|_\infty\le 96/\pi^5.$$

We evaluate (F.2) at 160 bits with $\delta=2^{-24}$ and upper cutoff 30. All arguments exceed -1.23. The omitted integrals are bounded as in Step 2, replacing 0.223 by 1.23 and using $\log 30-1.23>2$.

First, 100 equally spaced midpoint evaluations of f_L'' on $[v-1/4,v+1/4]$, each enlarged by $(96/\pi^5)(1/400+r_0)$, give $f_L''<-0.04$ throughout $[\lambda_{\mathrm{L}}-1/4,\lambda_{\mathrm{L}}+1/4]$. Taylor's formula gives the required quadratic gap there. Next cover distances from v between 0.249 and 1.001 on each side by 188 closed cells of width 0.004. On a cell with midpoint y and largest possible distance d from λ_{L}, the certificate verifies

$$f_L(v)-\Big\{f_L(y)+\frac{4}{\pi^3}(0.002)\Big\}>\frac{1}{50}\min(1,d^2).$$

Since $f_L(\lambda_{\mathrm{L}})\ge f_L(v)$, this proves the gap on the cells. They overlap the curvature region and extend past distance one from the mode. Monotonicity then gives a gap of at least

$1/50$ on both remaining tails. This proves (c). The record and the formal deduction from these inequalities are in Appendix I. □

F.4 Locating the finite modes

Compare f_N at a mode with its value at the integer m' nearest to $x\log x + \lambda_{\mathrm{L}} x$. Theorem 5.9(b) approximates both by f_L, whose quadratic drop away from λ_{L} forces the mode to lie near m'.

We first prove (a). So we assume (H), $E' < \Delta_L/2$ and $m_{\max} \le N/4$, with E_1 and E' as in the statement.

Let m' be the integer nearest to $x\log x + \lambda_{\mathrm{L}} x$. This number increases in $x \ge 1$ and is at least $\lambda_{\mathrm{L}} > -\frac{1}{2}$, so $m' \ge 0$. Also $|\lambda_{m'} - \lambda_{\mathrm{L}}| \le \frac{1}{2x}$, and since $f_L'(\lambda_{\mathrm{L}}) = 0$, $f_L(\lambda_{m'}) \ge f_L(\lambda_{\mathrm{L}}) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde m$ be a mode of f_N or $\tilde m = m_N$. We claim that $f_N(\tilde m) \ge f_N(m') - 16xH_N/N^2$. For a mode of f_N this is clear. For m_N we have $f_N(m_N) + f_N(N-m_N) = 2P_N(N-m_N) \ge 2P_N(N-m') \ge f_N(m')$. Here $N - m_N \ge \lceil N/2\rceil \ge \lceil N/4\rceil \ge m_{\max}$. So f_N is nonincreasing on $\{\lceil N/4\rceil, \dots, N\}$ by Lemma 5.4. There are at least $N/4$ integers from $\lceil N/4\rceil$ to $\lceil N/2\rceil$. So Lemma E.1 (with $j = \lceil N/4\rceil \ge m_{\max}$ and $b = \lceil N/2\rceil$) gives the second inequality below, and Lemma E.2 gives the third,

$$f_N(N-m_N) \le f_N(\lceil N/2\rceil) \le \frac{\mathbb{P}(M_N \ge \lceil N/4\rceil)}{N/4} \le \frac{16xH_N}{N^2}.$$

Now Theorem 5.9(b), used at $\tilde m$ and at m', gives

$$\begin{aligned} f_L(\lambda_{\tilde m}) &\ge x f_N(\tilde m) - E_1 \ge x f_N(m') - \frac{16x^2H_N}{N^2} - E_1 \\ &\ge f_L(\lambda_{m'}) - 2E_1 - \frac{16x^2H_N}{N^2} \ge f_L(\lambda_{\mathrm{L}}) - 2E'. \end{aligned}$$

The global gap in Lemma 5.10(c) and $2E' < \Delta_L$ give $(\lambda_{\tilde m} - \lambda_{\mathrm{L}})^2 \le 2E'/\Delta_L$. This proves (a).

For (b), suppose $\log N \ge 30000$ and $\log N \le x \le 2\log N$. The elementary bounds

$$H_N \le 1 + \log N, \qquad N = e^{\log N} \ge (\log N)^8/8! \ge 160(\log N)^6$$

give $e^{4x^2/N} \le 2$ and $\delta_B \le 40(\log N)^4/N$. Consequently each of $2.7x/N$, $x\delta_B$ and $16x^2H_N/N^2$ is at most $1/(2\log N) \le \log x/x$. The rounding term is also at most $\log x/x$, since $\|f_L''\|_\infty \le 16/\pi^4$. As $\log x > 10$,

$$E' \le \frac{47 + 14.4\log x}{x} < \frac{20\log x}{x} \le \frac{20\log 30000}{30000} < \frac{1}{100}.$$

The same bounds imply (H), $\varepsilon \log N \le 1/40$, $804x\log(x+2) \le N$ and $15400x\log^2(x+1) \le N/4$. Thus Lemma E.5 gives $m_{\max} \le N/4$. Part (a) now gives an error at most $\sqrt{2000x\log x} < 50\sqrt{x\log x}$, for every stated mode. At the crossing $x_N = 2\log N - \log\log N + O(1)$ for even N, by Theorem 4.6(b). So $\log N \le x_N \le 2\log N$ for large even N. □

G Proofs for the family estimates

We prove the estimates of Appendix E, using its pathwise construction of the chains.

Proof of Lemma E.1. By Lemma 5.4, $f_N(\ell) \ge f_N(b)$ for $j \le \ell \le b$ in the first case, and $f_N(\ell) \ge f_N(j)$ in the second. Sum over ℓ. □

Proof of Lemma E.2. We use the tree of Lemma 5.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \le \sum_{v \text{ marked}} |\mathcal{T}_v|$. We call a family large if its size is at least z, and small otherwise. If $M_t^{[k_0]} \ge z$, then some marked family is large, or the small marked families have total size at least z. We use the union bound for the first event and Markov's inequality for the second. Marks and tree are independent, so

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \ge z) + \frac{\varepsilon}{z} \sum_v \mathbb{E}\big[|\mathcal{T}_v|; |\mathcal{T}_v| < z\big].$$

Extend both sums to all nonroot vertices. By the first sum in Lemma 4.3, with t in place of N, on average $\frac{t}{d(d+1)}$ such vertices have a family of size d, for $1 \le d < t$. The first sum is therefore at most $\sum_{d \ge z} \frac{t}{d(d+1)} = \frac{t}{z}$, and the second is at most $\sum_{d=1}^{z-1} \frac{t}{d+1} = t(H_z - 1)$. These bounds also hold when $z > t$. Together they give $\frac{\varepsilon t H_z}{z}$. For the mean, only the vertices $v > k_0$ can be marked and $\mathbb{E}|\mathcal{T}_v| = t/v$, so $\mathbb{E}M_t^{[k_0]} \le \varepsilon \sum_{v=k_0+1}^{t} \frac{t}{v} \le \varepsilon t \log \frac{t}{k_0}$. □

Proof of Lemma E.3. By Lemma 5.3, $\mathbb{P}(|\mathcal{T}_V| = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2} \beta_j(d)$, and $1-(j-2)\varepsilon \le (1-\varepsilon)^{j-2} \le 1$. Apply the two sums of Lemma 4.3, and sum the upper bound over $d' \ge d$. □

Proof of Proposition E.4. In Lemma 5.3, keep only early first flips, namely $2 \le j \le J := \lfloor N/8 \rfloor$. The omitted terms with $j > J$ are nonnegative.

Step 1. Fix the first flip. Fix $2 \le j \le J$ and put $\Gamma_d = d - Y_d + Z_{N-d}^{(j)}$ for $1 \le d \le N-j+1$. The inner sum in Lemma 5.3 is $Q_j(m) = \mathbb{E}S_j$ with $S_j = \sum_d \beta_j(d) \mathbf{1}\{\Gamma_d = m\}$. Let $\mathcal{E}$ be the event $\{Z_{N-1}^{(j)} \le m-1,\ Y_{2m} \le m\}$. We claim that

$$S_j \ge \beta_j(m)\Big(\mathbf{1}_{\mathcal{E}} - \frac{2(j-2)}{N} Y_{2m}\Big).$$

Off $\mathcal{E}$ the right side is at most 0 and $S_j \ge 0$. So let $\mathcal{E}$ hold. As d grows by one, Y_d grows by 0 or 1 and $Z_{N-d}^{(j)}$ falls by 0 or 1. So every step of Γ lies in $\{-1, 0, 1\}$, and Γ cannot jump over m. It starts at $\Gamma_1 = 1 + Z_{N-1}^{(j)} \le m$ (note that $Y_1 = 0$). At $d^\circ = m + Y_{2m} \le 2m$ we have $\Gamma_{d^\circ} \ge d^\circ - Y_{d^\circ} \ge d^\circ - Y_{2m} = m$, since $Z^{(j)} \ge 0$ and Y is nondecreasing. (The index d° is allowed, because $2m + j \le N$.) So $\Gamma_d = m$ for some $d \le d^\circ$, and as β_j is nonincreasing, $S_j \ge \beta_j(d^\circ) = \beta_j(m + Y_{2m})$. It remains to compare $\beta_j(m+y)$ with $\beta_j(m)$ for $0 \le y \le m$. By (E.2),

$$\frac{\beta_j(m+y)}{\beta_j(m)} = \prod_{i=1}^{y}\Big(1 - \frac{j-2}{N-m-i}\Big) \ge 1 - \frac{2y(j-2)}{N}.$$

Here $N - m - i \ge N/2$, so each fraction is at most $\frac{2(j-2)}{N} \le \frac{1}{4}$, and a product of factors $1 - q_i$ with $0 \le q_i \le 1$ is at least $1 - \sum_i q_i$. This proves the claim. Now $\mathbf{1}_{\mathcal{E}} \ge 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z_{N-1}^{(j)} \ge m\}$. Taking means in the claim gives

$$Q_j(m) \ge \beta_j(m)\big[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z_{N-1}^{(j)} \ge m)\big] - \frac{2(j-2)}{N} \mathbb{E}Y_{2m}\, \beta_j(m).$$

Step 2. The chance that Y or Z is large. Lemma E.2 gives

$$\mathbb{P}(Y_{2m} > m) \le 2\varepsilon H_{m+1}, \qquad \mathbb{P}(Z_{N-1}^{(j)} \ge m) \le \frac{xH_m}{m}, \qquad \mathbb{E}Y_{2m} \le 2\varepsilon m \log N.$$

None of these bounds depends on j. Put $p = 2\varepsilon H_{m+1} + xH_m/m$. If $p > 1$, then $e_1'(m) > 1$ and there is nothing to prove. So let $p \le 1$.

Step 3. Sum over j. By Lemma 5.3, $f_N(m) \geq \sum_{j=2}^{J} \mathbb{P}(V=j)\, Q_j(m)$. With Steps 1 and 2 this gives

$$f_N(m) \geq (1-p) \sum_{j=2}^{J} \mathbb{P}(V=j)\, \beta_j(m) - \frac{4\varepsilon m \log N}{N} \sum_{j=2}^{J} \mathbb{P}(V=j)\,(j-2)\,\beta_j(m).$$

In the second sum we use $\mathbb{P}(V=j) \leq \varepsilon$ and the second sum in Lemma 4.3. So the second term is at most

$$\frac{4\varepsilon m \log N}{N} \cdot \frac{2x(N-m-1)}{m(m+1)(m+2)} \leq \frac{x}{m(m+1)}\, 8\varepsilon \log N.$$

The first sum is $\mathbb{P}(|\mathcal{T}_V| = m)$ without the terms $j > J$. So by Lemma E.3 it is at least $\frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon \sum_{j>J} \beta_j(m)$.

Step 4. The late first flips $j > J$. A family that starts after time $N/8$ rarely reaches size m. By (E.1), $\beta_j(m) \leq \frac{j-1}{N-m} r^{j-1}$ with $r = e^{-\eta}$ and $\eta = \frac{m-1}{N-1} \leq \frac{1}{4}$. We sum this over $j - 1 \geq J$. With $\sum_{i \geq J} i r^i \leq r^J \big(\frac{J}{1-r} + \frac{1}{(1-r)^2}\big)$, $1 - e^{-\eta} \geq \eta/2$ and $N - m \geq \frac{3}{4}N$ we get $\varepsilon \sum_{j>J} \beta_j(m) \leq \frac{4\varepsilon}{3N}\big(\frac{2J}{\eta} + \frac{4}{\eta^2}\big) e^{-\eta J}$. Since $\lfloor N/8 \rfloor \geq \frac{N-1}{9}$ for $N \geq 64$, we have $\eta J \geq \frac{m-1}{9}$. Also $J \leq N/8$ and $\eta \geq \frac{m-1}{N}$. So

$$\varepsilon \sum_{j>J} \beta_j(m) \leq x\Big(\frac{1}{3(m-1)} + \frac{16}{3(m-1)^2}\Big) e^{-(m-1)/9} \leq \frac{x}{m(m+1)}\, 20 m e^{-m/9}.$$

The last step uses $m+1 \leq 3(m-1)$ and $m+1 \leq 3(m-1)^2$, which hold for $m \geq 2$, and $e^{1/9} \leq \frac{9}{8}$, since $17 \cdot \frac{9}{8} \leq 20$.

Step 5. Combine the bounds. For $0 \leq p \leq 1$ and $q \geq 0$ we have $(1-p)(1-q) \geq 1-p-q$. So Steps 3 and 4 give

$$f_N(m) \geq \frac{x}{m(m+1)}\Big(1 - p - \frac{2x}{m+2} - 20 m e^{-m/9} - 8\varepsilon \log N\Big),$$

which is the bound with $e_1'(m)$. For $e_1' \leq e_1$ we use $\log(m+1) \geq \log 3 > 1$ and $\log N \geq \log 64 > 4$. So $H_m \leq 1 + \log m \leq 2\log(m+1)$, $H_{m+1} \leq 1 + \log N \leq \frac{5}{4} \log N$ and $\frac{2x}{m+2} \leq \frac{2x \log(m+1)}{m}$. Hence $e_1'(m) \leq \frac{4x \log(m+1)}{m} + \frac{21}{2}\varepsilon \log N + 20 m e^{-m/9} \leq e_1(m)$. □

Proof of Lemma E.5. We pick a point ℓ of order $x \log x$ at which $f_N(\ell)$ is at least half of $\frac{x}{\ell(\ell+1)}$. Since f_N is nondecreasing up to its mode, a mode far to the right of ℓ would force more mass into that interval than the tail bound for $\mathbb{P}(M_N \geq \ell)$ allows.

Since $\log(x+2) \geq \log 3 > 1$, put $\ell = \lceil 200 x \log(x+2) \rceil$, so that $200 \leq \ell \leq 201 x \log(x+2)$ and $4\ell \leq 804 x \log(x+2) \leq N$. Since $\log N \geq 4$, we have $\varepsilon \leq \frac{1}{160}$. We bound the five terms of $e_1'(\ell)$ in Proposition E.4. As $(1 + \log s)/s$ decreases for $s \geq 1$, and $\log 200 \leq 6$, $\log x \leq \log(x+2)$ and $\log\log(x+2) \leq \log(x+2)$, we get $\frac{xH_\ell}{\ell} \leq \frac{x(1+\log(200x\log(x+2)))}{200x\log(x+2)} \leq \frac{7+2\log(x+2)}{200\log(x+2)} \leq \frac{9}{200}$. Next $\frac{2x}{\ell+2} \leq \frac{1}{100}$, $2\varepsilon H_{\ell+1} \leq 2\varepsilon(1+\log N) \leq \frac{1}{80} + \frac{1}{20}$ and $8\varepsilon \log N \leq \frac{1}{5}$. Finally $s e^{-s/9}$ decreases for $s \geq 9$, so $20\ell e^{-\ell/9} \leq 4000 e^{-200/9} < 10^{-5}$. The five terms add up to less than $\frac{1}{2}$, so $f_N(\ell) \geq \frac{x}{2\ell(\ell+1)}$. If $m_{\max} \geq \ell$, Lemmas E.1 (with $j = \ell$, $b = m_{\max}$) and E.2 give $(m_{\max} - \ell + 1) f_N(\ell) \leq \mathbb{P}(M_N \geq \ell) \leq x H_\ell / \ell$. So $m_{\max} \leq (\ell+1)(1+2H_\ell)$ in every case. Here $\ell + 1 \leq 202 x \log(x+2)$, because $x \log(x+2) \geq 1$. Also $H_\ell \leq 1 + \log(201 x \log(x+2)) \leq 7 + 2\log(x+2) \leq 9\log(x+2)$, so $1 + 2H_\ell \leq 19 \log(x+2)$. Hence $m_{\max} \leq 3838\, x \log^2(x+2)$. Since $(x+1)^2 \geq x+2$, we have $\log(x+2) \leq 2\log(x+1)$, and so $m_{\max} \leq 15352\, x \log^2(x+1) \leq A_0 x \log^2(x+1)$. The last claim holds because $804 x \log(x+2) \leq 4 A_0 x \log(x+2)$. □

H Signs of the second difference

We prove the sign bounds of Section 2, with the notation of Section 5.

H.1 Averaging adjacent entries

Insert each η_i between its two neighbors in ζ. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The same argument applies when $\eta_i < 0$, proving the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

H.2 Signs at the first lattice point

Put $\xi_t = P_t(1)/P_t(0)$. By (2.1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{a}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{a\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/a$ for $t \geq 2$. For $t = 2$ this holds because $a < 1-\varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \leq \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{a}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \geq 2$, and $\xi_t < 1$ whenever $t \leq a/\varepsilon$.

Now (2.2) with $k = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{a}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - a = \varepsilon > 0$. If $t < a/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > a/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ξ_t increases. So $\operatorname{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \geq \frac{1}{3}$, then $a/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-,+)$. This proves the first claim.

For the second claim let $0 \leq \varepsilon \leq 1$. The proofs of (2.1) and (2.2) use only that step $t+1$ is $+1$ with probability $r_t(k)$ given $A_t = k$, so both hold for this ε. We compute $P_3(0) = \frac{1}{2}(1-\varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1-\varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$ and $P_3(0) = P_3(1) = \frac{1}{4}$. By (2.2) with $t = 3$ and $k = 1$, $s_4 = r_3(-1)D_3(0) = (\varepsilon - \frac{a}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \neq \frac{1}{5}$. (With P_t defined by (2.1), the same computation works for every real ε, because the other root $1 + 2^{-1/2}$ of s_3 is not $\frac{1}{5}$ either. We need only $0 < \varepsilon < \frac{1}{2}$.) □

H.3 At most five sign blocks

For $3 \leq t \leq N$ let $\mathcal{Z}_t$ be the list

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N - t - 1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

We check that $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ as in Lemma 2.2, for $3 \le t \le N-1$. The first $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N-t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t-2$ entries are $D_{t+1}(b)$ for $2 \le b \le t-1$. By (2.2),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big) D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, at the right places. The coefficients satisfy $r_t(b-2) \ge \varepsilon$ and $1 - r_t(b+1) \ge 1 - \varepsilon - a = \varepsilon$, since $2 \le b \le t-1$. The last $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 2.2, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 2.3 the sign pattern p of $(s_3, \dots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

For f_N we argue in the same way with $D_t^f(m) = f_t(m+1) - 2f_t(m) + f_t(m-1)$ and the lists

$$\mathcal{Z}_t^f = \big(s_N^f, \dots, s_{t+1}^f,\ s_t^f, D_t^f(2), \dots, D_t^f(t-1),\ d_t, d_{t+1}, \dots, d_{N-1}\big), \qquad 2 \le t \le N,$$

where $s_t^f = D_t^f(1)$ and $d_u = (1 - r_u(u+1)) D_u^f(u) = (\varepsilon - \frac{a}{u}) f_u(u-1)$. The entry $D_{t+1}^f(t)$ of $\mathcal{Z}_{t+1}^f$ equals $r_t(t-2) D_t^f(t-1) + d_t$ by (2.2), and the other entries are as before. The proof of Lemma 2.3 applies to s_t^f without change, with $\xi_t = f_t(1)/f_t(0)$, $\xi_2 = \frac{\varepsilon}{1-\varepsilon}$ and $s_2^f = 1 - 3\varepsilon$. The numbers d_u are negative for $u < a/\varepsilon$ and positive for $u > a/\varepsilon$. So $\mathrm{sp}(\mathcal{Z}_2^f)$ is a subsequence of $(+,-,+)$ followed by $(-,+)$, and $\mathcal{Z}_N^f = (D_N^f(1), \dots, D_N^f(N-1))$. As before $\mathrm{sp}(\mathcal{Z}_N^f) \preceq \mathrm{sp}(\mathcal{Z}_2^f)$, and a subsequence of $(+,-,+)$ followed by $(-,+)$ is a subsequence of Π_5 once equal neighbors are merged. This argument needs only $N \ge 2$, and for $N = 2$ the lists $\mathcal{Z}_2^f$ and $\mathcal{Z}_N^f$ are the same. □

H.4 Strict positivity between the sign changes

By symmetry $D_N(N-k_1) > 0 > D_N(N-k_0)$. Suppose that $D_N(k) \le 0$ for some k with $k_1 < k < N - k_1$.

The case $D_N(k) < 0$. Then the entries of $\mathcal{Z}_N$ at $k_0 < k_1 < k < N - k_1 < N - k_0$ have the signs $-,+,-,+,-$. So $(-,+,-,+,-) \preceq \mathrm{sp}(\mathcal{Z}_N) \preceq \Pi_5$ by Theorem 2.4. This is impossible, because a subsequence of Π_5 with five terms is Π_5 itself.

The case $D_N(k) = 0$. We use the lists $\mathcal{Z}_t$ of the proof of Theorem 2.4 and number the entries of $\mathcal{Z}_t$ from 1 to $2N-t-1$. So $D_t(b)$ stands at place $N-t+b$ for $1 \le b \le t-1$. For $3 \le t \le N-1$ let W_t be the list w of Lemma 2.2 for $\zeta = \mathcal{Z}_t$ and $\eta = \mathcal{Z}_{t+1}$. Then $\mathrm{sp}(W_t) = \mathrm{sp}(\mathcal{Z}_t) \preceq \Pi_5$. The entry of $\mathcal{Z}_{t+1}$ at place j is a combination of the entries of $\mathcal{Z}_t$ at the places j and $j+1$, with nonnegative coefficients. We call these two entries its parents. For $2 \le b \le t-1$ the parents of $D_{t+1}(b)$ are $D_t(b-1)$ and $D_t(b)$, and both coefficients are positive.

For $4 \le t \le N$ let B_t be the set of integers b with $\max(2, k-N+t) \le b \le \min(k, t-2)$. It is not empty, because $2 \le k \le N-2$. We prove by downward induction on t that there are places $p_1 < p_2 < p_4 < p_5$ in $\mathcal{Z}_t$ such that the entries there have the signs $-,+,+,-$, that $D_t(b) = 0$ for every $b \in B_t$, and that all the places $N-t+b$ with $b \in B_t$ lie strictly between p_2 and p_4. For $t = N$ we have $B_N = \{k\}$, and we take the places k_0, k_1, $N-k_1$ and $N-k_0$.

Let the claim hold for $t+1$, where $4 \le t \le N-1$, with the places $p_1' < p_2' < p_4' < p_5'$. Let $b \in B_{t+1}$. Then $D_{t+1}(b) = 0$ is a combination of its two parents with positive coefficients. So the parents both vanish, or they have strictly opposite signs. In the second case one

parent is negative. In W_t it stands next to $D_{t+1}(b)$, hence between the positive entries of $\mathcal{Z}_{t+1}$ at the places p_2' and p_4', and these stand between the negative entries at p_1' and p_5'. So $(-,+,-,+,-) \preceq \operatorname{sp}(W_t) \preceq \Pi_5$, which is impossible. Hence $D_t(b-1) = D_t(b) = 0$ for every $b \in B_{t+1}$. Every element of B_t is of the form b or $b-1$ with $b \in B_{t+1}$, so D_t vanishes on B_t.

The entry of $\mathcal{Z}_{t+1}$ at p_1' is negative, so one of its parents is negative. Let $p_1 \in \{p_1', p_1'+1\}$ be its place in $\mathcal{Z}_t$, and choose p_2, p_4 and p_5 in the same way. Then $p_1 \le p_1'+1 \le p_2' \le p_2$, and $p_1 \ne p_2$ because the signs differ. In the same way $p_4 < p_5$. Let q^- and q^+ be the smallest and the largest place $N-t-1+b$ with $b \in B_{t+1}$. The places $N-t+b$ with $b \in B_t$ lie between q^- and q^++1. Since $p_2 \le p_2'+1 \le q^-$ and $p_4 \ge p_4' \ge q^++1$, and the entries at p_2 and p_4 are not zero, these places lie strictly between p_2 and p_4. This proves the claim for t.

The induction step with $t = 4$ shows that both parents of $D_5(b)$ vanish for every $b \in B_5 \subseteq \{2,3\}$. The parents of $D_5(2)$ are s_4 and $D_4(2)$, and those of $D_5(3)$ are $D_4(2)$ and s_4. So $s_4 = D_4(2) = 0$. By (2.2) with $t = 3$ and $k = 2$, and since $D_3(1) = D_3(2) = s_3$, we have $D_4(2) = r_3(0)s_3 + \big(1 - r_3(3)\big)s_3 = 2\varepsilon s_3$. So $s_3 = 0$ as well, which contradicts Lemma 2.3. □

I Data and formal verification records

The source, records and rerun commands are in the tagged repository. Each numerical record includes source hashes, software versions, precision and acceptance checks. These records and the Lean sources are included in the submission archive.

- certify_elephant.py checks the series bound giving the constant 5200 in Theorem 4.5 and the narrow Landau-mode enclosure of Lemma 5.10.
- certify_landau_gap.py checks the global gap $\Delta_L = 1/50$, the Fourier decay bound $2|\theta|/5$ and the local-limit constants 47, 10.4 and 2.7 at 160 bits with python-flint 0.9.0. Its record is data/landau_gap_certificate.json.
- certify_center.py brackets the crossings for $10 \le N \le 2000$ and checks their lower endpoints against $2/(N+2)$. The rational witnesses in data/ give the finite input to Proposition 5.7.
- certify_threshold.py, with threshold_taylor.py, threshold_hankel.py and threshold_pointwise.py, checks the parameter cover in Table 4. Its nine records are data/threshold_*.txt. The mode `check` verifies that their accepted intervals cover the stated ranges.
- certify_transition.py encloses the parameter derivative of the central curvature and the fourth spatial derivative. Its records data/transition_derivative_* use 1100 moments at 1600 bits, Python 3.12.14 and python-flint 0.8.0.
- check_critical_shape.py evaluates the two threshold endpoints with 1800 moments at 2000 bits and the conditional slope with 1200 moments, using python-flint 0.9.0. Its record is data/critical_shape_check.json.
- certify_critical_constants.py encloses C_*, B_*, H_* and the two derived coefficients on I_* with 2000 moments at 2600 bits, using python-flint 0.9.0. Its record is data/critical_constants_certificate.json. Run both refined certificates with `python -O`.
- verify_elephant_modes.py produces the finite-length illustrations in Remark 3.6. They are not certificates. The phase diagram is generated by figures/make_phase_figure.py, with data in data/limit_phase_illustration.npz.

The formal library is kagey-problems/problem131/lean/PaperD. Its MAP.md and coverage_tables.tex identify each formal statement and its explicit hypotheses. The full source inventory, build log and axiom audit are in the verification record. The audit checks 376 modules and 23,912 declarations. The classical inputs are those in Table 3, together with Hoeffding's hypergeometric bound. The transition interface assumes the certified uniform Fourier envelopes. The written proofs establish these inputs for the walk. The module EJPGlobalModeGap.lean proves the global-gap and mode-stability deductions from the certified inequalities. The compact localization and maximum-stability lemmas are in EJPBiasBoundary.lean. They assume quasiconcavity and the strict endpoint comparisons. The density regularity and full boundary continuity proof remain on paper. The sharper numerical constants and explicit length threshold are not replayed in Lean.

References

[1] G. M. Schütz and S. Trimper, *Elephants can always remember: exact long-range memory effects in a non-Markovian random walk*, Phys. Rev. E **70** (2004), 045101(R). https://doi.org/10.1103/PhysRevE.70.045101.

[2] Z. Drezner and N. Farnum, *A generalized binomial distribution*, Comm. Statist. Theory Methods **22** (1993), 3051–3063. https://doi.org/10.1080/03610929308831202.

[3] C. C. Heyde, *Asymptotics and criticality for a correlated Bernoulli process*, Aust. N. Z. J. Statist. **46** (2004), 53–57. https://doi.org/10.1111/j.1467-842X.2004.00311.x.

[4] B. James, K. James and Y. Qi, *Limit theorems for correlated Bernoulli random variables*, Statist. Probab. Lett. **78** (2008), 2339–2345. https://doi.org/10.1016/j.spl.2008.01.104.

[5] L. Addario-Berry, L. Devroye, G. Lugosi and V. Velona, *Broadcasting on random recursive trees*, Ann. Appl. Probab. **32** (2022), 497–528. https://doi.org/10.1214/21-AAP1686.

[6] E. Althaus, L. Hartung and R. Steiner, *A random walk approach to broadcasting on random recursive trees*, J. Theor. Probab. **39** (2026), article no. 65. https://doi.org/10.1007/s10959-026-01519-z.

[7] C. F. Coletti, R. Gava and G. M. Schütz, *Central limit theorem and related results for the elephant random walk*, J. Math. Phys. **58** (2017), 053303. https://doi.org/10.1063/1.4983566.

[8] B. Bercu, *A martingale approach for the elephant random walk*, J. Phys. A: Math. Theor. **51** (2018), 015201. https://doi.org/10.1088/1751-8121/aa95a6.

[9] N. Kubota and M. Takei, *Gaussian fluctuation for superdiffusive elephant random walks*, J. Stat. Phys. **177** (2019), 1157–1171. https://doi.org/10.1007/s10955-019-02414-0.

[10] H. Guérin, L. Laulin, K. Raschel and T. Simon, *On the limit law of the superdiffusive elephant random walk*, Electron. J. Probab. **30** (2025), paper no. 102, 1–25. https://doi.org/10.1214/25-EJP1362.

[11] H. Guérin, L. Laulin and K. Raschel, *A fixed-point equation approach for the superdiffusive elephant random walk*, Ann. Inst. Henri Poincaré Probab. Stat. **62** (2026), 973–1005. https://doi.org/10.1214/24-AIHP1534.

[12] H. Trinas and G. Valle, *A superdiffusive local limit theorem for the elephant random walk and the breakdown of log-concavity*, preprint, 2026. arXiv:2609.14753.

[13] H. Murahara and G. Tokumitsu, *Optimal rates in local and central limit theorems for the elephant random walk*, preprint, 2026. arXiv:2610.35019.

[14] P. Vishwakarma, *Moments approach for the elephant random walk*, preprint, 2026. arXiv:2609.11035.

[15] P. Kagey, *Problem 131*, Open Problem Collection, https://peterkagey.com/problems/131/ (accessed October 2026).

[16] A. Pemmasani, *Uniform Bessel bounds for magnetization crossings in finite Ising chains*, preprint, 2026. arXiv:2610.06935.

[17] X. Fan and Y. Qi, *A generalized elephant random walk with time-dependent memory*, preprint, 2026. arXiv:2610.16065.

[18] R. Roy, M. Takei and H. Tanemura, *The elephant random walk in the triangular array setting*, J. Appl. Probab. **62** (2025), 997–1009. https://doi.org/10.1017/jpr.2024.106.

[19] R. Kürsten, *Random recursive trees and the elephant random walk*, Phys. Rev. E **93** (2016), 032111. https://doi.org/10.1103/PhysRevE.93.032111.

[20] E. Baur, *Percolation on random recursive trees*, Random Struct. Algorithms **48** (2016), 655–680. https://doi.org/10.1002/rsa.20603.

[21] J. Engländer and S. Volkov, *Turning a coin over instead of tossing it*, J. Theor. Probab. **31** (2018), 1097–1118. https://doi.org/10.1007/s10959-016-0725-1.

[22] E. Baur and J. Bertoin, *Elephant random walks and their connection to Pólya-type urns*, Phys. Rev. E **94** (2016), 052134. https://doi.org/10.1103/PhysRevE.94.052134.

[23] J. Bertoin, *Sizes of the largest clusters for supercritical percolation on random recursive trees*, Random Struct. Algorithms **44** (2014), 29–44. https://doi.org/10.1002/rsa.20448.

[24] T. Miyazaki and M. Takei, *Limit theorems for the 'laziest' minimal random walk model of elephant type*, J. Stat. Phys. **181** (2020), 587–602. https://doi.org/10.1007/s10955-020-02590-4.

[25] D. A. Kessler and H. Levine, *Scaling solution in the large population limit of the general asymmetric stochastic Luria–Delbrück evolution process*, J. Stat. Phys. **158** (2015), 783–805. https://doi.org/10.1007/s10955-014-1143-3.

[26] D. A. Kessler and H. Levine, *Large population solution of the stochastic Luria–Delbrück evolution model*, Proc. Natl. Acad. Sci. USA **110** (2013), 11682–11687. https://doi.org/10.1073/pnas.1309667110.

[27] P. Bokes, A. Hlubinová and A. Singh, *Reversible transitions in a fluctuation assay modify the tail of Luria–Delbrück distribution*, Axioms **12** (2023), 249. https://doi.org/10.3390/axioms12030249.

[28] A. G. Pakes, *Remarks on the Luria–Delbrück distribution*, J. Appl. Probab. **30** (1993), 991–994. https://doi.org/10.2307/3214530.

[29] M. Möhle, *Convergence results for compound Poisson distributions and applications to the standard Luria–Delbrück distribution*, J. Appl. Probab. **42** (2005), 620–631. https://doi.org/10.1239/jap/1127322016.

[30] J. Schweinsberg, *Dynamics of the evolving Bolthausen–Sznitman coalescent*, Electron. J. Probab. **17** (2012), paper no. 91, 1–50. https://doi.org/10.1214/EJP.v17-2378.

[31] J. Bertoin, *On the non-Gaussian fluctuations of the giant cluster for percolation on random recursive trees*, Electron. J. Probab. **19** (2014), paper no. 24, 1–15. https://doi.org/10.1214/EJP.v19-2822.

[32] A. Iksanov and M. Möhle, *A probabilistic proof of a weak limit law for the number of cuts needed to isolate the root of a random recursive tree*, Electron. Commun. Probab. **12** (2007), 28–35. https://doi.org/10.1214/ECP.v12-1253.

[33] A. Meir and J. W. Moon, *Cutting down recursive trees*, Math. Biosci. **21** (1974), 173–181. https://doi.org/10.1016/0025-5564(74)90013-3.

[34] C. Desmarais, C. Holmgren and S. Wagner, *Broadcasting-induced colorings of preferential attachment trees*, Random Struct. Algorithms **63** (2023), 364–405. https://doi.org/10.1002/rsa.21142.

[35] T. Kardas, V. Shneer and V. Wachtel, *Broadcasting on random trees: martingale methods and error bounds*, preprint, 2026. arXiv:2606.22960.

[36] A. Gut and U. Stadtmüller, *Bernoulli elephant random walks*, J. Theor. Probab. **39** (2026), article no. 60. https://doi.org/10.1007/s10959-026-01517-1.

[37] K. S. Kölbig and B. Schorr, *A program package for the Landau distribution*, Comput. Phys. Commun. **31** (1984), 97–111. https://doi.org/10.1016/0010-4655(84)90085-7.

[38] M. Yamazato, *Unimodality of infinitely divisible distribution functions of class L*, Ann. Probab. **6** (1978), 523–531. https://doi.org/10.1214/aop/1176995474.

[39] F. Johansson, *Numerical integration in arbitrary-precision ball arithmetic*, in: Mathematical Software – ICMS 2018, Lect. Notes Comput. Sci. **10931**, Springer, 2018, 255–263. https://doi.org/10.1007/978-3-319-96418-8_30.

[40] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), 1281–1292. https://doi.org/10.1109/TC.2017.2690633.

[41] S. Dharmadhikari and K. Joag-dev, *Unimodality, Convexity, and Applications*, Academic Press, Boston, 1988.

[42] Y. Katznelson, *An Introduction to Harmonic Analysis*, 3rd ed., Cambridge University Press, Cambridge, 2004. https://doi.org/10.1017/CBO9781139165372.

[43] R. Durrett, *Probability: Theory and Examples*, 5th ed., Cambridge University Press, Cambridge, 2019. https://doi.org/10.1017/9781108591034.

[44] H. Scheffé, *A useful convergence theorem for probability distributions*, Ann. Math. Stat. **18** (1947), 434–438. https://doi.org/10.1214/aoms/1177730390.

[45] F. W. J. Olver et al. (eds.), *NIST Digital Library of Mathematical Functions*, Release 1.2.8 of 2026-09-15, https://dlmf.nist.gov.

[46] M. Dekking and D. Kong, Multimodality of the Markov binomial distribution, *J. Appl. Probab.* **48** (2011), 938–953. https://doi.org/10.1239/jap/1324046011.

[47] H. Trinas, The prodigal elephant returns: last-return laws in the superdiffusive regime, arXiv:2611.01279 (2026). https://arxiv.org/abs/2611.01279.